

MAR IVANIOS COLLEGE

(An Autonomous College Affiliated to the University of Kerala)

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DEPARTMENT OF STATISTICS

Internship Report on

**A STATISTICAL STUDY ON THE ANNUAL MARKET
PRICE, PRODUCTION OF EGG, MILK AND MEAT IN THE
STATE OF KERALA AND INFLUENCING FACTORS**

COURSE CODE: APSTD

A STATISTICAL STUDY ON THE ANNUAL MARKET PRICE, PRODUCTION OF EGG, MILK AND MEAT IN THE STATE OF KERALA AND INFLUENCING FACTORS

Internship report submitted to the University of Kerala in partial fulfillment of the requirements for the award of the Degree of Master of Science in Statistics with Specialization in Data Analytics

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Declaration

We, **Aiswarya Mohan, M J Khadeeja , Renimol Shaji , Saranya A P, Shilpa Manuel, Silpa A and Varsha M** declare that this internship report on the study on the Annual market price, production of egg, milk and meat in the state of Kerala and influencing factors is a group work by Aiswarya Mohan(1121101),M J Khadeeja(1121105), Renimol Shaji(1121107),Saranya A P(1121108),Shilpa Manuel(1121109), Silpa A (1121110) and Varsha M (1121111) except where due reference is made, and it has never been submitted and will not be submitted, either in part or full, for the award of any Degree at this or any other University or Institution of Higher Learning.

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Abstract

This report describes Internship completed in the Month of May,2023 carried out at The Directorate of Animal Husbandry, Vikas Bhavan, Thiruvananthapuram. Included is a statement of Internship objectives, description of methods, rationale and reports based on statistical analysis and study on Internship Projects. The report consists of 9 Chapters including :- (1) A Statistical Study on Trends In Market Prices Of Egg Varieties in the State of Kerala, (2) A Statistical study on Monthly Average Price Of Milk Of Cow, Goat, and Buffalo, (3) A Statistical Study On Price Of Meats, (4) Regression Analysis on the Annual Cattle Milk, Poultry Meat and Egg Production in the State of Kerala from the year 2011-2020, (5) Time-series Analysis on the Annual Cattle Meat Production in the State of Kerala from the year 1992-2021, (6) A Statistical Study on Diseases Treated in Cattles, (7) A Statistical Study on Seasonal Milk Production in Kerala. The above-mentioned are critical studies based on data equipped by The Directorate of Animal Husbandry.

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Chapter-1

Introduction

1.1 The Directorate of Animal Husbandry

The Department of Animal Husbandry was formed in 1956 and started functioning with very few institutions and staff. Today it is one of the leading development departments in Kerala and Kerala is the only state which has Artificial insemination and veterinary centres in all the Villages(panchayaths). Apart from these, the dairy Farms, Poultry Farms, Pig Farms, Goat rearing units, turkey rearing centres, duck Breeding Farm, Biological Production Unit, Diagnostic Laboratories, Farmer Training Centres etc are also working under this department. The main objectives of the Department of Animal Husbandry is to increase the production of food products such as milk, egg and meat. Through this, increasing the per capita availability of milk, egg and meat thus by increasing and ensuring the employment opportunities of the weaker sections and gradually increasing their income by expanding livestock and poultry farming and are establishing ways to make this possible by leading various projects. In addition to this, the department volunteers importance in facilitating better animal health care through alleviating the suffering of sick animals, prevention and control of animal diseases, Improving immunity and treatment facilities, and expansion of knowledge dissemination training facilities and research facilities.

Animal welfare programs in the state are mainly carried out by institutions like a multi-specialty hospital under this department, 14 district veterinary centres 51 Veterinary Polyclinics, 213 Veterinary Hospitals, 865 Veterinary Dispensaries, 4 Mobile Dispensaries, 5 Mobile Veterinary Hospitals, 9 Mobile Farm Aid Units, 9 intensive livestock development projects, 1342 Vibrant Livestock Development Scheme sub-centres, 6 Regional Poultry Breeding Centres, 2 District Poultry Breeding Centres, and 1 Central Hatchery.

The Directorate of Animal Husbandry is the state headquarters of the Animal Husbandry Department and is headed by the Director of Animal Husbandry. Directorate of Animal Husbandry coordinates all the veterinary services and animal welfare activities of the state.

The Director of Animal Husbandry is the overall head or appellate authority for the technical, administrative and financial control of all animal veterinary

establishments in the Department of Husbandry, Government of Kerala. He/she is assisted by Additional Directors, Joint Directors, Deputy Directors, Assistant Directors, Veterinary Surgeons for technical functioning. Administrative Officer, Administrative Assistants, Minister's staff support the Director in administrative matters while Finance Officer, Accounts Officer and other staff provide necessary support to the Director in financial affairs.

Chapter-2

Objectives

The main objectives of the study are:

- To analyse the trend of Monthly average market price of different type of eggs in all districts of Kerala, to fit the best model and to predict the price for the future.
- To analyse the trend of Monthly average market price of milk of cow, goat and buffalo in all districts of Kerala, to fit the best model and to predict the price for the future.
- To analyse the trend of Monthly average market price of different types of Meats in all districts of Kerala, to fit the best model and to predict the price for the future.
- To fit a best model for the different factors affecting Annual Milk production of Cattle, Annual Poultry Meat production and Annual Egg production using regression analysis.
- To analyse the trend of Annual Meat production of Cattle, to fit the best model and to predict the production for the future.
- To analyse the trend of total diseases treated in Cattle, to fit the best model and to predict the chances of occurrence of diseases in Cattle for future.
- To analyse the trend of season wise Milk production in Cattle, Buffalo and Goat, to fit the best model and to predict the season wise milk production for the future.

CHAPTER-3

Methodology

3.1 Time Series Analysis

A time series is a sequence of data points, measured typically at successive time spaced at uniform time intervals. There are mainly two types, stationary and non-stationary time series.

3.1.1 Stationary Time Series

A time series X_t is said to be strictly stationary if the joint probability distribution of observations $(X_t, X_{t+1}, \dots, X_{t+n})$ is exactly same as the joint probability distribution of the observations $(X_{t+h}, X_{t+h+1}, \dots, X_{t+h+n})$ where n, h and $(t, t+1, \dots, t+n)$ are arbitrary positive integers. The process X_t is said to be weakly stationary if it has a constant mean, finite variance and its autocovariance function (t, s) depends only on the time lag $|t - s|$. A strictly stationary process with finite variance is always weakly stationary.

3.1.2 Time Series Plot

After getting a data, the most important step in time series analysis is to plot observations against time. This graph should show up important features of time series such as trend, seasonality, outliers and discontinuity.

3.1.3 Autocorrelation Function (ACF)

Autocorrelation function of a stationary time series X_t , $\rho(k)$ at lag k is defined as the correlation at the lag k between X_t and X_{t+k} . Thus, autocorrelation at lag k is given by

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

where $\gamma(k) = \text{cov}(X_t, X_{t+k})$.

3.1.4 Partial Autocorrelation Function (PACF)

In time series analysis, the partial autocorrelation function (PACF) plays an important role in data analysis aimed at identifying the extent of the lag in an autoregressive model. The partial autocorrelation ρ_{kk} measures correlation between (time series) observations that are k time periods apart after controlling for correlations at intermediate lags. In other words, in time series data a large proportion of correlation between X_t and X_{t-k} may be due to the correlations they have with intervening lags $X_{t-1}, X_{t-2}, \dots, X_{t-k+1}$. The partial correlation removes the influence of these intervening variables.

3.1.5 Akaike's Information Criterion (AIC)

The Akaike's Information Criterion is a measure of relative quality of statistical model for a given set of data. As such, AIC provides means for model selection. AIC is given by

$$AIC = \log \hat{\sigma}_k^2 + \frac{n + 2k}{n}$$

where $\hat{\sigma}_k^2 = \frac{SSE(k)}{n}$; SSE(k) is the residual sum of squares under the model with

k regression coefficients and n is the sample size. AIC methodology attempts to find the model that best explains the data with a minimum of free parameters.

3.1.6 Autoregressive (AR) model

An autoregressive model of order p , AR(p) is given by

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \dots + \phi_p x_{t-p} + w_t$$

where x_t is stationary, w_t is a white noise with mean 0 and variance σ_w^2 , $\phi_1, \phi_2, \dots, \phi_p$ are constants called autoregressive parameters, ($\phi_p \neq 0$). The validity of this time series is only when series is made stationary. In compact form, it is written

as,

$$w_t = \varphi(B)x_t$$

where

$$\varphi(B) = 1 - \varphi_1(B) - \varphi_2(B^2) - \dots - \varphi_p(B^p)$$

and B is the backshift operator defined as $Bx_t = x_{t-1}$.

The random walk (RW) model is a special case of the AR(1) model which is non stationary. It can be expressed as

$$x_t = x_{t-1} + w_t$$

This time series is purely predicted as a stochastic model with time dependency based entirely on the previous time point t-1. On applying first differencing, random walk would result in a white noise time series (which would be stationary).

3.1.7 Moving Average (MA) Model

The moving average model of order q, MA (q) is given by

$$x_t = w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \dots + \theta_q w_{t-q}$$

where w_t is a white noise with mean 0 and variance σ_w^2 , $\theta_1, \theta_2, \dots, \theta_q$ ($\theta_q \neq 0$) are moving average parameters. Its compact form is

$$x_t = \theta(B)w_t$$

where

$$\theta(B) = 1 + \theta_1(B) + \theta_2(B^2) + \dots + \theta_q(B^q)$$

3.1.8 General Form of the ARIMA Model

For reasons to be given below, it is sometimes useful to consider a slight extension of the ARIMA model, by adding a constant term θ_0 , yielding the more general form

$$(B)z_t = (B)\nabla^d z_t = \theta_0 + \theta(B)a_t$$

where

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$$

$$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q$$

In what follows:

1. $\phi(B)$ will be called the autoregressive operator; it is assumed to be stationary, that is, the roots of $\phi(B)=0$ lie outside the unit circle.
2. $(B) = (B)\nabla^d$ will be called the generalized autoregressive operator; it is a nonstationary operator with d of the roots of $\phi(B)=0$ equal to unity, that is, d unit roots.
3. $\theta(B)$ will be called the moving average operator; it is assumed to be invertible, that is, the roots of $\theta(B)=0$ lie outside the unit circle.

When $d = 0$, this model represents a stationary process. The requirements of stationarity

and invertibility apply independently, and, in general, the operators $\phi(B)$ and $\theta(B)$ will not be of the same order.

Some Important Special Cases of the ARIMA Model

We examined some important special cases of the model, corresponding to the stationary situation, $d = 0$. The following models represent some special cases of the nonstationary model ($d \geq 1$), which seem to be common in practice.

1. The (0, 1, 1) process:

$$\begin{aligned}\nabla z_t &= a_t - \theta_1 a_{t-1} \\ &= (1 - \theta_1 B)\end{aligned}$$

corresponding to $p = 0, d = 1, q = 1, \phi(B)=1, \theta(B)=1 - \theta_1 B$.

2. The (0, 2, 2) process:

$$\begin{aligned}\nabla^2 z_t &= a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2} \\ &= (1 - \theta_1 B - \theta_2 B^2)\end{aligned}$$

corresponding to $p = 0, d = 2, q = 2, \phi(B)=1, \theta(B)=1 - \theta_1 B - \theta_2 B^2$.

3. The (1, 1, 1) process:

$$\nabla z_t - \phi_1 \nabla z_{t-1} = a_t - \theta_1 a_{t-1}$$

3.1.9 Forecasting Time Series

The use at time t of available observations from a time series to forecast its value at some future time $t + l$ can provide a basis for

- (1) Economic and business planning
- (2) Production planning
- (3) Inventory and production control
- (4) Control and optimization of industrial processes.

As originally described by Holt et al. (1963), Brown (1962), and the Imperial Chemical Industries (ICI) monograph on short term forecasting (Coutie, 1964), forecasts are usually needed over a period known as the lead time, which varies with each problem. For example, the lead time in the inventory control problem was defined by Harrison (1965) as a period that begins when an order to replenish stock is placed with the factory and lasts until the order is delivered into stock.

3.2 Regression Analysis

Regression analysis is a statistical technique for investigating and modeling the relationship between variables. A regression model that involves two or more regressors (independent variables) is called a multiple linear regression model. The model with k regressors is given by the relation

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

where y is the response/dependent variable, X_j 's are the regressors and $\beta_j, j = 0, 1, 2, \dots, k$ are called the regression coefficients which represents the expected change in the response y per unit change in X_j when all the remaining regressors are held constant and ϵ is the error term.

The major assumptions that we consider in regression analysis are as follows:

- The relationship between the response and the regressors is linear, at least approximately.
- The errors are normally distributed.

- The error term ϵ has zero mean.
- The error term ϵ has constant variance

The parameters of the model are estimated using least square estimation and the adequacy of the fitted model can be checked by analysing the residuals of the model. Residuals can be viewed as the deviation between the data and the fit. Thus, any departures from the assumptions on the errors should show up in the residuals.

3.2.1 Multicollinearity

When there are near-linear dependencies among the regressors, the problem of multicollinearity said to exist. In such cases the inferences based on the regression model can be misleading. This problem may be due to the data collection method employed, an over defined model or may be due to the wrong choice of model. Variance Inflation Factor (VIF) values are usually used to check the presence of multicollinearity.

It is given by

$$VIF = \frac{1}{R_j^2}$$

where R_j^2 is the coefficient of determination obtained when x_j is regressed on the remaining regressors.

VIF for each term in the model measures the combined effect of the dependences among the regressors on the variance of that term. Usually VIF values beyond 10 indicate that associated regression coefficients are poorly estimated because of multicollinearity.

3.2.2 Principal component analysis

The principal component analysis is concerned with explaining the variance covariance structure of a set of variables through a linear combination of these variables.

Main objectives are;

- Data reduction
- Interpretation

If there is almost as much information in a smaller number k , ($k < p$) of the linear combination as there original p variables, then k linear combinations can be replaced initial p variables. These linear combinations are called principal components.

Let $\underline{X} = (X_1 \ X_2 \ X_3 \ \dots \ X_p)^T$ be a random vector with variance covariance matrix, Σ

Consider the linear combinations,

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p = \mathbf{a}_1^T \mathbf{X}$$

$$Y_2 = a_{21}X_1 + a_{22}X_2 + \dots + a_{2p}X_p = \mathbf{a}_2^T \mathbf{X}$$

$$Y_3 = a_{31}X_1 + a_{32}X_2 + \dots + a_{3p}X_p = \mathbf{a}_3^T \mathbf{X}$$

.

$$Y_p = a_{p1}X_1 + a_{p2}X_2 + \dots + a_{pp}X_p = \mathbf{a}_p^T \mathbf{X}$$

Then variance, $V(Y_i) = \mathbf{a}_i^T \Sigma \mathbf{a}_i$

And $\text{Cov}(Y_i, Y_j) = \text{Cov}(\mathbf{a}_i^T \mathbf{X}, \mathbf{a}_j^T \mathbf{X})$

$$= \mathbf{a}_i^T \Sigma \mathbf{a}_j$$

Then the principal components are uncorrelated linear combinations $Y_1, Y_2, \dots, Y_p$ with maximum variances. First principal component is the normalised linear combination (coefficient vector of unit length) with maximum variance.

Second principal component is the normalised linear combination having second largest variance and uncorrelated with the first principal components.

The first principal component is $\mathbf{a}_1^T \mathbf{X}$ which maximise $\text{var}(\mathbf{a}_1^T \mathbf{X})$ subjected to the condition $\mathbf{a}_1^T \mathbf{a}_1 = 1$

Second principal component is another linear combination $\mathbf{b}_1^T \mathbf{X}$ which maximise $\text{var}(\mathbf{b}_1^T \mathbf{X})$ subjected to the conditions $\mathbf{b}_1^T \mathbf{b}_1 = 1$ and $\text{cov}(\mathbf{a}_1^T \mathbf{X}, \mathbf{b}_1^T \mathbf{X}) = 0$

Similarly, the i^{th} principal component is the normalised linear component $\mathbf{a}_i^T \mathbf{X}$ which maximise $\text{var}(\mathbf{a}_i^T \mathbf{X})$ and are uncorrelated with previous $i-1$ principal components.

3.3 Data collection

Data Collection (Secondary Data)

Secondary data refers to data that is collected by someone other than the primary user. Common sources of secondary data for social science include censuses, information collected by government departments, organizational records and data that was originally collected for other research purposes.

Tabulation and data visualization

The acquired data was categorized and sorted into a convenient format using MS Excel. Just like the collection and analysis of data, presentation of the collected information in a form which is attractive and informative is also an important function of statistics. Tabular presentation is one form of doing it. But for those who do not have an affinity for numbers, tables will have very little attraction. Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data.

Tools used: R, Python Programming, Microsoft Excel.

Chapter-4

A Statistical Study On Trends In Market Prices Of Egg Varieties In The State Of Kerala

4.1 State monthly market price of eggs in Kerala from 1999 -2000 to 2021-2022

year	WHITE EGG	BROWN EGG	DUCK EGG	HATCHING EGG	QUAIL EGG
1999-2000	1.45	1.96	2.45	4.45	0.93
2000-01	1.52	2.54	2.61	4.59	0.99
2001-02	1.42	2.1	2.52	4.6	0.99
2002-03	1.37	2.07	2.66	4.22	1.09
2003-04	1.46	2.24	2.95	4.28	1.02
2004-05	1.55	2.42	3.1	3.93	0.97
2005-06	1.43	2.32	2.97	3.63	0.95
2006-07	1.61	2.41	3.06	3.43	0.97
2007-08	1.91	2.72	3.44	4.09	1.03
2008-09	2.21	2.98	3.8	4.76	1.19
2009-10	2.65	3.38	4.23	5.15	1.36
2010-11	2.84	3.89	4.84	6.82	1.56
2011-12	2.99	4.22	5.12	6.7	1.66
2012-13	3.52	4.72	6.24	6.58	1.86
2013-14	3.89	5.32	6.87	6.97	2.21
2014-15	3.9	5.59	6.92	6.98	2.28
2015-16	4.2	6.2	7.44	7.25	2.38
2016-17	4.42	6.32	7.91	7.91	2.36
2017-18	4.81	7.42	10.1	11.87	2.91
2018-19	5.01	7.5	10.45	12.51	3.18
2019-20	5.2	7.75	10.62	13.24	3.25
2020-21	5.5	7.92	10.92	13.5	2.58
2021-22	5.6	8.1	11.2	13.55	2.99

The above table contains the monthly market price of different types of eggs in Kerala in the years of 1999 to 2022

4.2 Analysis On Annual Price Of White Eggs

First we are considering the monthly market price of white eggs in the years 1999 to 2022. The graph (figure 1.1) below shows the price distribution of white eggs for these years.

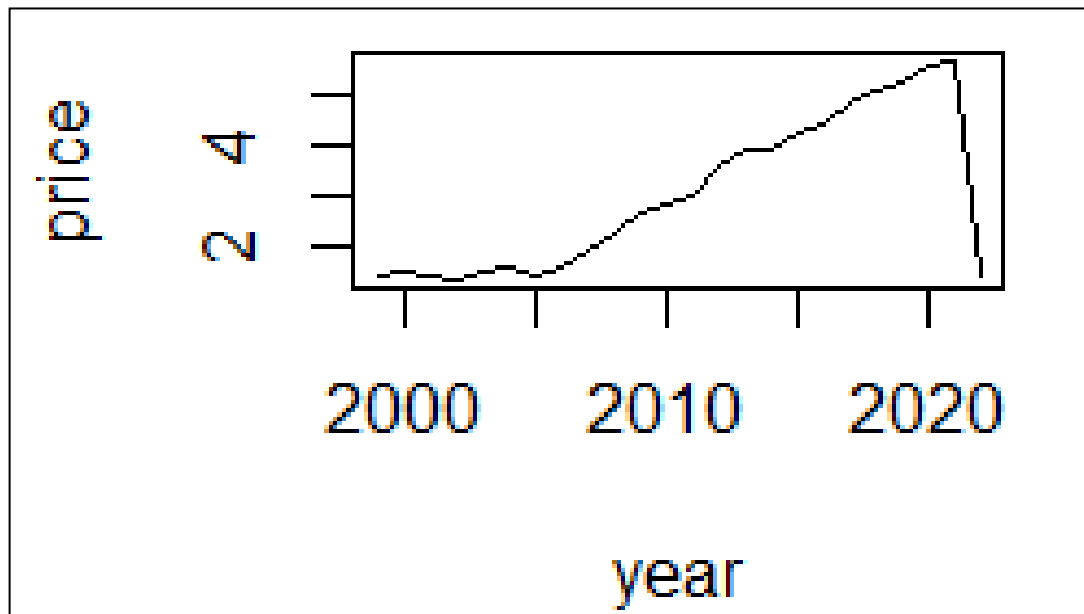


Figure 1.1

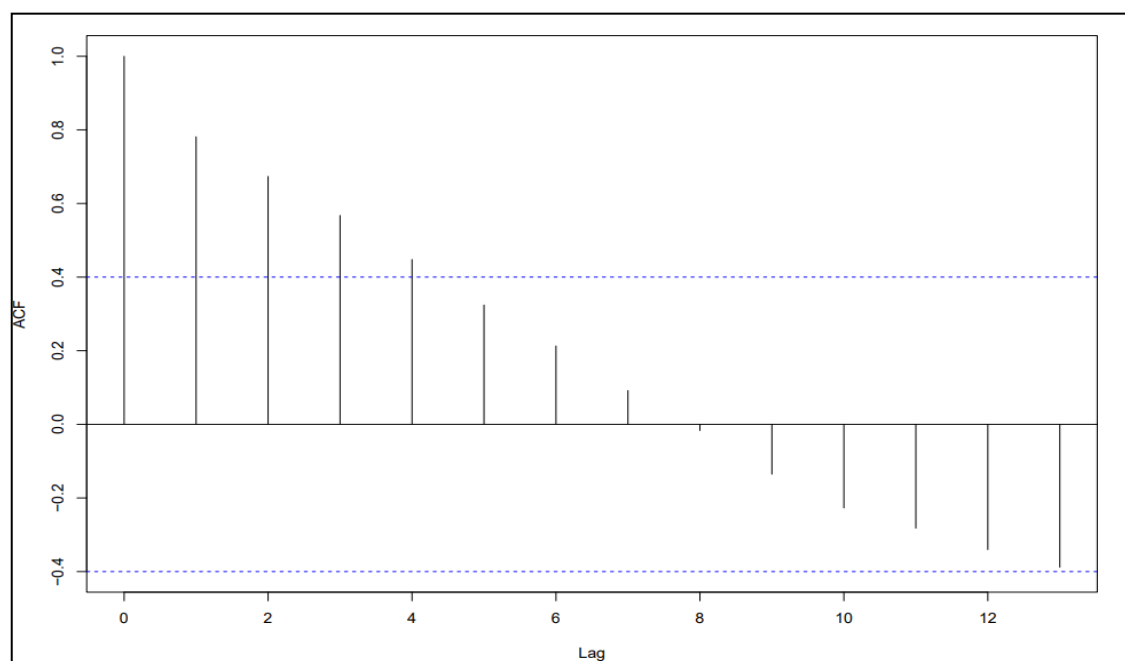


Figure 1.2

In order to test the stationarity of the data ACF test is used and the above plot (figure 1.2) is obtained. From the figure we can see that lot of the data points lies outside the error band which indicates that the data is not stationary.

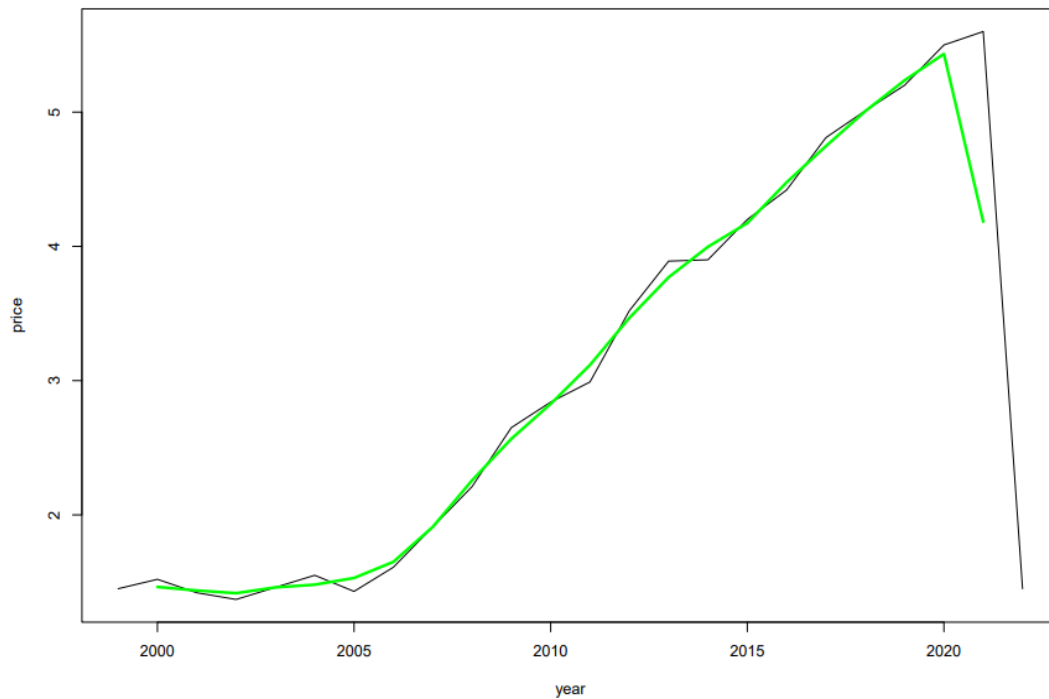


Figure 1.3

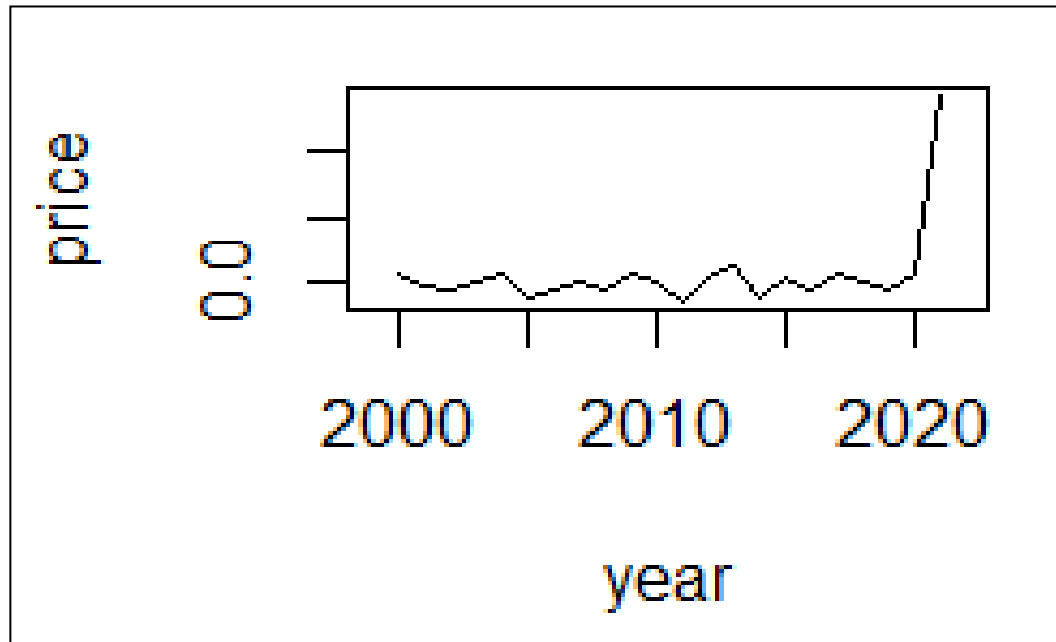


Figure 1.4

Figure 1.3 and Figure 1.4 gives the trend and seasonality of the data. It is clear that there is an increasing trend in the data till 2020 and then there is a decrease in trend .i.e, the price of the white egg increases from 2000 to 2020 and there is a decrease in price during 2020 to 2022. From figure 1.4 we can see that the seasonality is negligible.

➤ Fitting an ARIMA Model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is $ARIMA(0,2,1)$.i.e, $p=0, d=2$ and $q=1$.

- $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- $d=2$ tells that we need to take second order difference in order to obtain stationarity.
- $q=1$ tells us that this model takes into account the error term from 1 preceding/lagged value.

- Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated `Box.test()` function is used. The observed P-value = 0.9533 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e our model is good.

➤ Forecasting with ARIMA

We use the ARIMA to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The price of the white eggs for the next 8 years (2023 to 2030) are given below and the figure 1.5 gives the graphical representation of the same, with a 95% confidence.

	Point Forecast	Lo 95	Hi 95
2023	5.802895	5.477080	6.128710
2024	6.005790	5.458545	6.553035
2025	6.208685	5.430291	6.987079
2026	6.411580	5.386227	7.436933

2027	6.614475	5.325381	7.903569
2028	6.817370	5.247952	8.386788
2029	7.020265	5.154465	8.886065
2030	7.223160	5.045522	9.400798

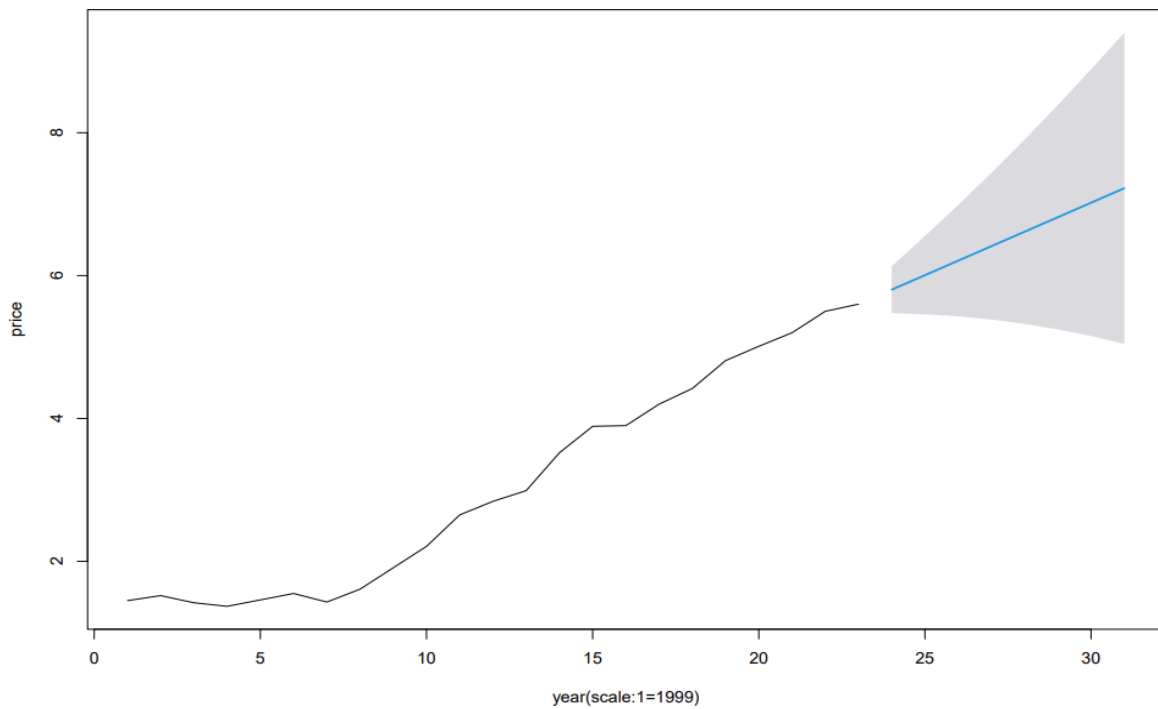


Figure 1.5

➤ RESULTS

- ☐ The best model that fits the data is ARIMA (0,2,1).
- ☐ For the next 8 years (2023 to 2030) there is a steady increase in the price of white eggs.

4.3 Analysis On Annual Price of Brown eggs

Now consider the monthly market price of brown eggs in the years 1999 to 2022. The graph (figure 2.1) below shows the price distribution of brown eggs for these years.

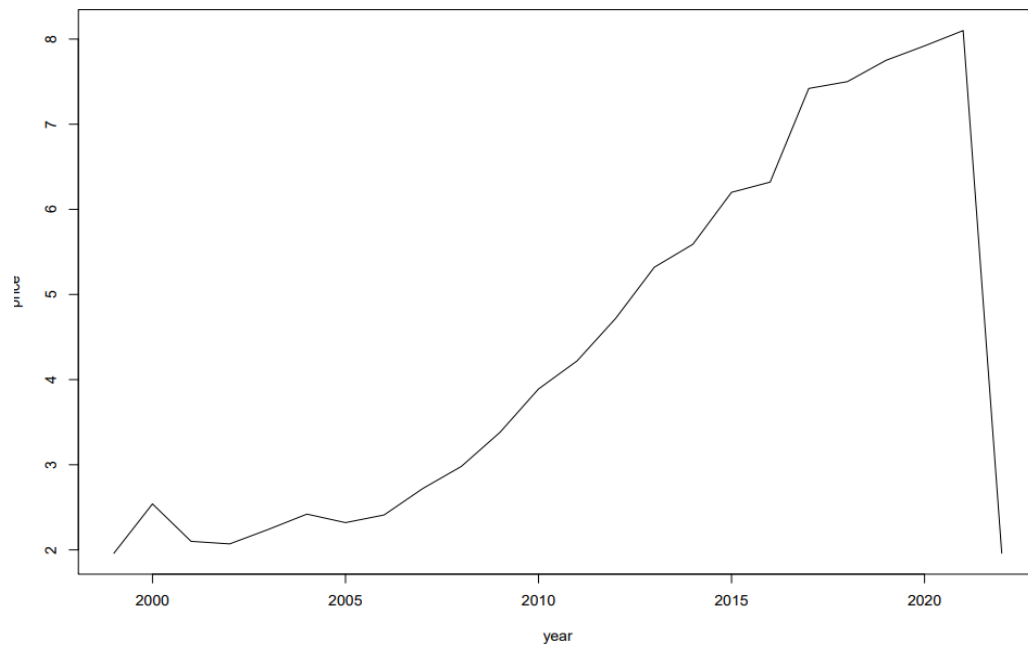


Figure 2.1

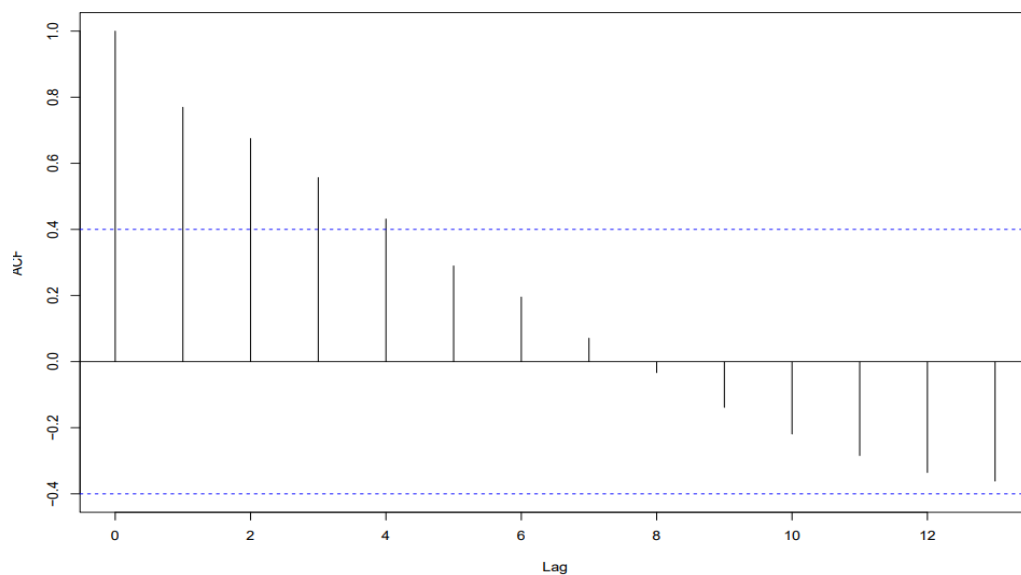


Figure 2.2

Using the ACF test the stationarity of the data is tested and the above plot (figure 2.2) is obtained. From the figure we can see that lot of the data points lies outside the error band which indicates that the data is not stationary. In the beginning we can see some autocorrelation between the time periods.

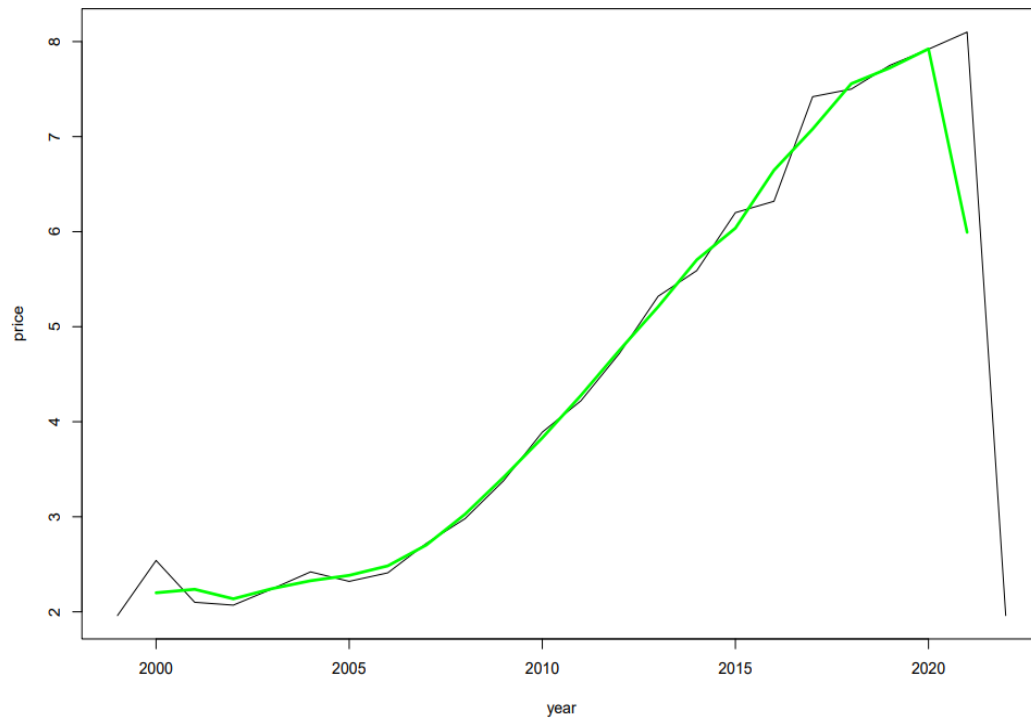


Figure 2.3

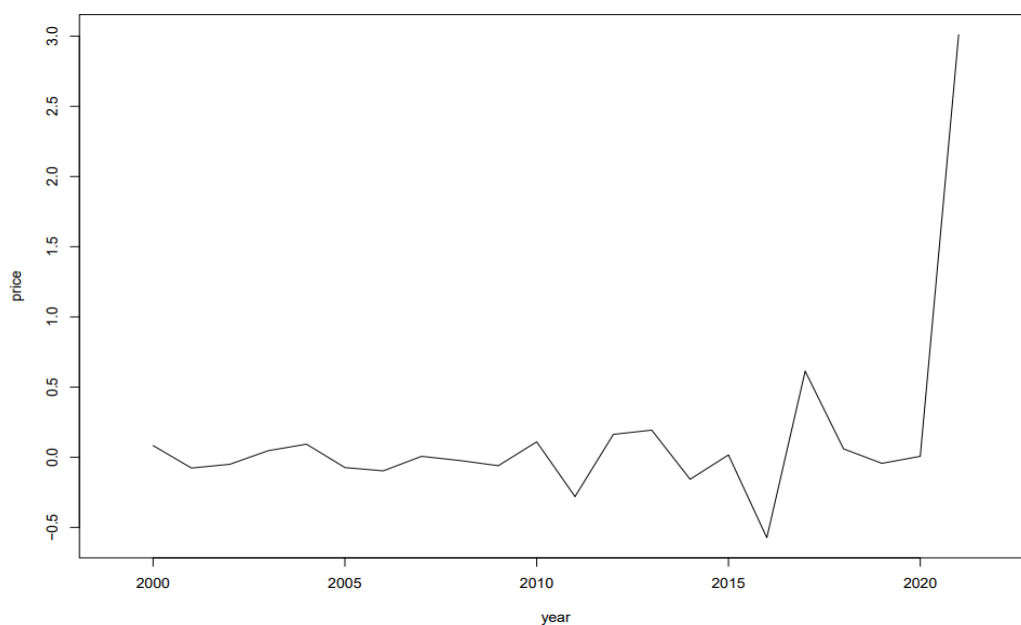


Figure 2.4

Figure 2.3 and Figure 2.4 give the trend and seasonality of the data. It is clear that there is an increasing trend in the data till 2020 and then there is a decrease in trend .i.e, the price of the brown egg increases from 2000 to 2020 and there is a decrease in price during 2020 to 2022. From figure 2.4 we can see that the seasonality is negligible.

➤ **Fitting an ARIMA Model**

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is $ARIMA(0,1,0)$. i.e, $p=0, d=1$ and $q=0$.

- ☐ $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- ☐ $d=1$ tells that we need to take first order difference in order to obtain stationarity.
- ☐ $q=0$ tells us that this model takes into account the error term from adjacent preceding/lagged value.

- ☐ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated `Box.test()` function is used. The observed P-value = 0.3153 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e our model is good.

➤ **Forecasting with ARIMA**

We use the ARIMA to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The price of the brown eggs for the next 8 years (2023 to 2030) are given below and the figure 2.5 gives the graphical representation of the same, with a 95% confidence.

	Point Forecast	Lo 95	Hi 95
2023	8.379091	7.774386	8.983795
2024	8.658182	7.803001	9.513363
2025	8.937273	7.889894	9.984652
2026	9.216364	8.006955	10.425773
2027	9.495455	8.143294	10.847615
2028	9.774545	8.293328	11.255763
2029	10.053636	8.453739	11.653534
2030	10.332727	8.622365	12.043090

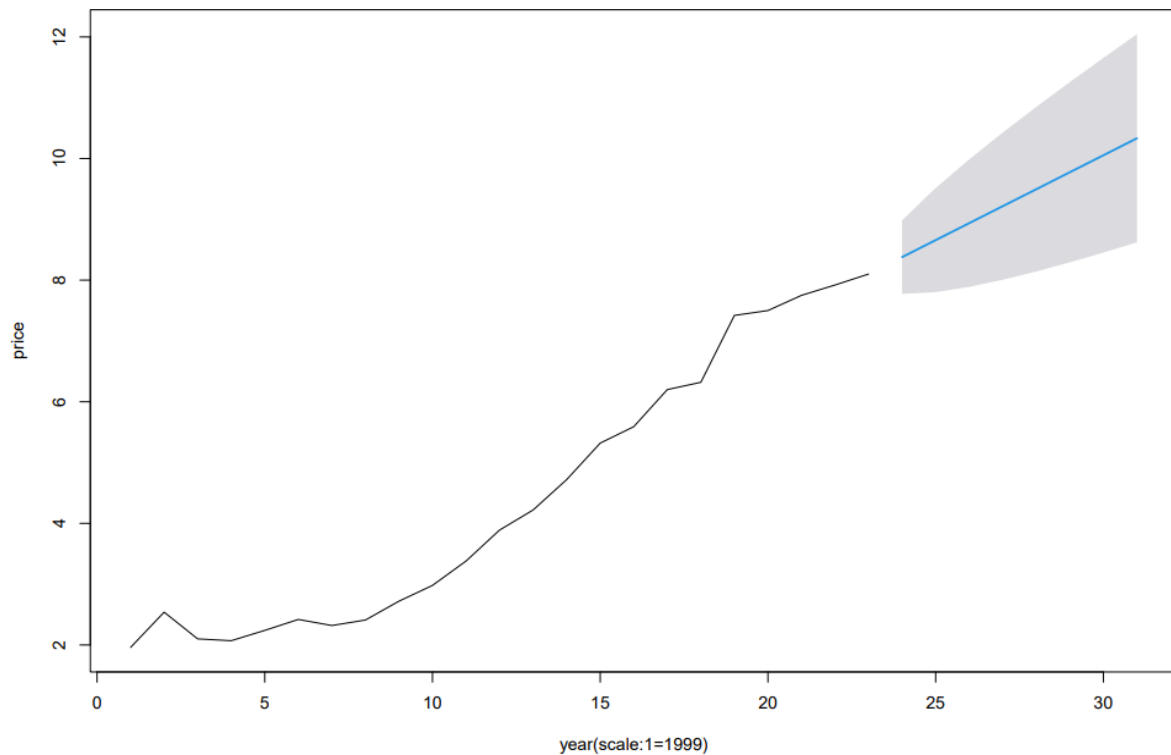


Figure 2.5

➤ RESULTS

- ☐ The best model that fits the data is ARIMA (0,1,0).
- ☐ For the next 8 years (2023 to 2030) there is a steady increase in the price of brown eggs.

4.4 Analysis On Annual Price of Hatching eggs

Consider the monthly market price of hatching eggs in the years 1999 to 2022. The graph (figure 3.1) below shows the price distribution of hatching eggs for these years.

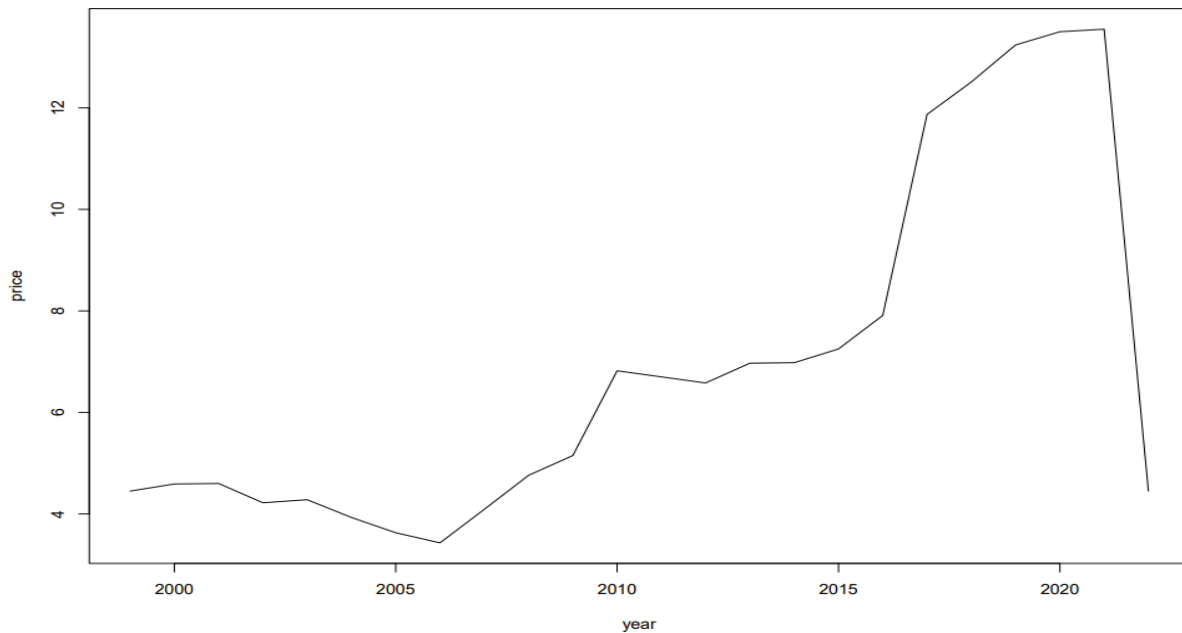


Figure 3.1

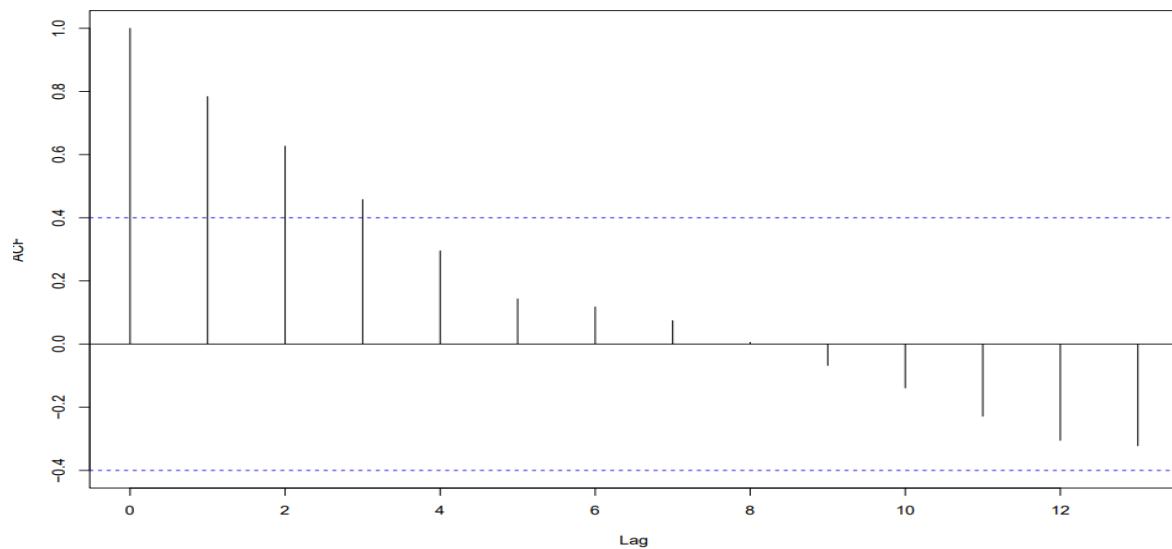


Figure 3.2

Using the ACF test the stationarity of the data is tested and the above plot (figure 3.2) is obtained. From the figure we can see that lot of the data points lies outside the error band which indicates that the data is not stationary. In the beginning we can see some autocorrelation between the time periods.

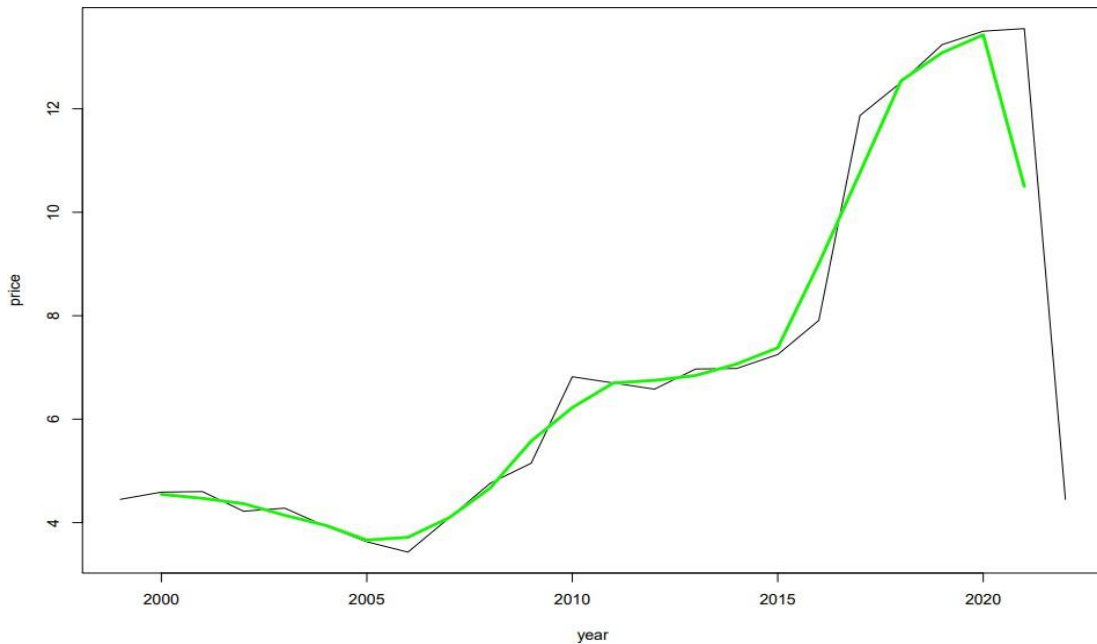


Figure 3.3

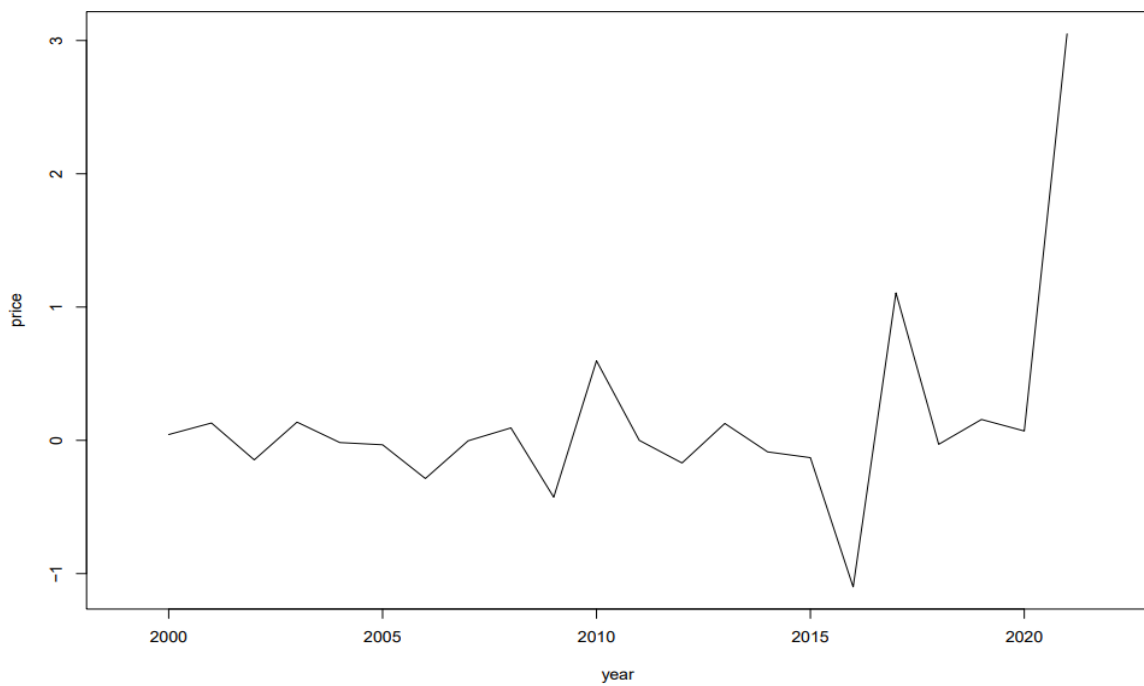


Figure 3.4

Figure 3.3 and Figure 3.4 gives the trend and seasonality of the data. We can see that there is a decreasing trend in the data till 2005-06, then the trend increases till 2009-10, then a small decrease in 2015 and last an increasing trend till 2020 from where there is another decrease .i.e, the price of the hatching egg alternates between decreasing and increasing in the years 2005, 2010, 2015, 2020 respectively. From figure 3.4 we can see that seasonality is negligible .

➤ Fitting an ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is ARIMA (0,1,0).i.e, $p=0, d=1$ and $q=0$

- $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- $d=1$ tells that we need to take first order difference to attain stationarity.
- $q=0$ tells us that this model takes into account the error term from adjacent preceding/lagged value.

- Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated `Box.test()` function is used. The observed P-value = 0.8995 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e our model is good.

➤ Forecasting with ARIMA

We use the ARIMA to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The price of the hatching eggs for the next 8 years (2023 to 2030) are given below and the figure 3.5 gives the graphical representation of the same, with a 95% confidence.

	Point Forecast	Lo 95	Hi 95
2023	13.96364	12.15586	15.77141
2024	14.37727	11.82070	16.93385
2025	14.79091	11.65975	17.92206
2026	15.20455	11.58900	18.82009
2027	15.61818	11.57588	19.66049
2028	16.03182	11.60370	20.45994
2029	16.44545	11.66254	21.22837
2030	16.85909	11.74594	21.97225

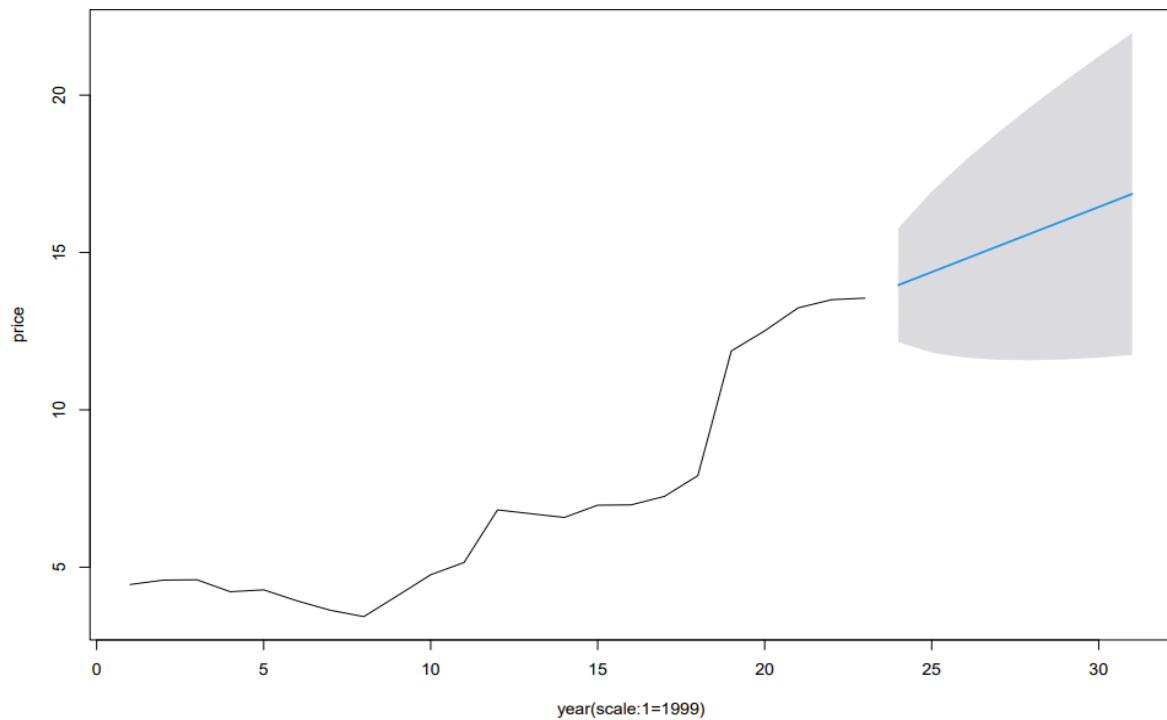


Figure 3.5

➤ RESULTS

- ☐ The best model that fits the data is ARIMA (0,1,0).
- ☐ For the next 8 years (2023 to 2030) there is a steady increase in the price of hatching eggs

4.5 Analysis On Annual Price of Duck eggs

Consider the monthly market price of duck eggs in the years 1999 to 2022. The graph (figure 4.1) below shows the price distribution of duck eggs for these years.

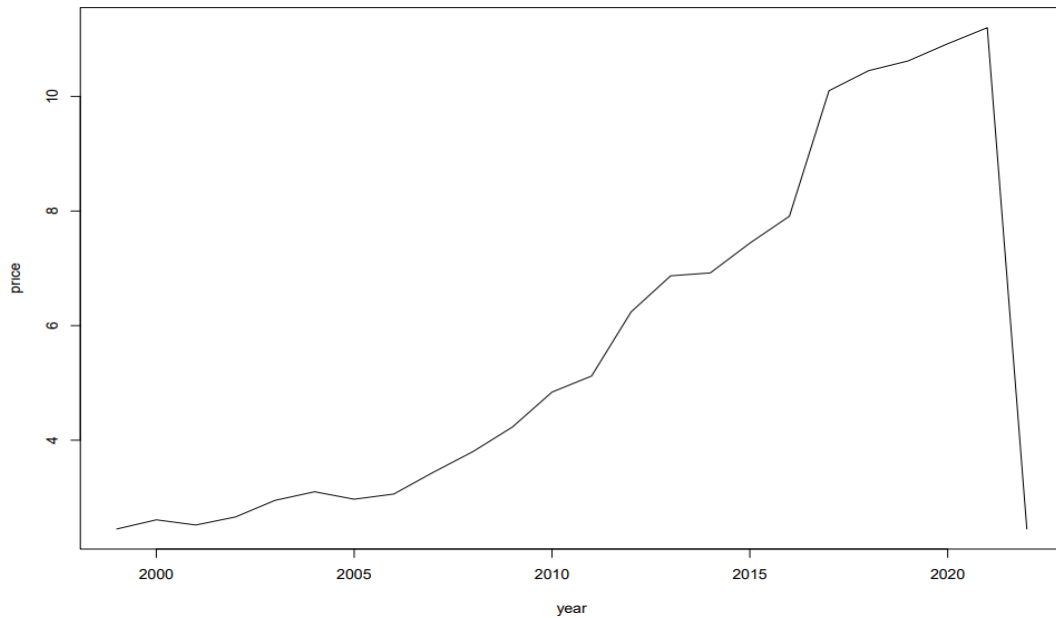


Figure 4.1

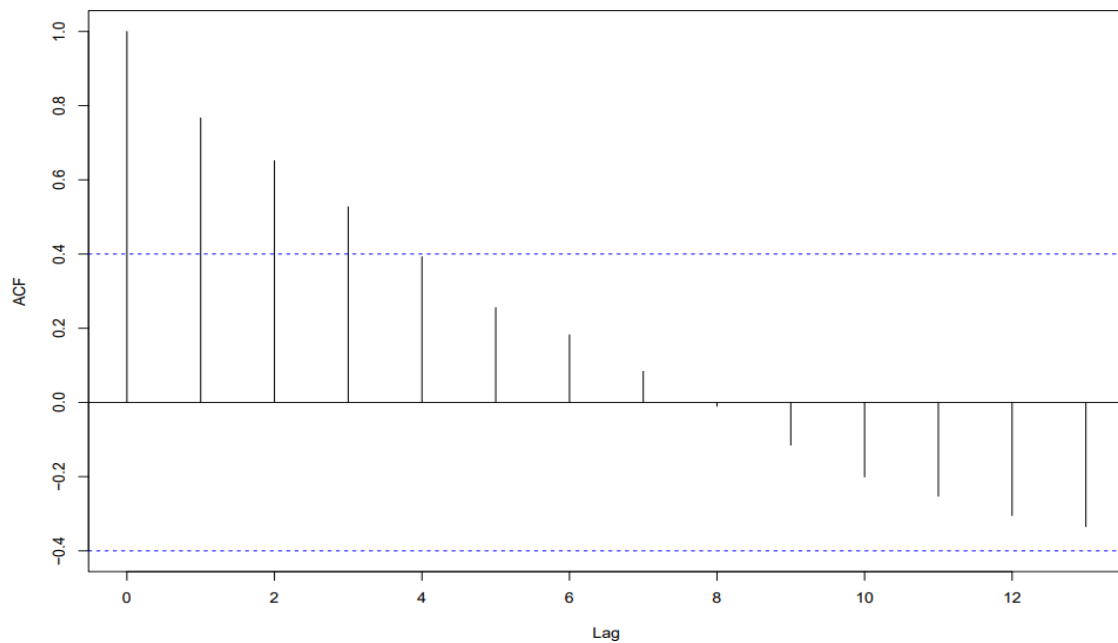


Figure 4.2

Using the ACF test the stationarity of the data is tested and the above plot (figure 4.2) is obtained. From the figure we can see that lot of the data points lies outside the error band which indicates that the data is not stationary. In the beginning we can see some autocorrelation between the time periods.

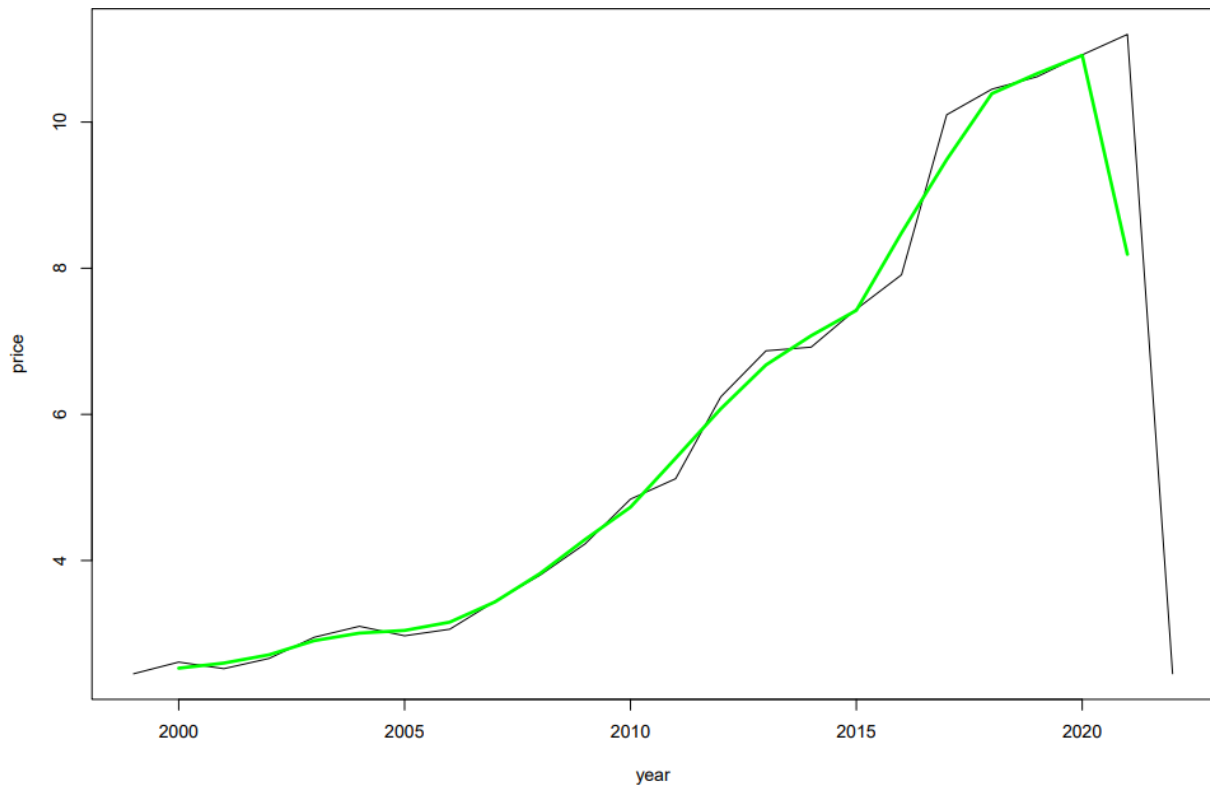


Figure 4.3

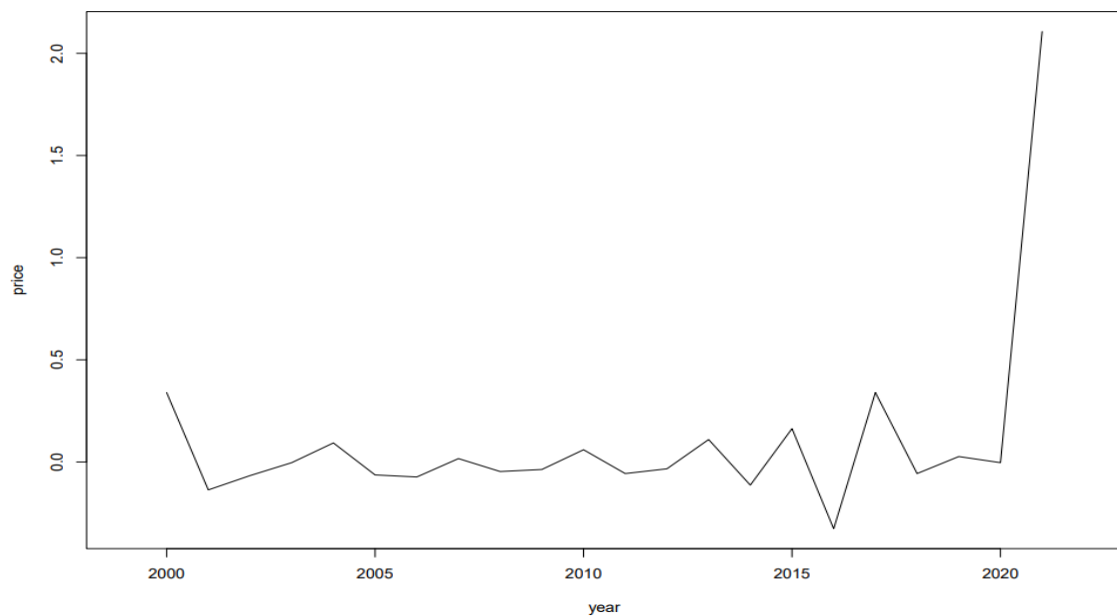


Figure 4.4

Figure 4.3 and Figure 4.4 gives the trend and seasonality of the data. It is clear that there is an increasing trend in the data till 2020 and then there is a decrease in trend .i.e, the price of the duck egg increases from 2000 to 2020 and there is a decrease in price during 2020 to 2022. From figure 4.4 we can see that the seasonality is negligible.

➤ Fitting an ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is ARIMA (0,1,0).i.e, $p=0, d=1$ and $q=0$.

- ☐ $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- ☐ $d=1$ tells that we need to take first order difference to reach stationarity.
- ☐ $q=0$ tells us that this model takes into account the error term from adjacent preceding/lagged value.

- ☐ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated `Box.test()` function is used. The observed P-value = 0.8995 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e our model is good.

➤ Forecasting with ARIMA

We use the ARIMA model to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The price of the duck eggs for the next 8 years (2023 to 2030) are given below and the figure 4.5 gives the graphical representation of the same, with a 95% confidence

	Point Forecast	Lo 95	Hi 95
2023	11.59773	10.65337	12.54208
2024	11.99545	10.65994	13.33097
2025	12.39318	10.75751	14.02885
2026	12.79091	10.90220	14.67961
2027	13.18864	11.07700	15.30027
2028	13.58636	11.27318	15.89955
2029	13.98409	11.48557	16.48261
2030	14.38182	11.71079	17.05285

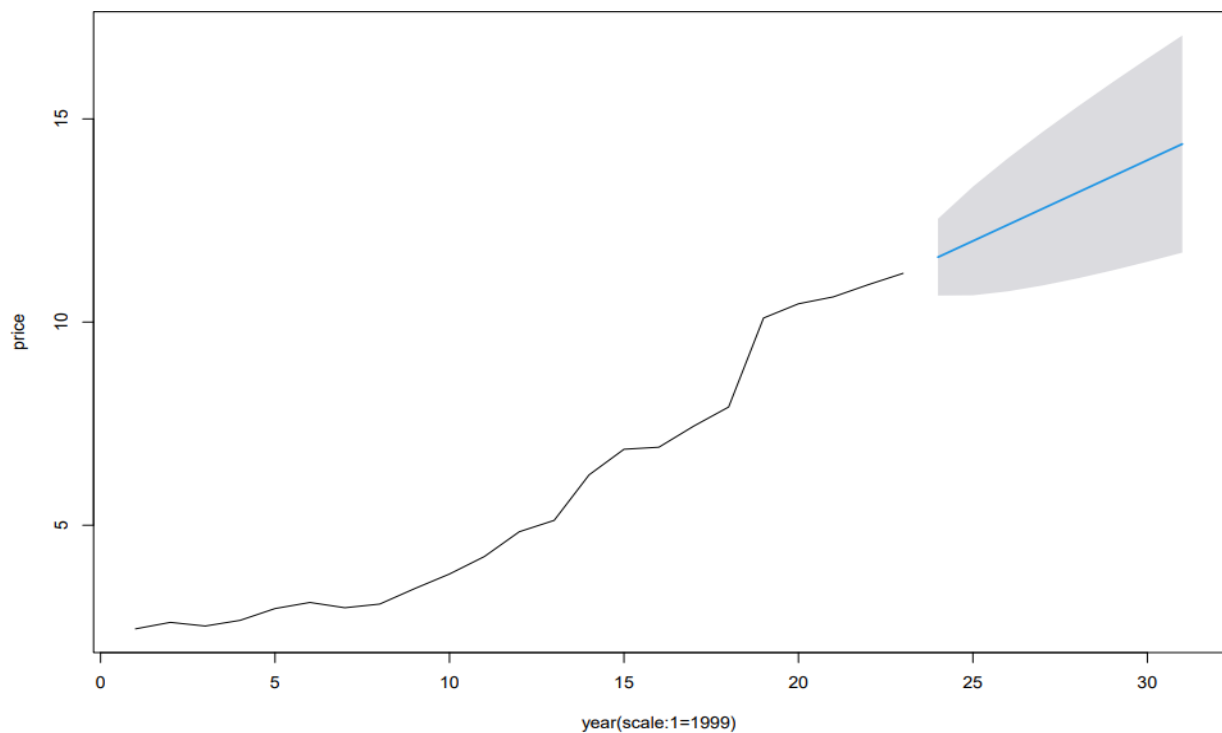


Figure 4.5

➤ RESULTS

- ☐ The best model that fits the data is ARIMA (0,1,0).
- ☐ For the next 8 years (2023 to 2030) there is a steady increase in the price of duck eggs.

4.6 Analysis On Annual Price of Quail eggs

Consider the monthly market price of quail eggs in the years 1999 to 2022. The graph (figure 5.1) shows the price distribution of quail eggs during these years.

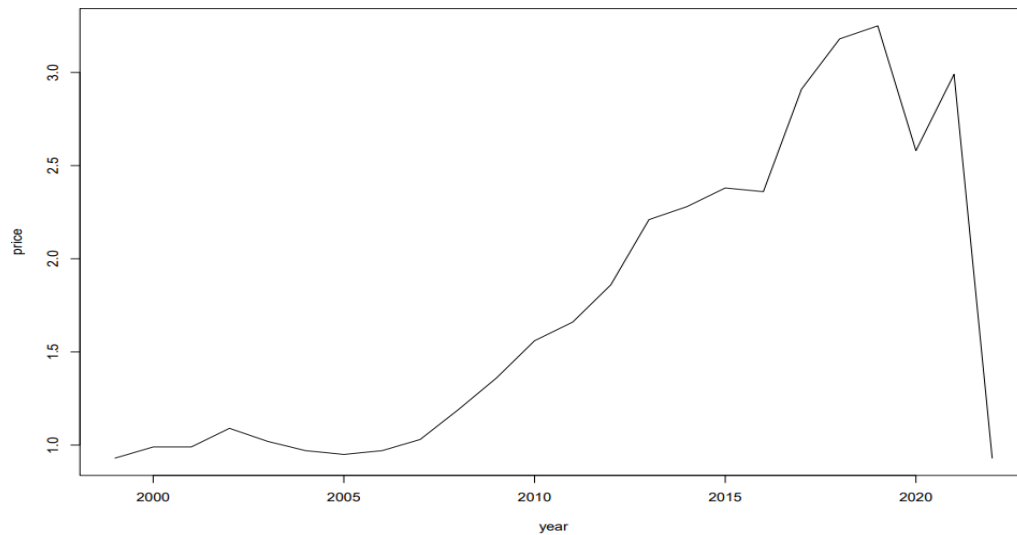


Figure 5.1

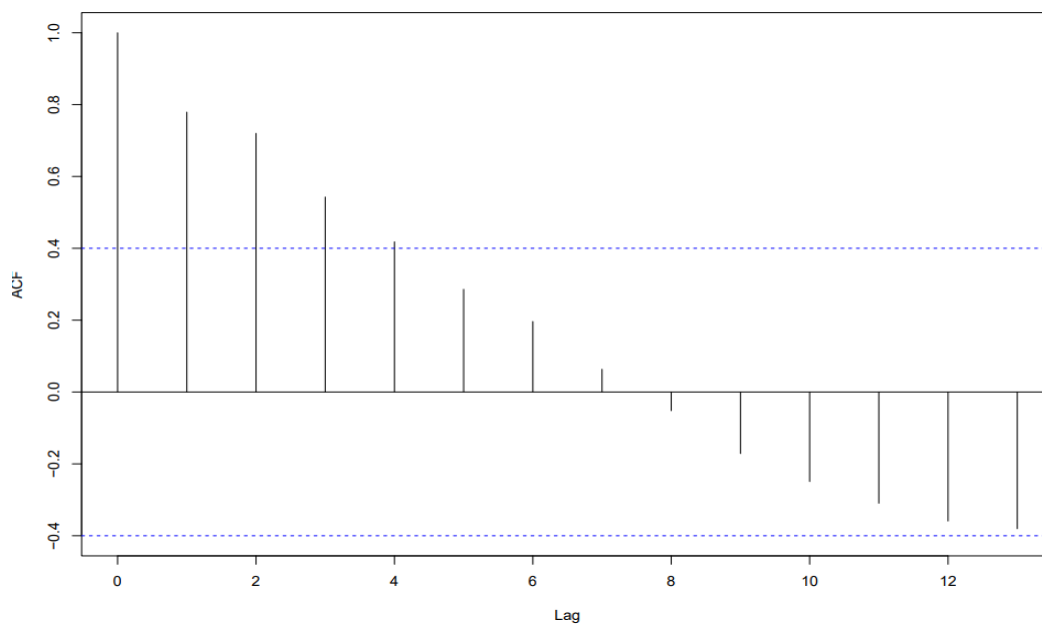


Figure 5.2

Using the ACF test the stationarity of the data is tested and the above plot (figure 5.2) is obtained. From the figure we can see that lot of the data points lies outside the error band which indicates that the data is not stationary. In the beginning we can see some autocorrelation between the time periods.

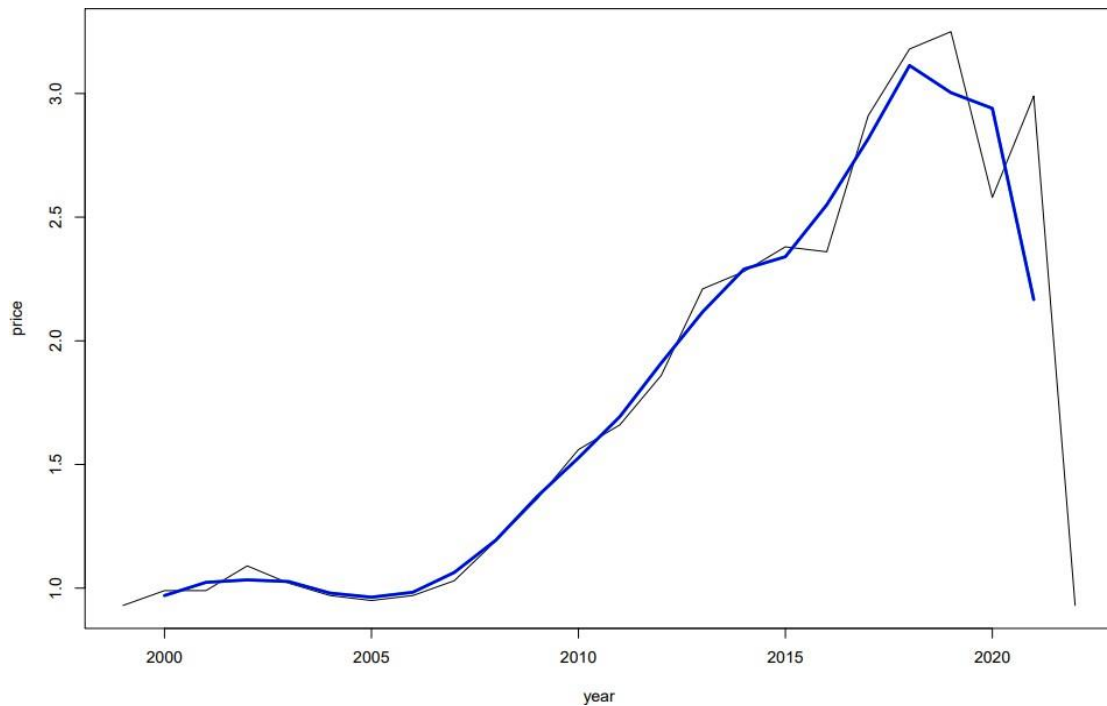


Figure 5.3

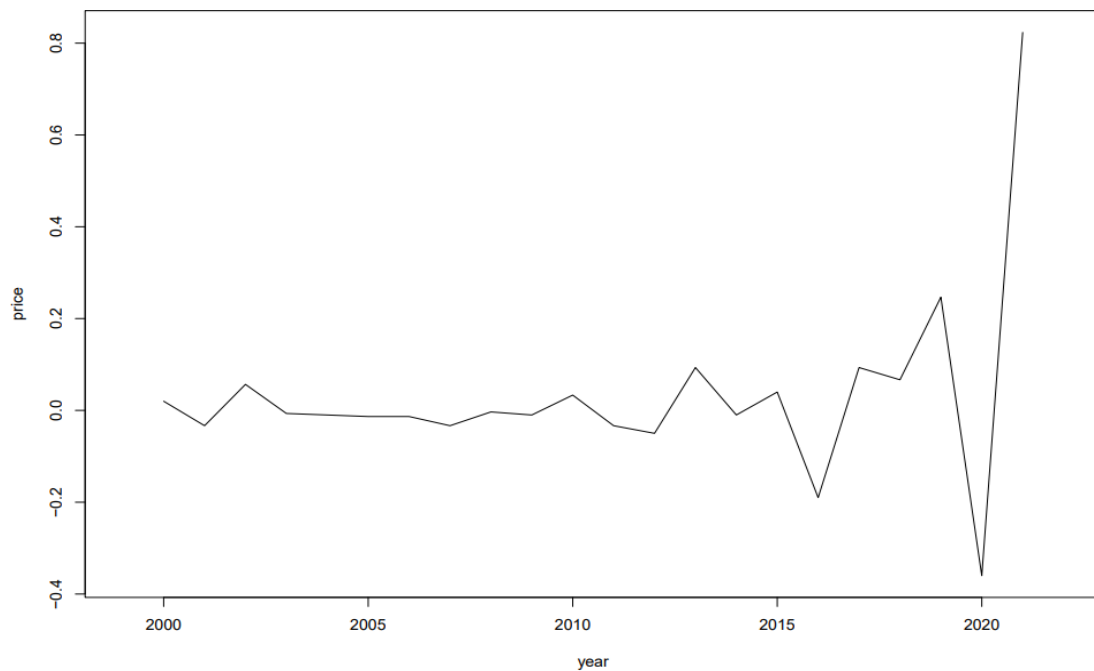


Figure 5.4

Figure 5.3 and Figure 5.4 gives the trend and seasonality of the data. It is clear that there is an increasing trend in the data till 2018 and then there is a decrease in trend. i.e, the price of the quail egg increases from 2000 to 2018 and there is a decrease in price during 2018 to 2022. From figure 5.4 we can see that the seasonality is negligible.

➤ Fitting an ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is ARIMA (0,1,0).i.e, $p=0, d=1$ and $q=0$.

- ☐ $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- ☐ $d=1$ tells that we need to take first order difference to obtain stationarity.
- ☐ $q=0$ tells us that this model takes into account the error term from adjacent preceding/lagged value.

- ☐ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated `Box.test()` function is used. The observed P-value 0.3778 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e our model is good.

➤ Forecasting with ARIMA

We use the ARIMA model to forecast future time points in the series. For this we use `forecast ()` function from the “forecast” package. The price of the quail eggs for the next 8 years (2023 to 2030) are given below and the figure 5.5 gives the graphical representation of the same, with a 95% confidence.

	Point Forecast	Lo 95	Hi 95
2023	3.083636	2.632599	3.534673
2024	3.177273	2.539410	3.815136
2025	3.270909	2.489690	4.052128
2026	3.364545	2.462471	4.266620
2027	3.458182	2.449632	4.466731
2028	3.551818	2.447007	4.656629
2029	3.645455	2.452123	4.838787
2030	3.739091	2.463365	5.014817

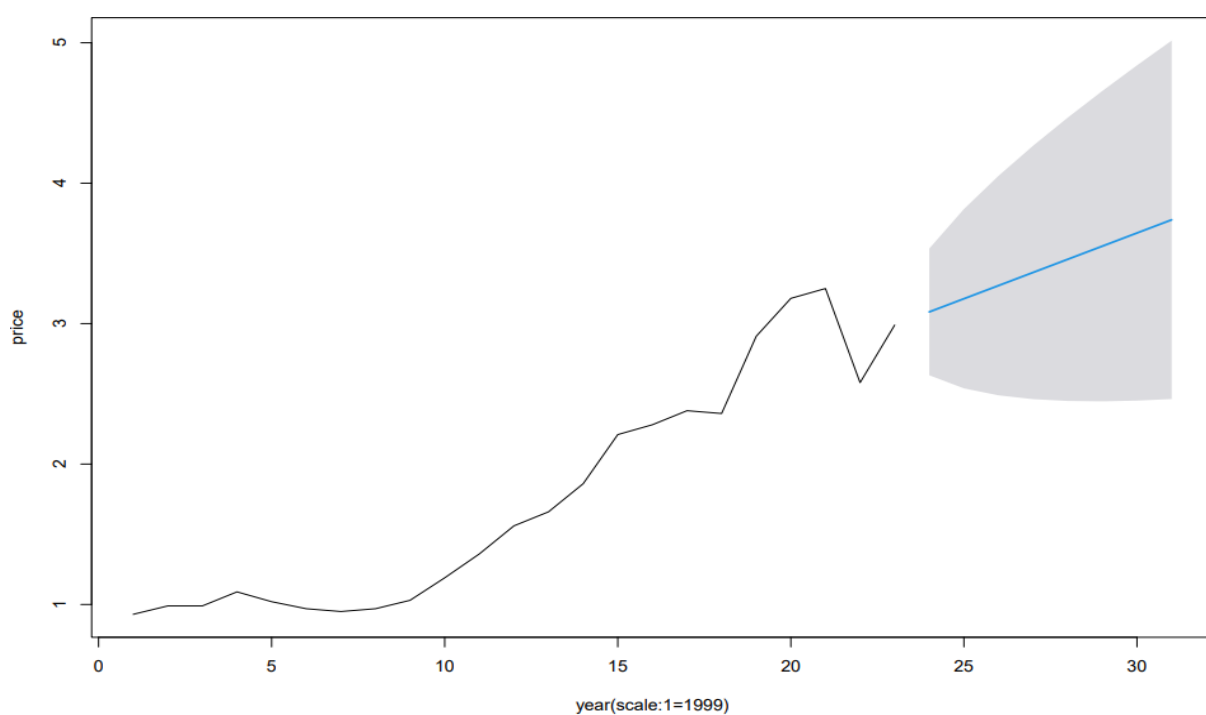


Figure 5.5

➤ RESULTS

- ☐ The best model that fits the data is ARIMA (0,1,0).
- ☐ For the next 8 years (2023 to 2030) there is a steady increase in the price of quail eggs.

4.7 Conclusion

Analysing the trends in the monthly market price of variety of eggs and predicting the price for future:

Here we have a sample of 5 varieties of eggs namely white egg, brown egg, hatching egg, duck egg and quail egg and the monthly market prices for them in Kerala from 1999 to 2022. For each varieties trends are analysed, models are fitted and prices are predicted for next 8 years separately. Then the following conclusions are reached:

- In the case of white egg we could clearly see an increasing trend in the data. On fitting the best model for the data is found to be ARIMA (0,2,1). Also for the next 8 years the model gave a steady increase in the prices.
- The data of brown eggs also provided an increasing trend in the prices. The best model that fits the data was found to be ARIMA(0,1,0). For the next 8 years the model predicted a steady increase in the price of brown egg.
- In the case of hatching egg the data had a trend which was alternating between decreasing and increasing. On fitting the best model for the data is found to be ARIMA (0,1,0). Also for the next 8 years the model gave a steady increase in the prices.
- The data of duck eggs provided an increasing trend in the prices. The best model that fits the data was found to be ARIMA (0,1,0). For the next 8 years the model predicted a steady increase in the price of duck egg.
- In the case of quail egg we have a relatively increasing trend in the data. On fitting the best model for the data is found to be ARIMA (0,1,0). Also for the next 8 years the model gave a steady increase in the prices of quail egg.

Chapter-5

A Statistical Study On Monthly Average Price Of Milk Of Cow, Goat And Buffalo

Monthly Average Price of milk of Cattle, Buffalo and Goat

Year	Monthly Average Price of MILK (Cow) / litre (Price in Rs. per litre)	Monthly Average price of Milk (Buffalo)/ litre (Prices in Rs. Per litre)	Monthly Average Price of Milk (GOAT)/litre (Prices in Rs. Per litre)
1999-2000	12.66	14.56	11.68
2000-01	12.83	15.32	12.34
2001-02	12.67	14.91	12.51
2002-03	12.84	16.31	13.27
2003-04	13.13	16.77	14.09
2004-05	13.81	17.8	14.75
2005-06	14.71	19.03	15.8
2006-07	15.05	19.61	16.56
2007-08	16.7	21.3	17.94
2008-09	19.76	24.28	21.26
2009-10	20.66	25.23	21.96
2010-11	23.37	27.61	25.51
2011-12	27.49	32.88	31.41
2012-13	31.23	37.25	37.93
2013-14	33.74	40.11	41.31
2014-15	37.15	44.86	48.54
2015-16	38.84	45.2	52.01
2016-17	41.07	58.54	64.48
2017-18	44.29	62.18	75.25
2018-19	45.19	68.23	81.43
2019-20	48.08	74.73	84.45
2020-21	49.76	81.44	90.32
2021-22	50.68519	81.6466	90.54321

5.1 Monthly Average Price of milk (cow)/litre (prices in Rs. Per litre)

We are considering the Monthly Average Price of milk of cow from the year 1999 to 2021.

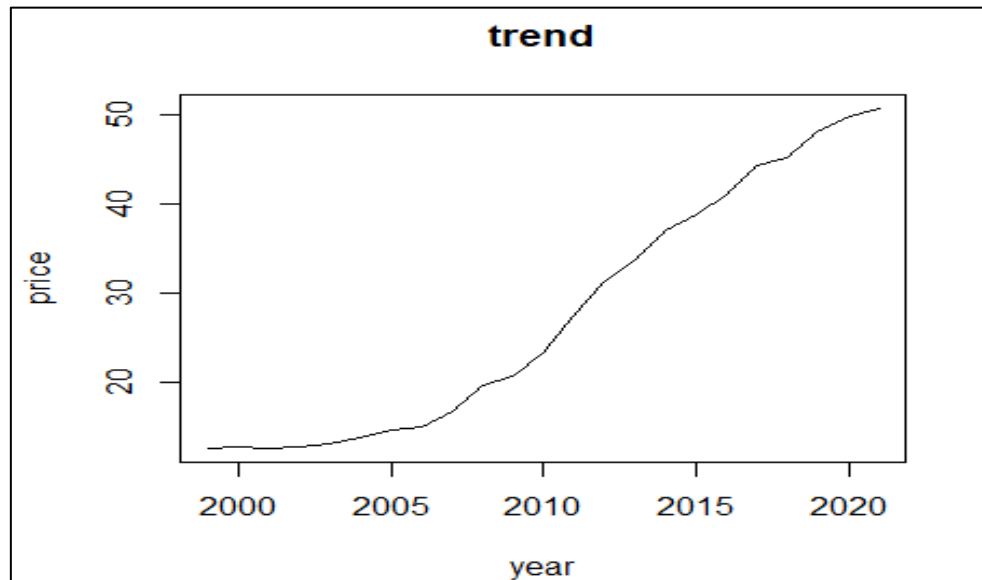


Figure 1.1

- The figure 1.1 shows the price distribution of milk (cow)/litre (prices in Rs. per litre). In order to test the stationarity of the data, ACF test is used ,

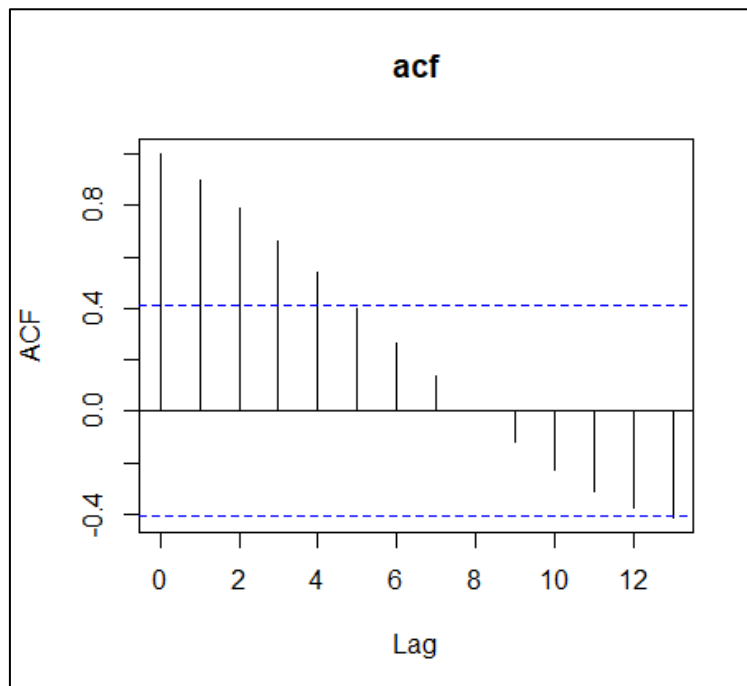


Figure 1.2

By applying ACF test, figure 1.2 is obtained. From the figure 1.2 we can see that lot of data points lies outside the error bands which indicate that the data is not stationary.

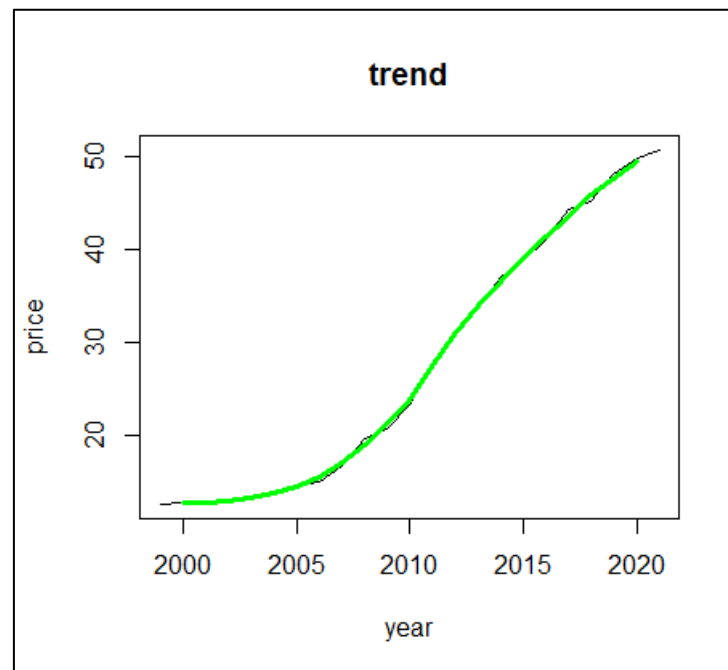


Figure 1.3

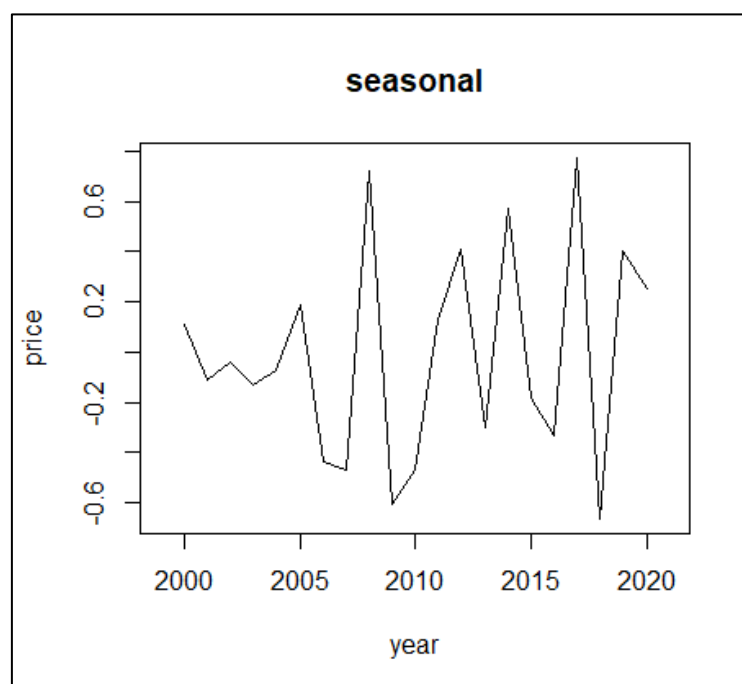


Figure 1.4

- The figure 1.3 gives the trend of the data and the figure 1.4 gives the seasonality of the data. The figure 1.3 indicates that there is a steady trend from 1999 to 2005 and an increase in trend from 2005 to 2021. i.e., the price of milk of cow shows a constant trend from 1999 to 2005. For the next 16 years, price increases to a highest extend. From figure 1.4 we can conclude that the seasonality is negligible.

➤ Fitting an ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is ARIMA (0,2,1). i.e., $p=0$, $d=2$ and $q=1$.

- $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- $d=2$ tells that the time series is not stationary so we need to take second order difference.
- $q=1$ tells us that this model takes in to account the error term from 1 preceding/lagged value.

□ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated, `Box.test()` function is used. The observed p -value = 0.7459 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e., our model is good.

➤ Forecasting with ARIMA

We use the ARIMA to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The price of the milk (cow) for the next 8 years (2022 to 2029) is given below and the figure 1.5 gives the graphical representation of the same, with a 95% confidence

Year	Forecast	Lo 95	Hi 95
2022	52.17217	50.03702	54.30732
2023	53.65915	49.82694	57.49136
2024	55.14614	49.43879	60.85349
2025	56.63312	48.86035	64.40589
2026	58.12010	48.09987	68.14034
2027	59.60709	47.16786	72.04632
2028	61.09407	46.07419	76.11395
2029	62.58105	44.82760	80.33451

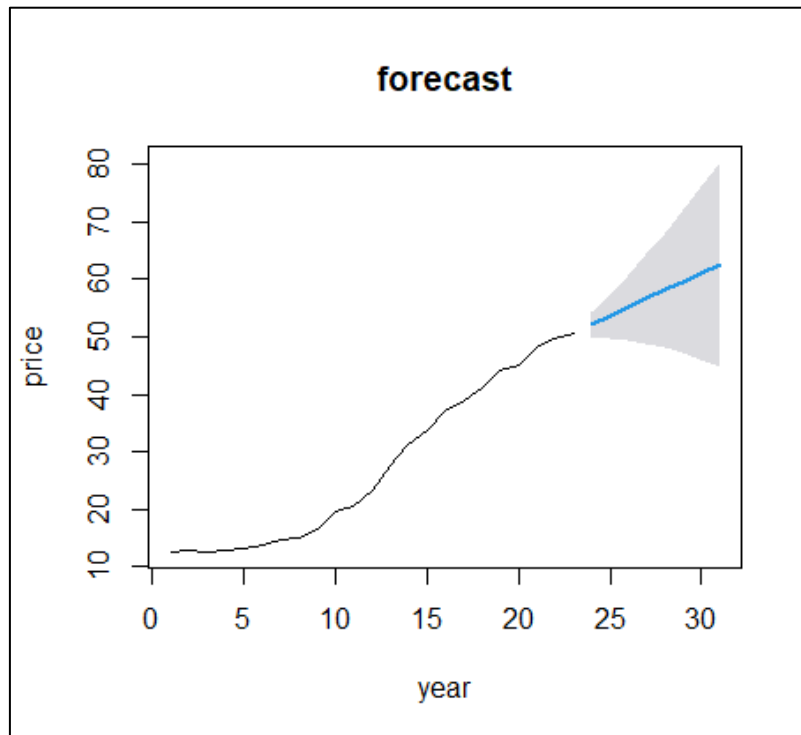


Figure 1.5

➤ Finding the difference between the actual data and the forecasted data of price of cow milk

To find the difference in the actual data and the forecasted data we choose the price of milk in Kerala from 1999 to 2018, and forecast the data for coming 3 years.

Thus, the following was obtained,

Year	Forecasted data	Original data	Difference
2019	52.17217	48.08	4.09217
2020	53.65915	49.76	3.89915
2021	55.14614	50.68519	4.46095

➤ RESULTS

- ❖ The best model that fits the data is ARIMA (0,2,1).
- ❖ For the next 8 years (2022 to 2029), there is a steady increase in the price of milk of cow.

5.2 Monthly Average Price of milk (buffalo)/litre (prices in Rs. Per litre)

We are considering the Monthly Average Price of milk of buffalo from the year 1999 to 2021.

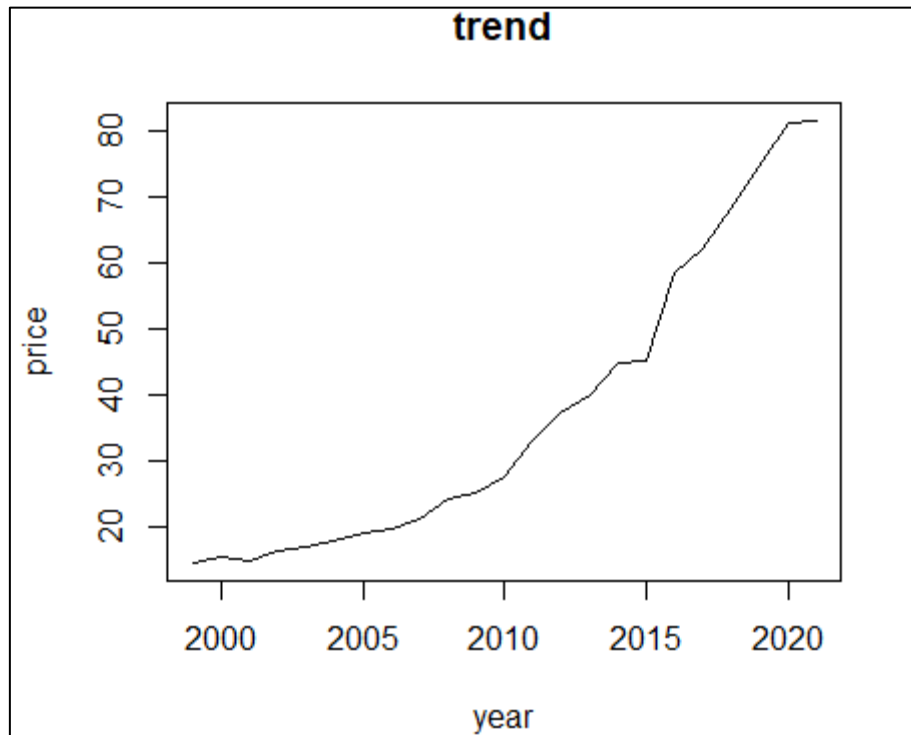


Figure 2.1

- The figure 2.1 shows the price distribution of milk of buffalo/litre (prices in Rs. per litre), In order to test the stationarity of the data ACF test is used

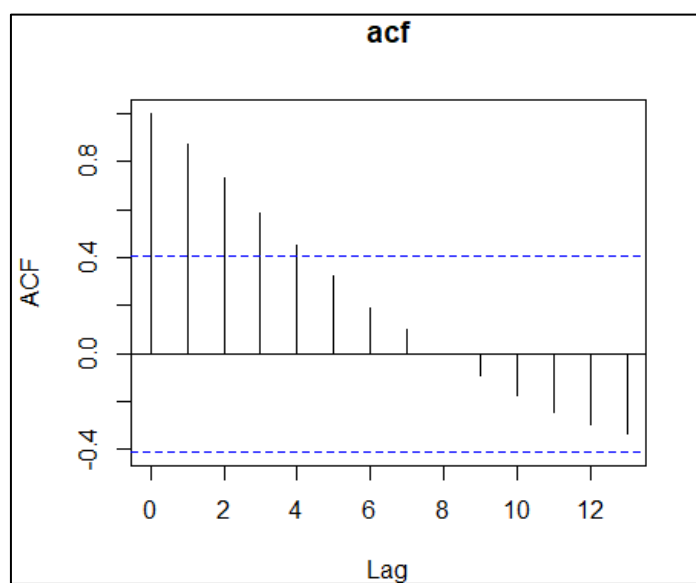


Figure 2.2

By applying ACF test, figure 2.2 is obtained. From the figure 2.2 we can see that lot of points lies outside the error bands which indicate that the data is not stationary.

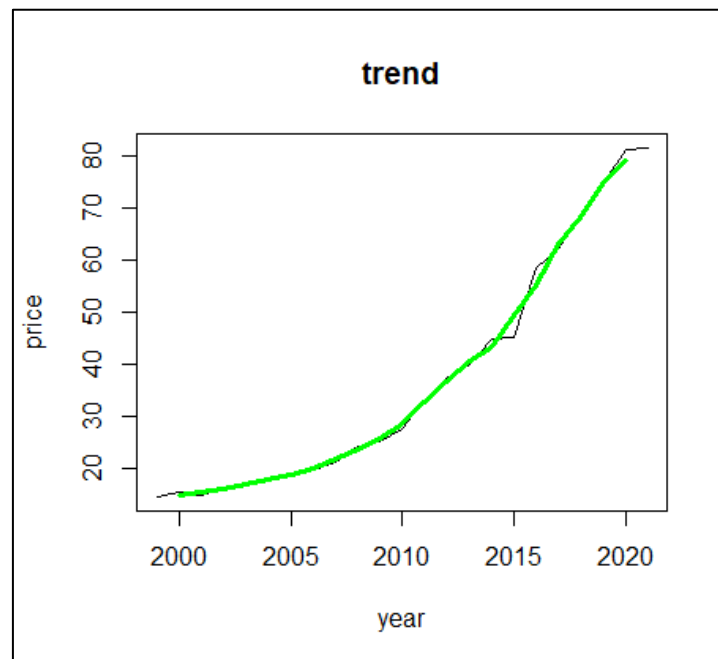


Figure 2.3

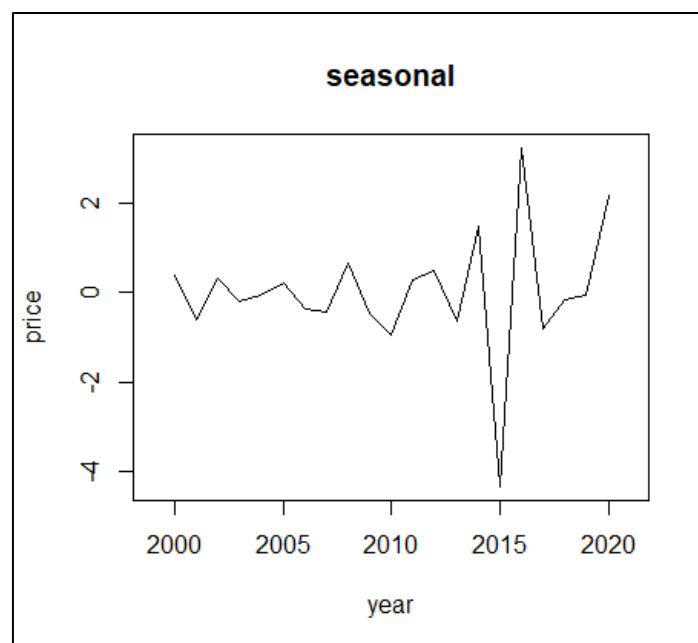


Figure 2.4

- The figure 2.3 gives the trend of the data and the figure 2.4 gives the seasonality of the data. The figure 2.3 indicates that there is a steady trend from 1999 to 2005, increase in trend from 2005 to 2020 and a constant trend is observed in 2021 i.e., the price of milk remains almost constant from the year 1999 to 2005. Then we can observe an increase in price from 2005 to 2020 and again it becomes constant. From figure 2.4 we can conclude that the seasonality is negligible.

➤ Fitting an ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is ARIMA (0,2,2). i.e., $p=0$, $d=2$ and $q=2$.

- $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- $d=2$ tells that the time series is not stationary so we need to take second order difference.
- $q=2$ tells us that the moving average terms capture the dependency of the current value on past forecast errors.

□ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated, `Box.test()` function is used. The observed p -value = 0.8309 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e., our model is good.

➤ Forecasting with ARIMA

We use the ARIMA to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The price of the milk (buffalo) for the next 8 years (2022 to 2029) are given below and the figure 2.5 gives the graphical representation of the same, with a 95% confidence.

Year	Forecast	Lo 95	Hi 95
2022	89.08165	83.44807	94.71523
2023	93.47750	85.97504	100.97995
2024	97.87335	87.65728	108.08941
2025	102.26919	88.71132	115.82706
2026	106.66504	89.27350	124.05658
2027	111.06089	89.42406	132.69772
2028	115.45674	89.21342	141.70005
2029	119.85258	88.67602	151.02915

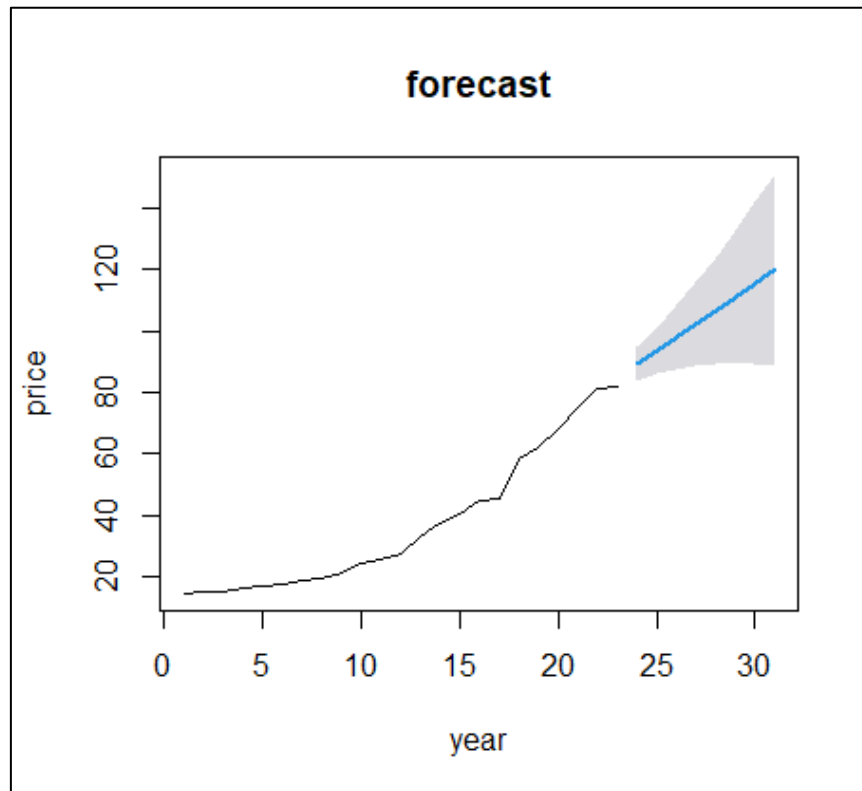


Figure 2.5

➤ Finding the difference between the actual data and the forecasted data of price of buffalo milk

To find the difference in the actual data and the forecasted data we choose the price of milk in Kerala from 1999 to 2018, and forecast the data for coming 3 years.

Thus, the following was obtained,

Year	Forecasted data	Original data	Difference
2019	89.08165	74.73	14.35
2020	93.47750	81.44	12.037
2021	97.87335	81.6466	16.22675

➤ RESULTS

- ❖ The best model that fits the data is ARIMA (0,2,2).
- ❖ For the next 8 years (2022 to 2029), there is a steady increase in price of milk of buffalo.

5.3 Monthly Average Price of milk (Goat)/litre (prices in Rs. Per litre)

We are considering the Monthly Average Price of milk of goat from the year 1999 to 2021.

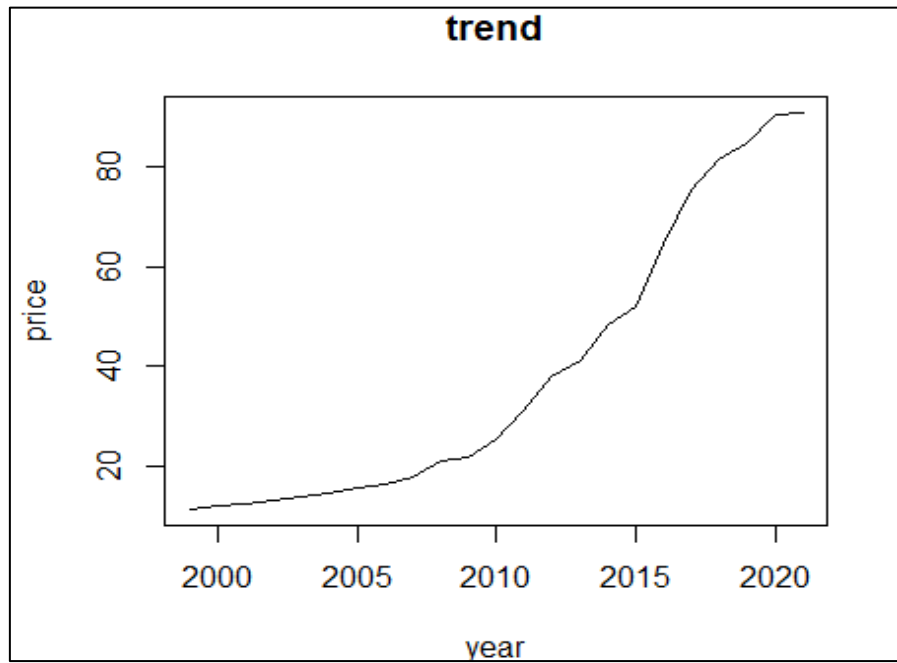


Figure 3.1

- The figure 3.1 shows the price distribution of milk of goat/litre (prices in Rs. per litre). In order to test the stationarity of the data ACF test is used,

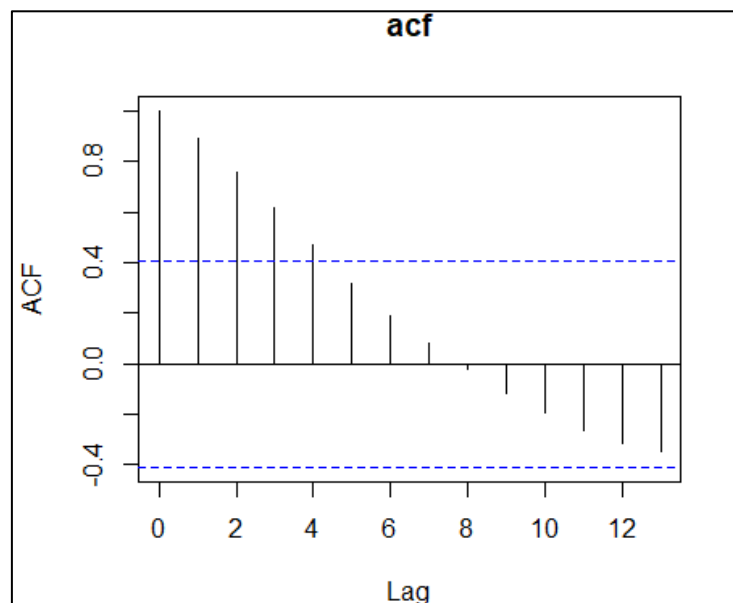


Figure 3.2

By applying ACF test, figure 3.2 is obtained. From the figure 3.2 we can see that lot of points lies outside the error bands which indicate that the data is not stationary.

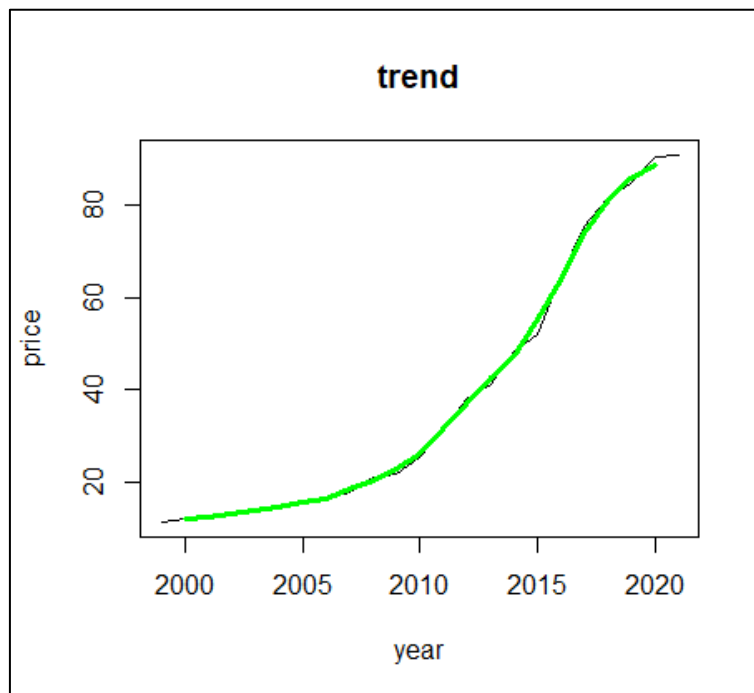


Figure 3.3

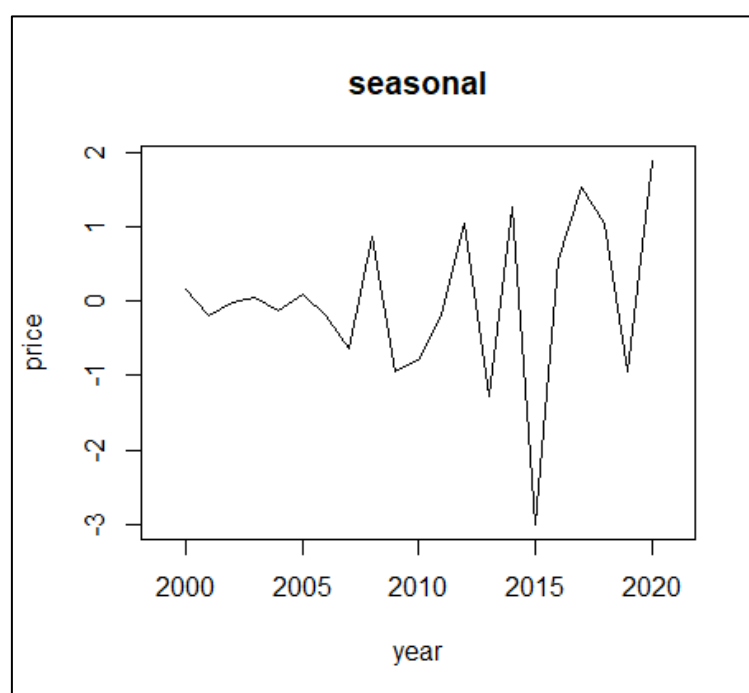


Figure 3.4

- The figure 3.3 gives the trend of the data and the figure 3.4 gives the seasonality of the data. The figure 3.3 indicates that there is a steady trend from 1999 to 2005, increase in trend from 2005 to 2020 and a constant trend is observed in 2021 i.e., the price of milk remains almost constant from the year 1999 to 2005

Then we can observe an increase in price from 2005 to 2020 and again it become constant. From figure 2.4 we can conclude that the seasonality is negligible.

➤ Fitting an ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that the model that best fits the data is ARIMA (0,2,1). i.e., $p=0$, $d=2$ and $q=1$

- $p=0$ tells that we need to take into account the Y value adjacent to the given time point t .
- $d=2$ tells that the time series is not stationary so we need to take second order difference.
- $q=1$ tells us that this model takes in to account the error term from 1 preceding/lagged value.

□ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated, `Box.test()` function is used. The observed p-value = 0.8933 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e., our model is good.

➤ Forecasting with ARIMA

We use the ARIMA to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The price of the milk (goat) for the next 8 years (2022 to 2029) are given below and the figure 3.5 gives the graphical representation of the same, with a 95% confidence.

Year	Forecast	Lo 95	Hi 95
2022	93.28937	87.41997	99.15876
2023	96.03552	85.30132	106.76973
2024	98.78168	82.60257	114.96079
2025	101.52784	79.31264	123.74304
2026	104.27400	75.46564	133.08236
2027	107.02015	71.09765	142.94266
2028	109.76631	66.24075	153.29187
2029	112.51247	60.92266	164.10228

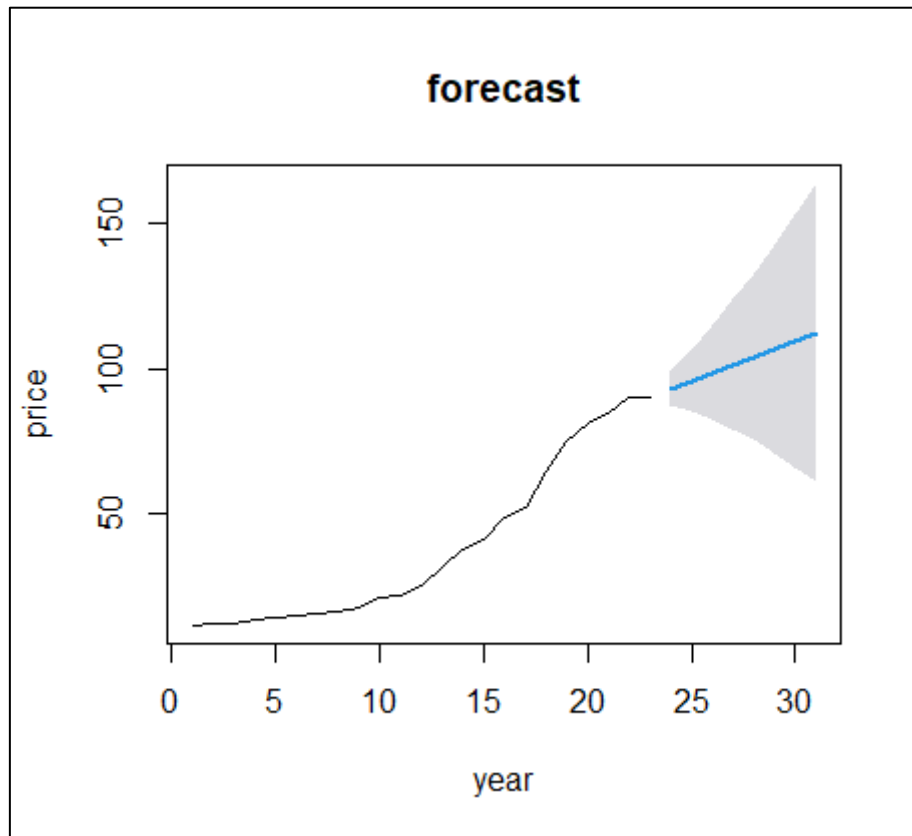


Figure 3.5

➤ Finding the difference between the actual data and the forecasted data of price of goat milk

To find the difference in the actual data and the forecasted data we choose the price of milk in Kerala from 1999 to 2018, and forecast the data for coming 3 years.

Thus, the following was obtained,

Year	Forecasted data	Original data	Difference
2019	93.28937	84.45	8.83937
2020	96.03552	90.32	5.571552
2021	98.78168	90.54321	8.23847

➤ RESULTS

- ❖ The best model that fits the data is ARIMA (0,2,1).
- ❖ For the next 8 years (2022 to 2029), there is a steady increase in price of milk of goat.

5.4 Conclusion

We analyse the trends in the monthly average price of milk of cow, buffalo and goat in the year 1999 to 2022, and predict the price for future.

Here we have a sample of average price of milk of cow, buffalo and goat from 1999 to 2021. For each variety, trends are analysed, models are fitted and the price are predicted for the next 8 years separately. Then the following conclusions are reached.

- In the case of cow, we could clearly see an increasing trend in the data. ARIMA (0,2,1) is the best model that fits the data. Also, for the next 8 years, the model gave a steady increase in the price of cow milk.
- In the case of buffalo, we could see an increasing trend in the data. ARIMA (0,2,2) is the best model that fits the data. Also, for the next 8 years the model gave a steady increase in the price of buffalo milk.
- In the case of Goat, we could clearly see an increasing trend in the data. ARIMA (0,2,1) is the best model that fits the data. Also, for the next 8 years the model predicts steady increase in the price of goat milk.

Chapter-6

A Statistical Study on Price of Meats

To analyse the trend of Monthly average market price of different types of Meats in all districts of Kerala, to fit the best model and to predict the price for the future.

❖ **Monthly average market price of different types of meats in all districts of Kerala during the year 1999-2021**

Year	Price of Broiler Poultry Meat	Price of Desi Poultry Meat	Price of Cattle Meat	Price of Buffalo Meat	Price of Goat Meat	Price of Pork Meat	Price of Duck Meat	Price of Quail Meat
1999-2000	59.03	67.36	48.21	49.04	107.08	56.85	64.7	141.19
2000-2001	57.63	73.4	50.16	50.38	109.44	59.97	64.7	141.19
2001-2002	55.25	71.03	51.31	52.02	113.06	62.58	67.95	200
2002-2003	55.25	71.03	51.31	52.02	113.06	62.58	67.77	159.26
2003-2004	55.04	74.59	54.47	55.37	126.81	64.34	68.04	164.33
2004-2005	58.63	84.64	58.18	59.42	137.54	66.7	72.54	163.13
2005-2006	60.5	88.62	69.98	71.55	145.66	73.79	81.12	131.81
2006-2007	60.02	92.45	80.95	83.02	156.9	78.4	80.42	142.54
2007-2008	69.56	111.37	89.38	92.38	163.15	87.9	98.19	176.74
2008-2009	98.23	145.7	105.71	107.89	176.88	105.83	124.29	196.48
2009-2010	112.85	158.77	121.73	126.62	210.59	122.62	150.68	183.86
2010-2011	118.12	182.76	138.67	141.37	252.04	137.68	160.94	201.53
2011-2012	116.51	193.44	155.52	156.57	306.28	153.12	171.81	243.9
2012-2013	129.11	207.42	178.44	179.66	378.77	180	206.2	264.44
2013-2014	145.2	239.28	193.29	195.53	403.1	201.82	228.03	349.12
2014-2015	147.12	270.57	221.27	233.39	437.22	220.47	233.67	374.58
2015-2016	163.78	344.62	252.79	259.27	454.37	236.97	251.51	416.45
2016-2017	157.58	325.77	270.73	275.06	495.1	226.84	291.94	380.28
2017-2018	162.23	352.81	284.37	290.27	524.63	257.02	311.58	449.19
2018-2019	168.44	355.3	316.73	321.31	555.59	265.18	330.6	465.93
2019-2020	176.83	362.34	350.5	354.36	595.93	279.55	353.83	475.57
2020-2021	179.61	388.65	361.52	368.09	612.75	281.43	354.17	479.37
2021-2022	192.62	390.53	363.04	371.06	623.7	283.47	363.7	480.32

6.1 Analysis of Broiler Poultry Meat

❖ Trend

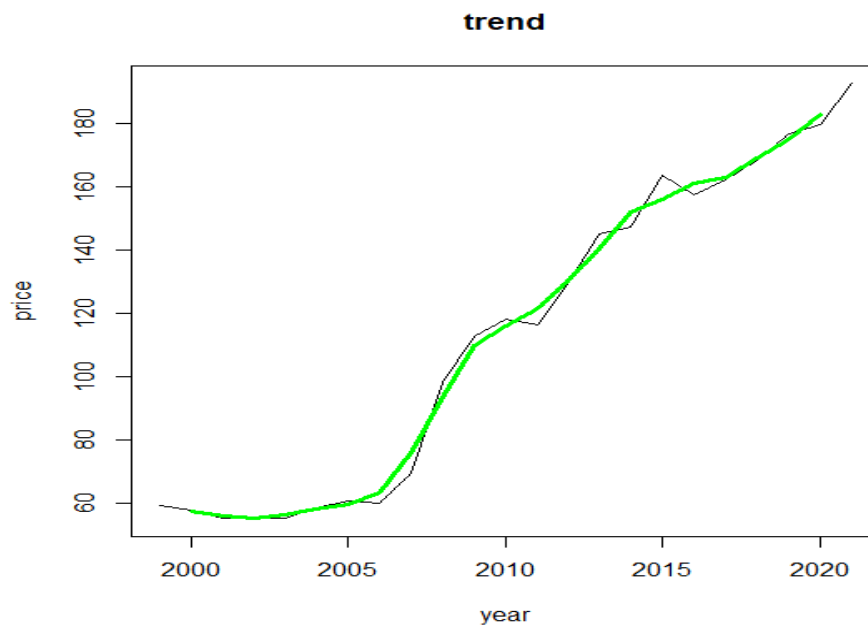


Figure 1.1

❖ Seasonality

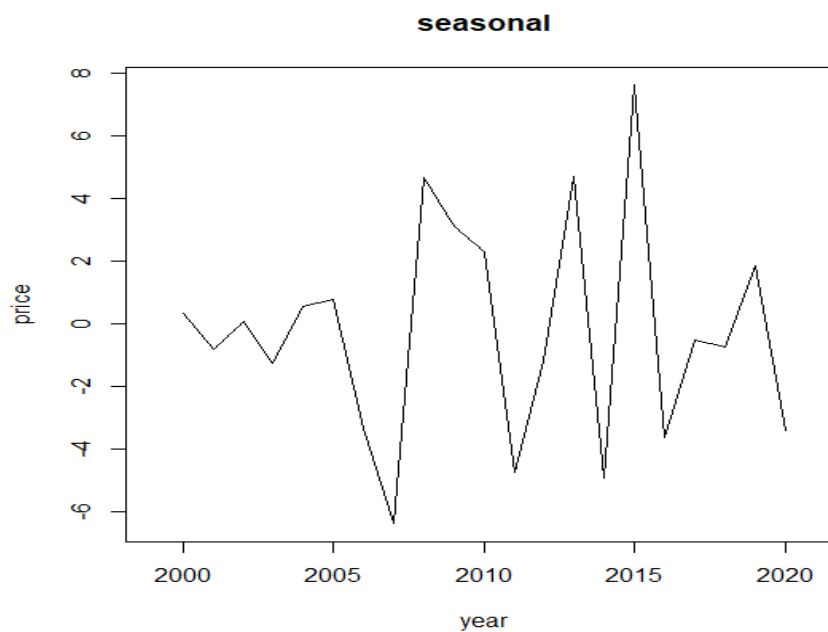


Figure 1.2

The above graph (fig 1.1) shows the TREND of the price distribution of broiler poultry meat during the year 1999-2021. We can observe upward trend in the price. From this graph, we can see a clear inflation from the year 1999-2021.

Figure 1.2 shows the seasonality trend of the price distribution of broiler poultry meat. From this, it is clear that the seasonality is negligible.

❖ Auto-correlation Function

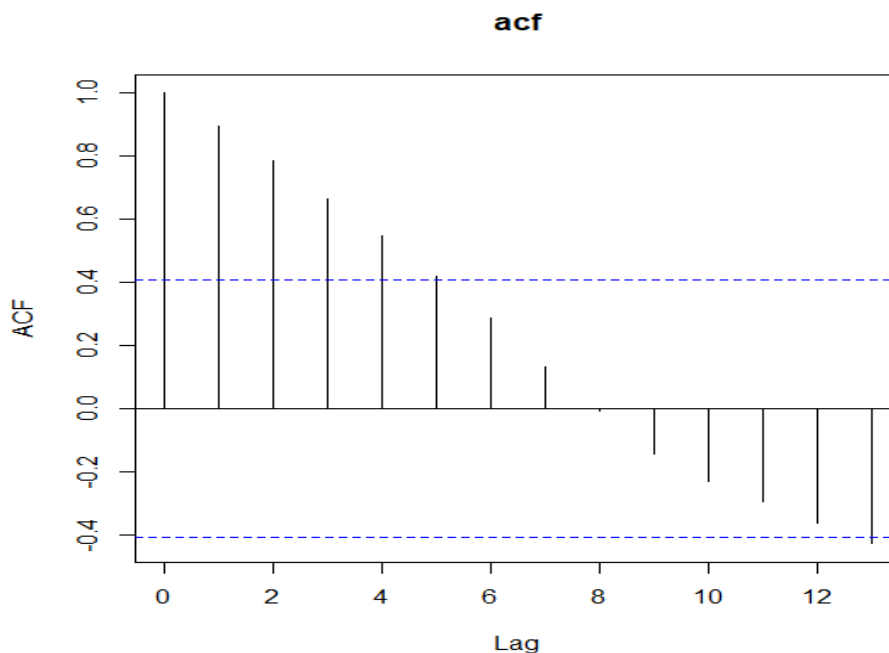


Figure 1.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 1.3), we can see that some of the points falls above the significance range (dotted horizontal blue lines), so can conclude that our data is not stationary (or non-stationary series).

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,1,0) is the best model that fits our data.

ARIMA (0,1,0) => $p=0, d=1$ and $q=0$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t.
- ❖ $d=1$ means that, the time series is not stationary and requires differencing once to make it stationary.
- ❖ $q=0$ means that, there is no lagged forecast error in the prediction equation. (this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model).

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.6835 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors,

i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast() function from the “forecast” package.

The price of the broiler poultry meat for the next 9 years (2022 to 2030) are given below and the figure 1.4 gives the graphical representation of the same, with a 95% level of confidence.

Years	Point Forecast	Lo 95	Hi 95
2022	198.6923	182.6033	214.7813
2023	204.7645	182.0113	227.5178
2024	210.8368	182.9699	238.7038
2025	216.9091	184.7311	249.0871
2026	222.9814	187.0053	258.9574
2027	229.0536	189.6438	268.4635
2028	235.1259	192.5584	277.6934
2029	241.1982	195.6916	286.7047
2030	247.2705	199.0035	295.5374

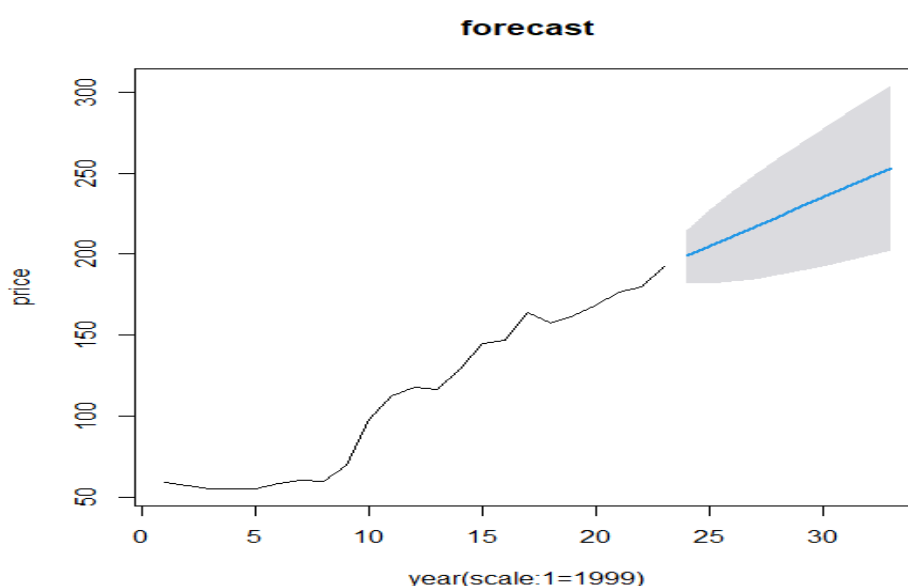


Figure 1.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ **Finding difference between the actual data and forecasted data**

To find the difference in the actual data and the forecasted data we choose the price of broiler poultry meat in Kerala from the year 1999 to 2017 and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	167.9639	168.44	0.4767
2019	173.6967	176.83	3.1333
2020	179.4300	179.61	0.18
2021	185.1633	192.62	7.4567

There is only a slight difference in actual value and forecasted value.

➤ **RESULTS**

In the case of Broiler poultry meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,0). Also for the next 9 years the price of broiler poultry meat increases.

6.2 Analysis of Desi Poultry Meat

❖ Trend

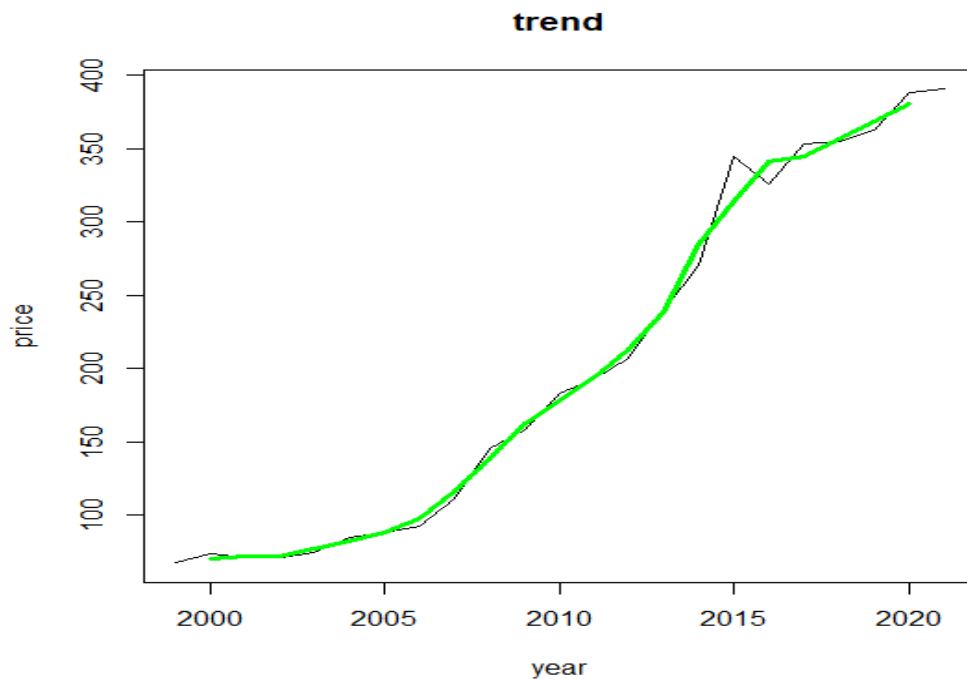


Figure 2.1

❖ Seasonality

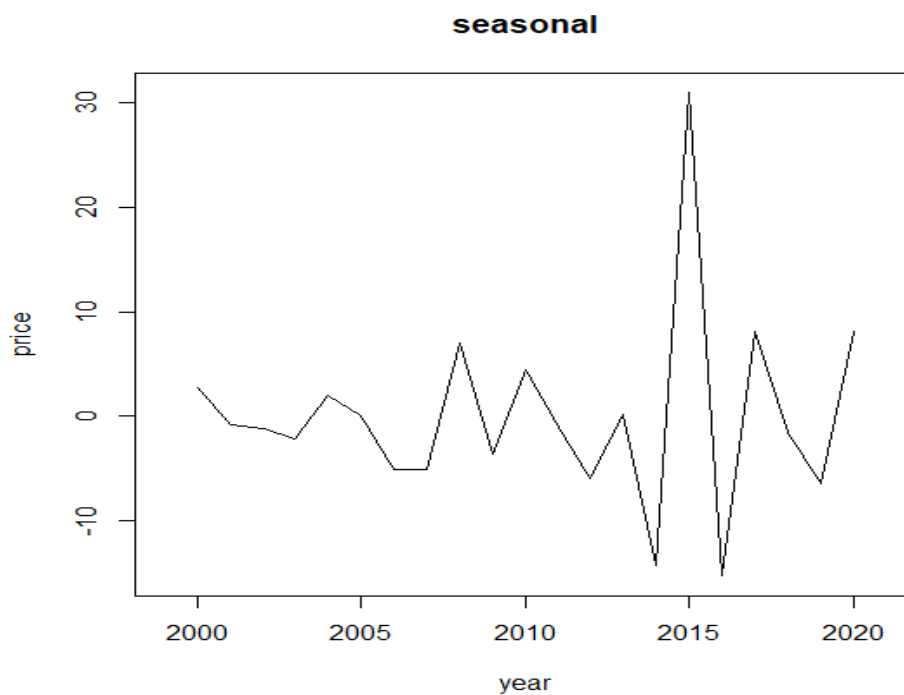


Figure 2.2

The above graph (fig 2.1) shows the TREND of the price distribution of desi poultry meat during the year 1999-2021. We can observe upward trend in the price. From this

graph, we can see a clear inflation from the year 1999-2021. Figure 2.2 shows the seasonality trend of the price distribution of desi poultry meat. From this it is clear that the seasonality is negligible.

❖ Auto-correlation Function

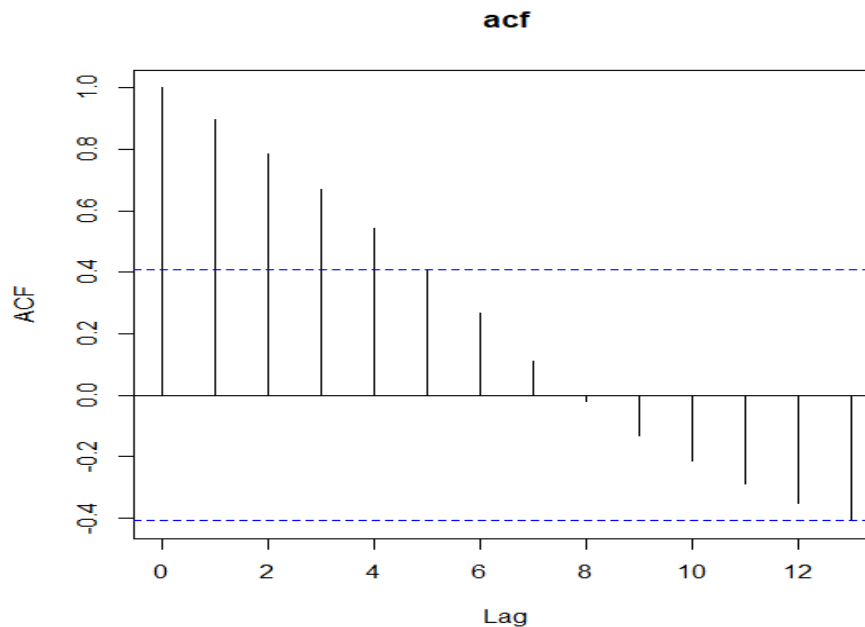


Figure 2.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 2.3), we can see that some of the points falls above the significance range (dotted horizontal blue lines),so we can conclude that our data is not stationary(or non-stationary series).

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,1,0) is the best model that fits our data.

ARIMA (0,1,0) => $p=0$, $d=1$ and $q=0$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t.
- ❖ $d=1$ means that, the time series is not stationary and requires differencing once to make it stationary.
- ❖ $q=0$ means that, there is no lagged forecast error in the prediction equation. (this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model)

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.3695 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast() function from the “forecast” package.

The price of the desi poultry meat for the next 9 years (2022 to 2030) are given below and the figure 2.4 gives the graphical representation of the same, with a 95% level of confidence.

Year	Point Forecast	Lo 95	Hi 95
2022	405.2195	368.5596	441.8795
2023	419.9091	368.0641	471.7541
2024	434.5986	371.1017	498.0956
2025	449.2882	375.9683	522.6081
2026	463.9777	382.0036	545.9519
2027	478.6673	388.8691	568.4655
2028	493.3568	396.3637	590.3500
2029	508.0464	404.3563	611.7364
2030	522.7359	412.7560	632.7158

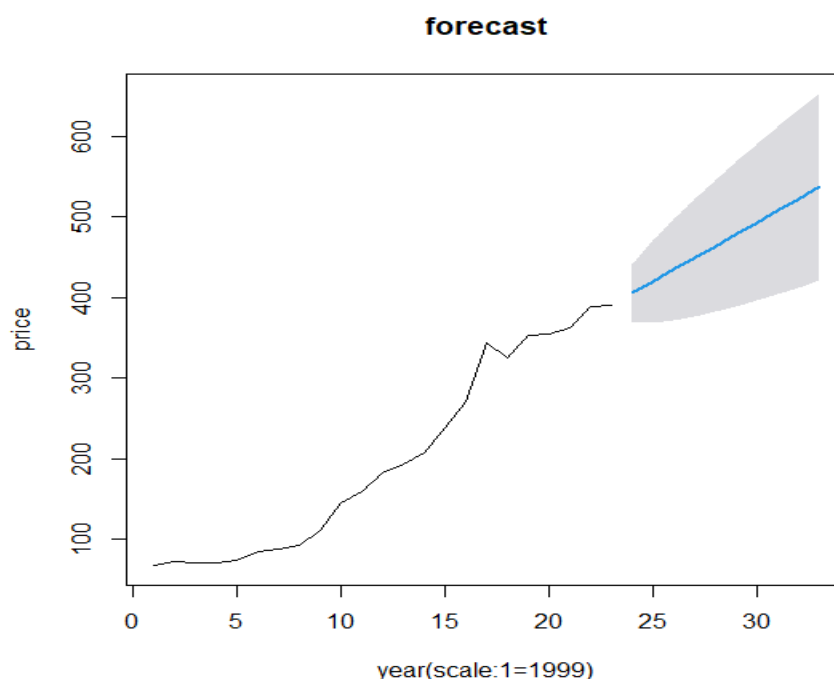


Figure 2.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ **Finding difference between the actual data and forecasted data**

To find the difference in the actual data and the forecasted data, we choose the price of desi poultry meat in Kerala from the year 1999 to 2017, and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	368.6683	355.3	13.3683
2019	384.5267	362.34	6.044
2020	400.3850	388.65	11.735
2021	416.2433	390.53	25.7133

There is only a slight difference in actual value and forecasted value.

➤ **RESULTS**

In the case of Desi poultry meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,0). Also for the next 9 years the price of desi poultry meat increases.

6.3 Analysis of Cattle Meat

❖ Trend

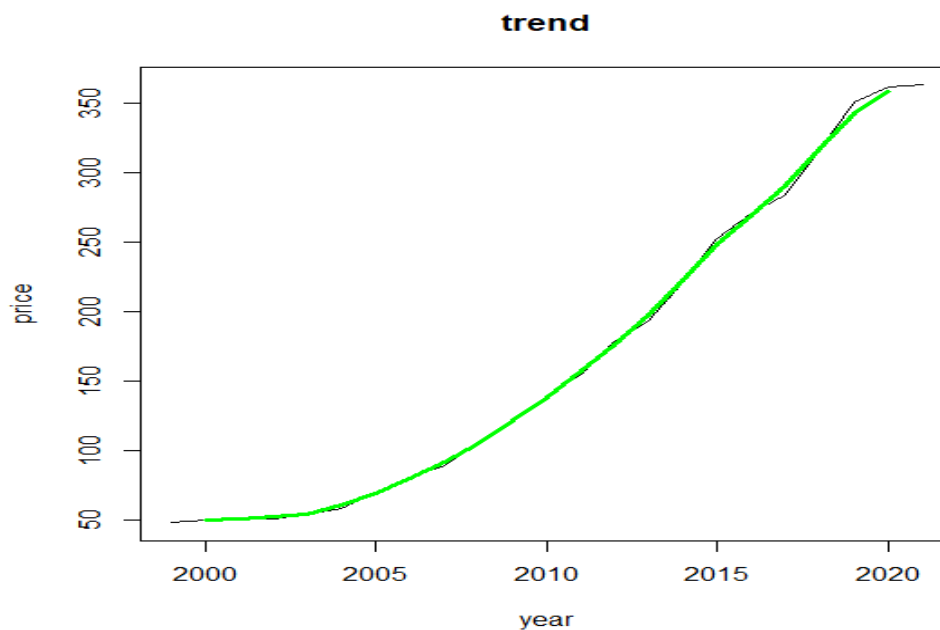


Figure 3.1

❖ Seasonality

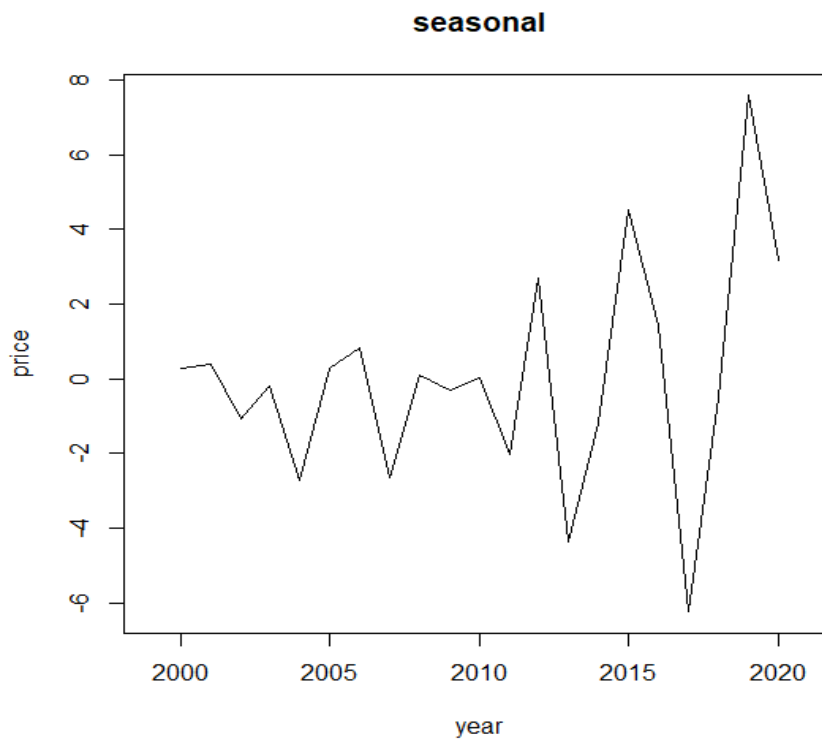


Figure 3.2

The above graph (fig 3.1) shows the TREND of the price distribution of Cattle meat during the year 1999-2021. We can observe upward trend in the price. From this

graph, we can see a clear inflation from the year 1999-2021. Figure 3.2 shows the seasonality trend of the price distribution of Cattle meat. From this it is clear that the seasonality is negligible.

❖ Auto-correlation Function

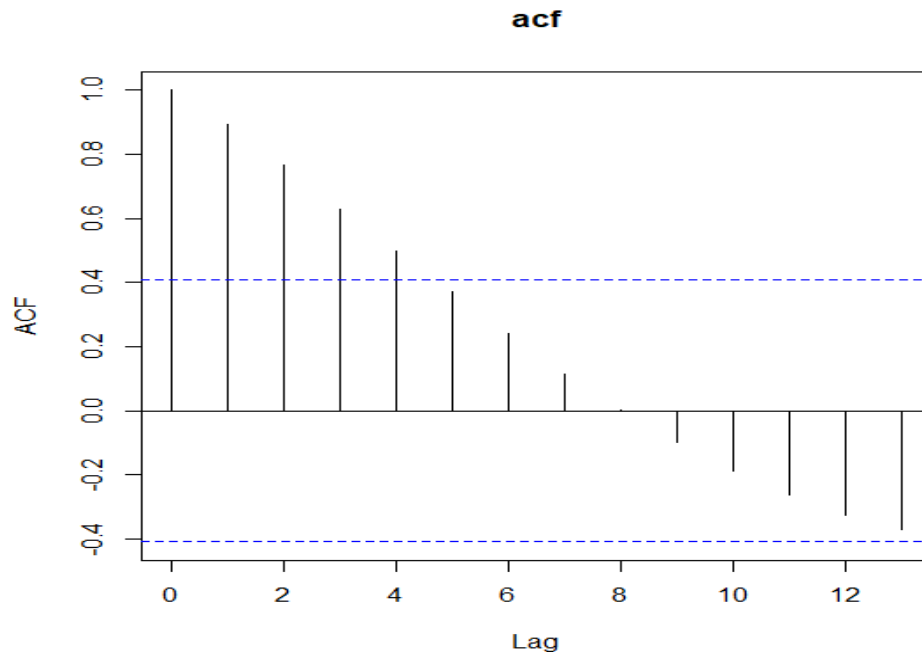


Figure 3.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 3.3) , we can see that some of the points falls above the significance range (dotted horizontal blue lines),so we can conclude that our data is not stationary(or non- stationary series).

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,2,0) is the best model that fits our data.

ARIMA (0,2,0) => $p=0, d=2$ and $q=0$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t.
- ❖ $d=2$ means that, the time series is not stationary, so we need to take second order difference.
- ❖ $q=0$ means that, there is no lagged forecast error in the prediction equation. (this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model)

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.9809 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast () function from the “forecast” package. The price of the Cattle meat for the next 9 years (2022 to 2030) are given below and the figure 4.1.3(d) gives the graphical representation of the same, with a 95% level of confidence.

Year	Point Forecast	Lo 95	Hi 95
2022	364.56	347.46140	381.6586
2023	366.08	327.84637	404.3136
2024	367.60	303.62291	431.5771
2025	369.12	275.46713	462.7729
2026	370.64	243.83341	497.4466
2027	372.16	209.04978	535.2702
2028	373.68	171.36667	575.9933
2029	375.20	130.98319	619.4168
2030	376.72	88.06246	665.3775

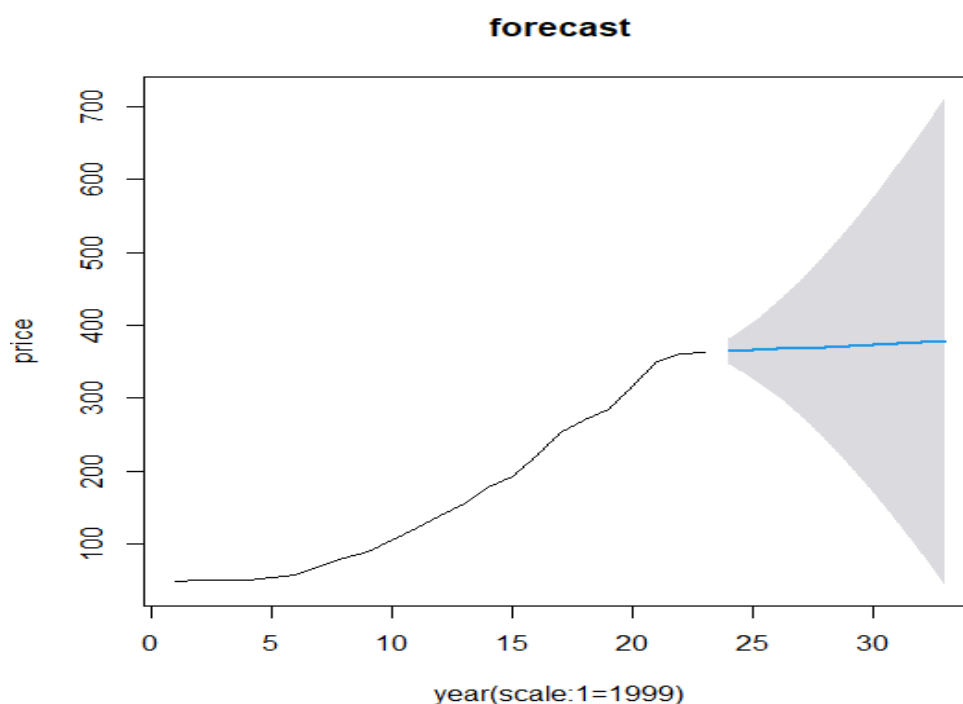


Figure 3.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ **Finding difference between the actual data and forecasted data**

To find the difference in the actual data and the forecasted data, we choose the price of Cattle meat in Kerala from the year 1999 to 2017, and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	298.01	316.73	18.72
2019	311.65	350.5	38.85
2020	325.29	361.52	36.23
2021	338.93	363.04	24.11

There is only a slight difference in actual value and forecasted value.

➤ **RESULTS**

In the case of Cattle meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,2,0). Also for the next 9 years the price of the meat increases.

6.4 Analysis of Buffalo Meat

❖ Trend

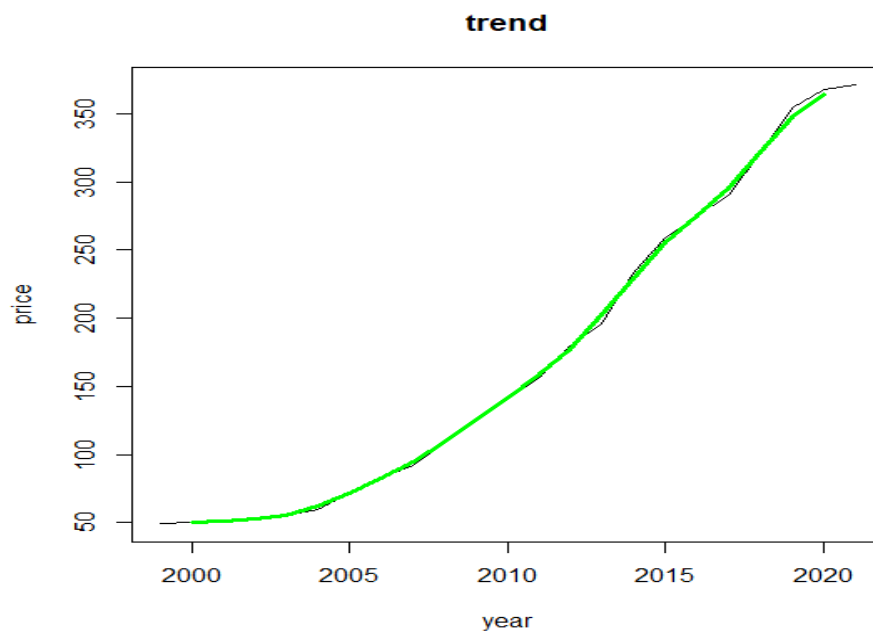


Figure 4.1

❖ Seasonality

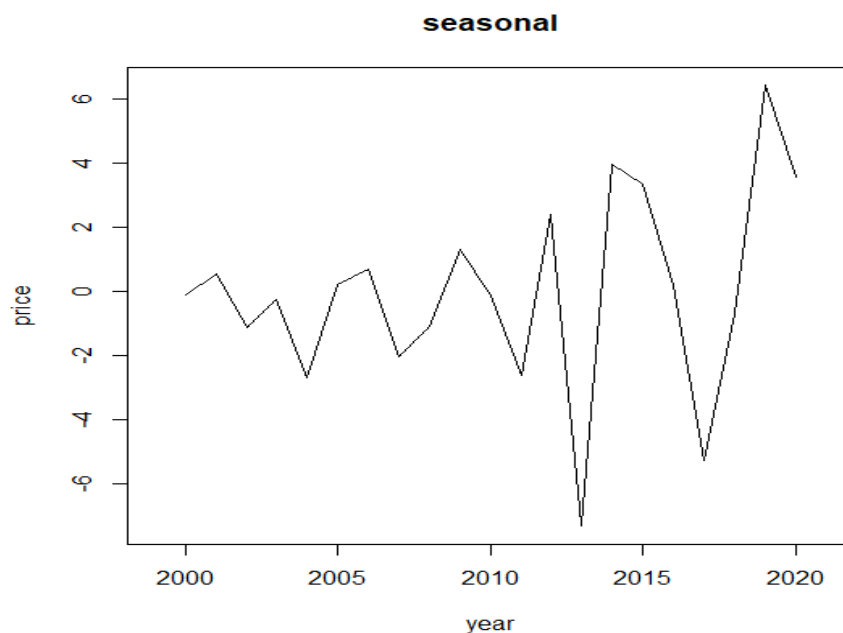


Figure 4.2

The above graph (fig 4.1) shows the TREND of the price distribution of Buffalo meat during the year 1999-2021. We can observe upward trend in the price. From this graph, we can see a clear inflation from the year 1999-2021. Figure 4.2 shows the seasonality trend of the price distribution of Cattle meat. From this it is clear that the seasonality is negligible.

❖ Auto-correlation Function

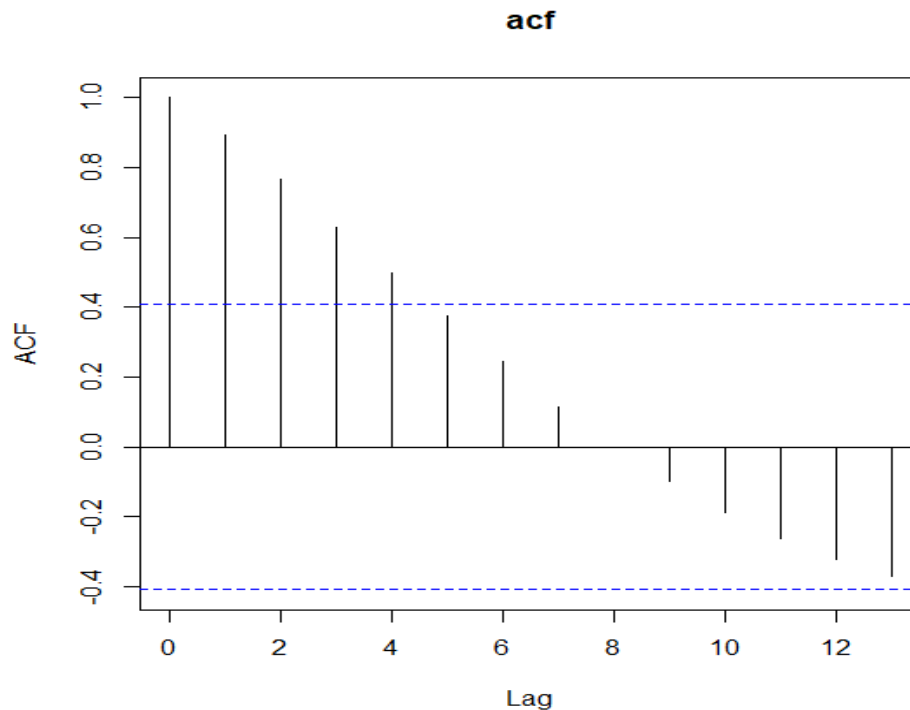


Figure 4.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 4.3), we can see that some of the points falls above the significance range (dotted horizontal blue lines), so we can conclude that our data is not stationary (or non-stationary series).

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,2,0) is the best model that fits our data.

ARIMA (0,2,0) => $p=0$, $d=2$ and $q=0$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t .
- ❖ $d=2$ means that, the time series is not stationary, so we need to take second order difference.
- ❖ $q=0$ means that, there is no lagged forecast error in the prediction equation. (this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model)

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.2785 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast () function from the “forecast” package. The price of the Buffalo meat for the next 9 years (2022 to 2030) are given below and the figure 4.1.4(d) gives the graphical representation of the same, with a 95% level of confidence.

Year	Point Forecast	Lo 95	Hi 95
2022	374.03	356.20885	391.8511
2023	377.00	337.15070	416.8493
2024	379.97	313.28937	446.6506
2025	382.94	285.32955	480.5504
2026	385.91	253.74483	518.0752
2027	388.88	218.87708	558.8829
2028	391.85	180.98733	602.7127
2029	394.82	140.28309	649.3569
2030	397.79	96.93439	698.6456

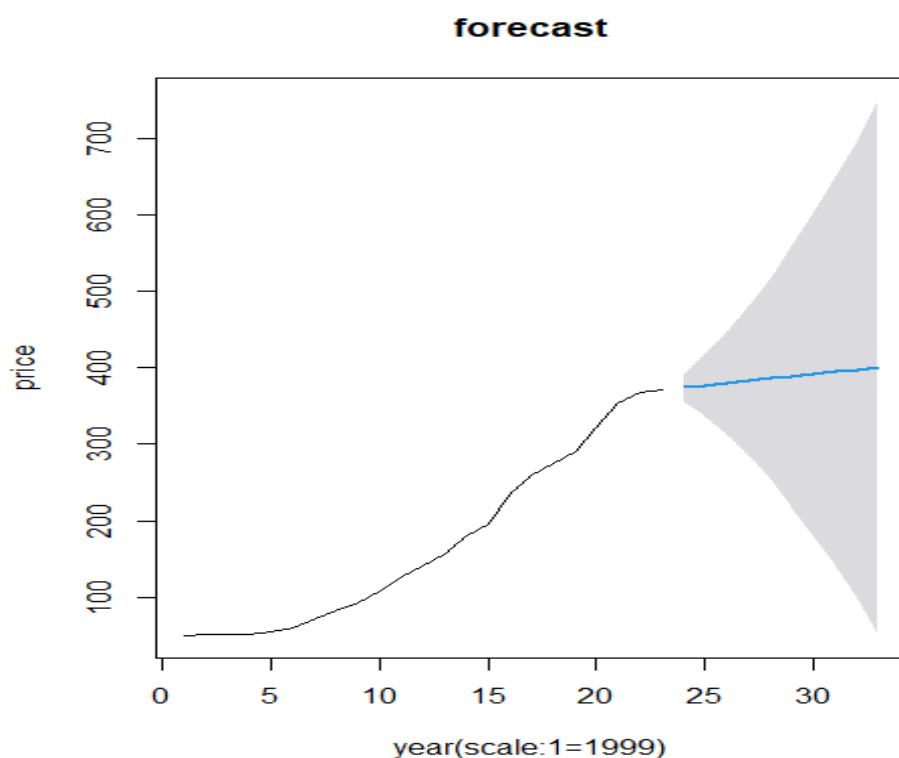


Figure 4.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ **Finding difference between the actual data and forecasted data**

To find the difference in the actual data and the forecasted data, we choose the price of Buffalo meat in Kerala from the year 1999 to 2017, and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	307.8001	321.31	13.5099
2019	325.3301	354.36	29.0299
2020	342.8602	368.09	25.2298
2021	360.3903	371.06	10.6697

There is only a slight difference in actual value and forecasted value.

➤ **RESULTS**

In the case of Buffalo meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA(0,2,0). Also for the next 9 years the price of the meat increases

6.5 Analysis of Goat Meat

❖ Trend

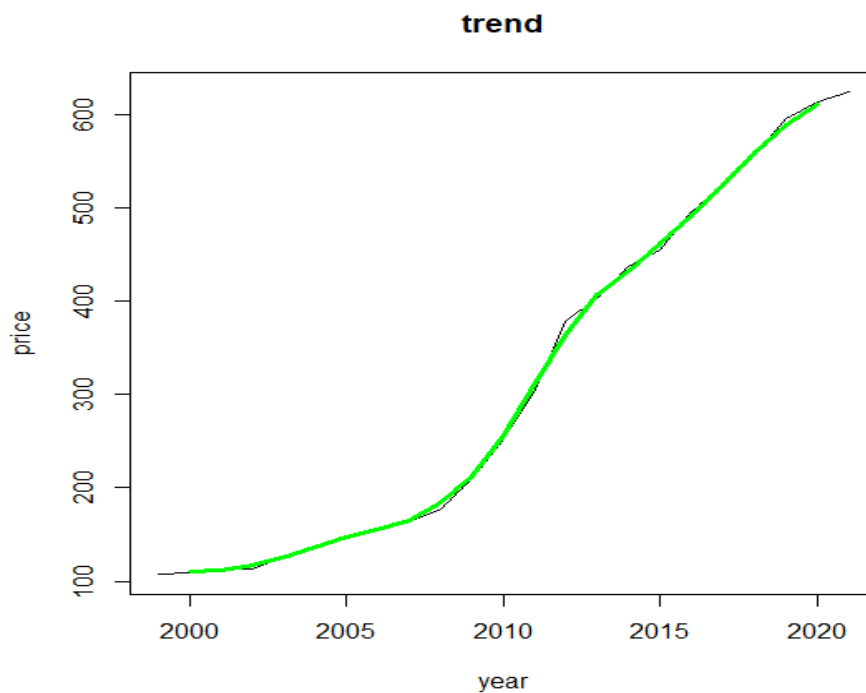


Figure 5.1

❖ Seasonality

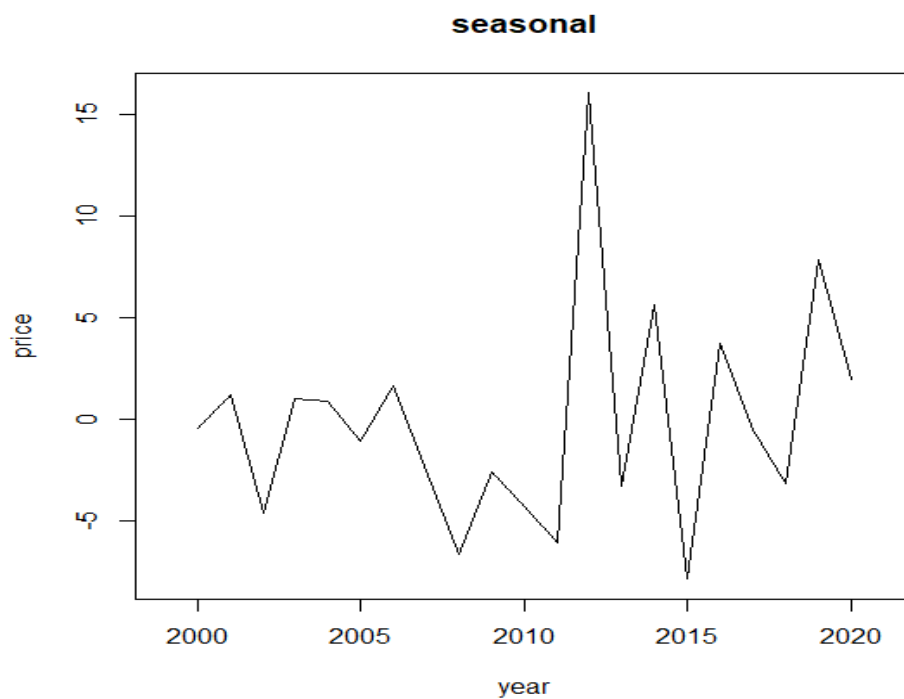


Figure 5.2

The above graph (fig 5.1) shows the TREND of the price distribution of Goat meat during the year 1999-2021. We can observe upward trend in the price. From this graph, we can see a clear inflation from the year 1999-2021.

Figure 5.2 shows the seasonality trend of the price distribution of Goat meat. From this it is clear that the seasonality is negligible.

❖ Auto-correlation Function

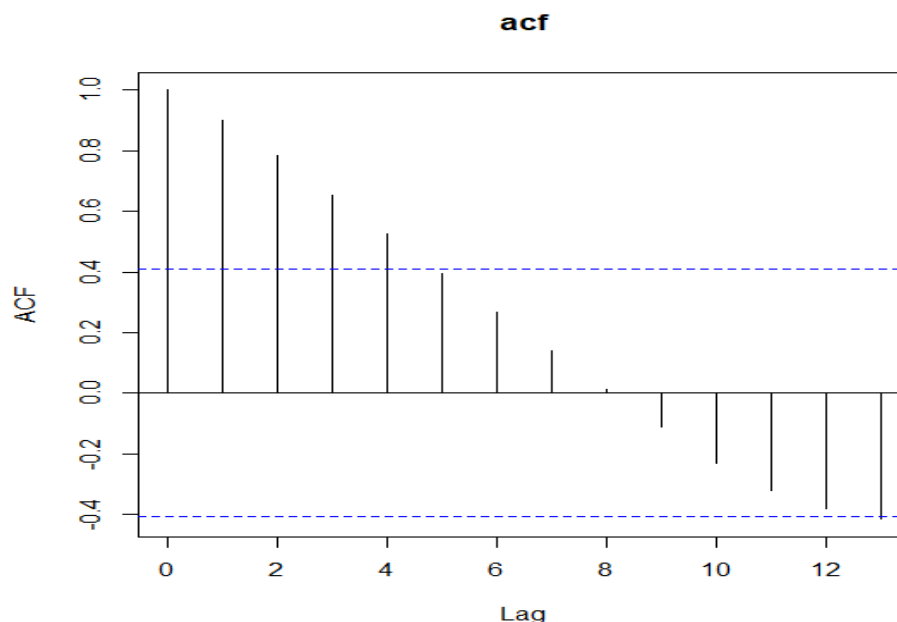


Figure 5.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 5.3), we can see that some of the points falls above the significance range (dotted horizontal blue lines), so we can conclude that our data is not stationary (or non-stationary series).

❖ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,2,0) is the best model that fits our data.

ARIMA (0,2,0) => $p=0$, $d=2$ and $q=0$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t .
- ❖ $d=2$ means that, the time series is not stationary, so we need to take second order difference.
- ❖ $q=0$ means that, there is no lagged forecast error in the prediction equation. (this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model)

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.1421 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast() function from the “forecast” package. The price of the Goat meat for the next 9 years (2022 to 2030) are given below and the figure 4.1.5(d) gives the graphical representation of the same, with a 95% level of confidence.

Year	Point Forecast	Lo 95	Hi 95
2022	634.65	603.4297	665.8703
2023	645.60	575.7894	715.4106
2024	656.55	539.7345	773.3655
2025	667.50	496.4996	838.5004
2026	678.45	446.9145	909.9856
2027	689.40	391.5777	987.2223
2028	700.35	330.9469	1069.7531
2029	711.30	265.3855	1157.2145
2030	722.25	195.1914	1249.3086

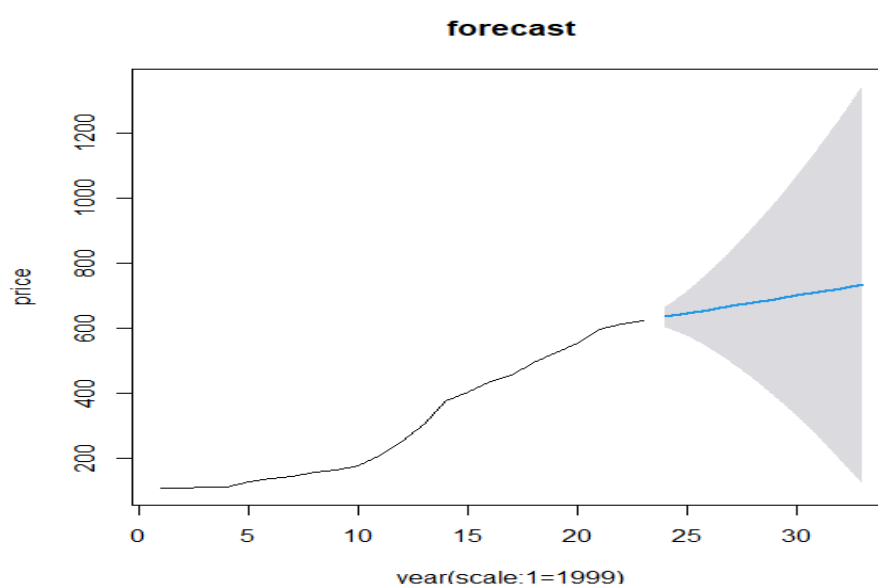


Figure 5.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ Finding difference between the actual data and forecasted data

To find the difference in the actual data and the forecasted data, we choose the price of Goat meat in Kerala from the year 1999 to 2017, and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	554.16	555.59	1.43
2019	583.69	595.93	12.24
2020	613.22	612.75	0.47
2021	642.75	623.7	19.05

There is only a slight difference in actual value and forecasted value.

➤ RESULTS

In the case of Goat meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,2,0). Also for the next 9 years the price of the meat increases.

6.6 Analysis Of Pork Meat

❖ Trend

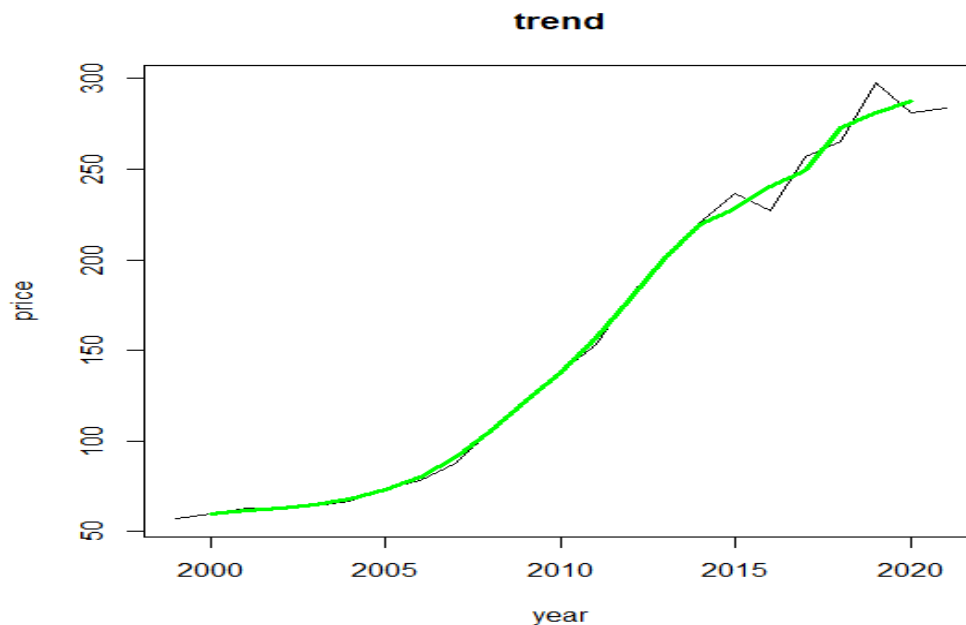


Figure 6.1

❖ Seasonality

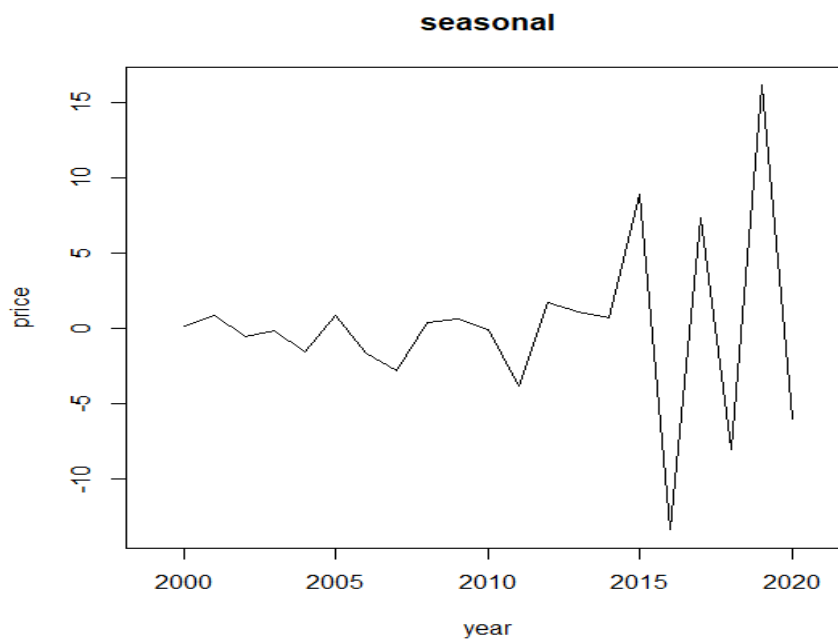


Figure 6.2

The above graph (fig 6.1) shows the TREND of the price distribution of Pork meat during the year 1999-2021. We can observe upward trend in the price. From this graph, we can see a clear inflation from the year 1999-2021. Figure 6.2 shows the seasonality trend of the price distribution of Pork meat. From this it is clear that the seasonality is negligible.

❖ Auto-correlation Function

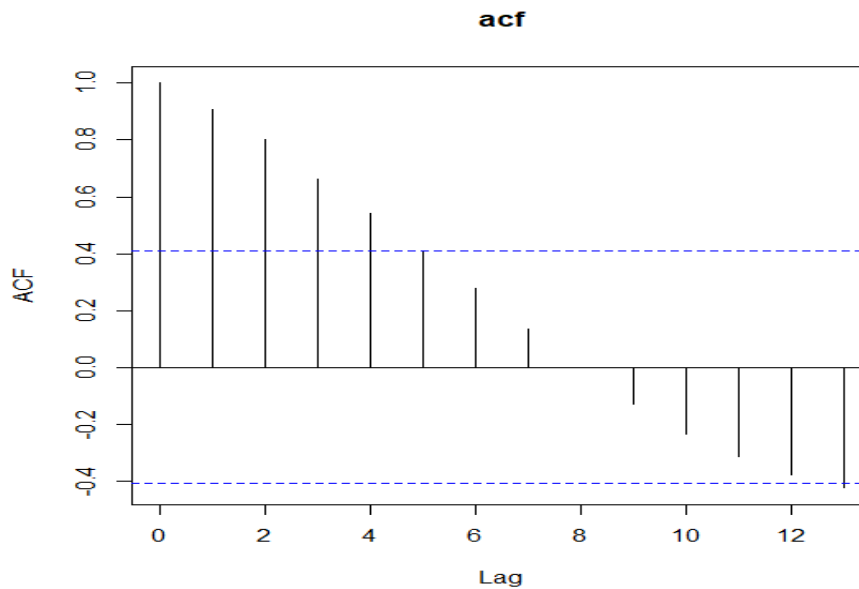


Figure 6.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 6.3), we can see that some of the points falls above the significance range (dotted horizontal blue lines), thus we can conclude that our data is not stationary (or non-stationary series).

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,1,0) is the best model that fits our data.

ARIMA (0,1,0) => $p=0, d=1$ and $q=0$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t .
- ❖ $d=1$ means that, the time series is not stationary and requires differencing once to make it stationary.
- ❖ $q=0$ means that, there is no lagged forecast error in the prediction equation. (this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model).

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.26 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast() function from the “forecast” package.

The price of the Pork meat for the next 9 years (2023 to 2030) are given below and the figure 6.4 gives the graphical representation of the same, with a 95% level of confidence.

Year	Point Forecast	Lo 95	Hi 95
2022	293.7709	269.8359	317.7059
2023	304.0718	270.2226	337.9210
2024	314.3727	272.9161	355.8293
2025	324.6736	276.8037	372.5436
2026	334.9745	281.4543	388.4948
2027	345.2755	286.6470	403.9040
2028	355.5764	292.2503	418.9024
2029	365.8773	298.1789	433.5756
2030	376.1782	304.3732	447.9831

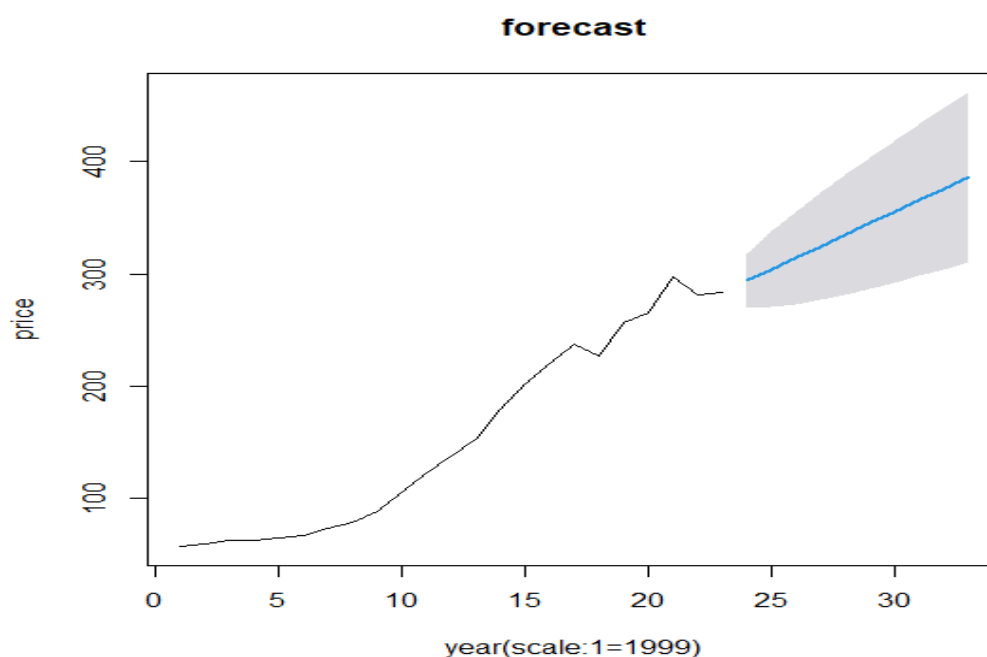


Figure 6.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ **Finding difference between the actual data and forecasted data**

To find the difference in the actual data and the forecasted data, we choose the price of Pork meat in Kerala from the year 1999 to 2017, and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	272.9071	265.18	7.7271
2019	288.7941	279.55	9.2441
2020	304.6812	281.43	23.2515
2021	320.5682	283.47	37.0982

There is only a slight difference in actual value and forecasted value.

➤ **RESULTS**

In the case of Pork meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,0). Also for the next 9 years the price of the meat increases.

6.7 Analysis of Duck Meat

❖ Trend

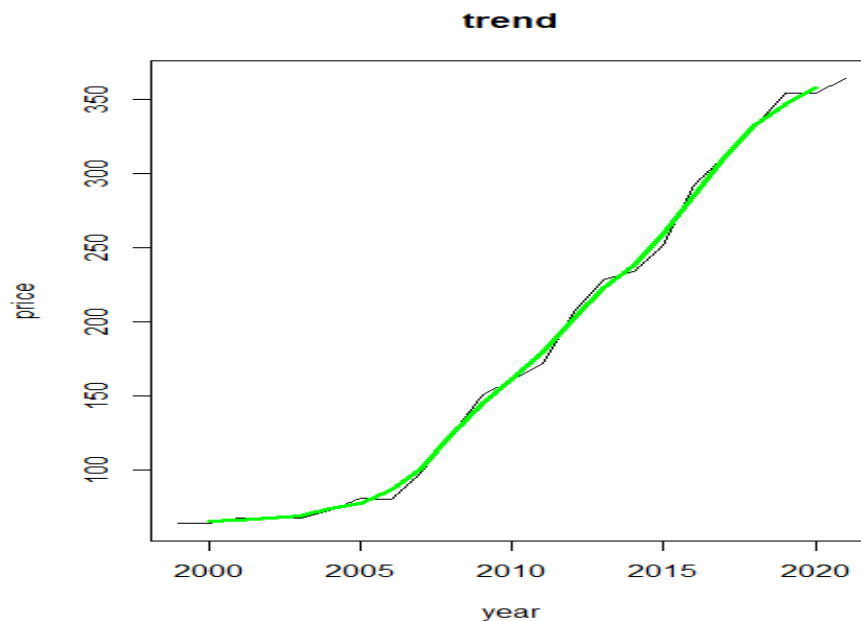


Figure 7.1

❖ Seasonality

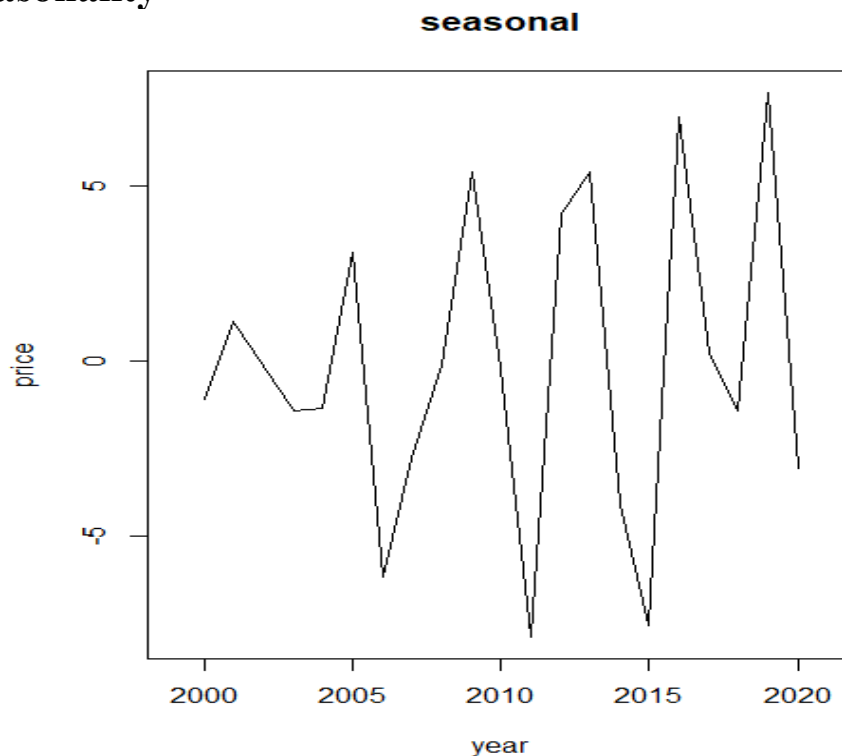


Figure 7.2

The above graph (fig 7.1) shows the TREND of the price distribution of Duck meat during the year 1999-2021. We can observe upward trend in the price. From this graph, we can see a clear inflation from the year 1999-2021.

Figure 7.2 shows the seasonality trend of the price distribution of Duck meat. From this it is clear that the seasonality is negligible.

❖ Auto-correlation Function

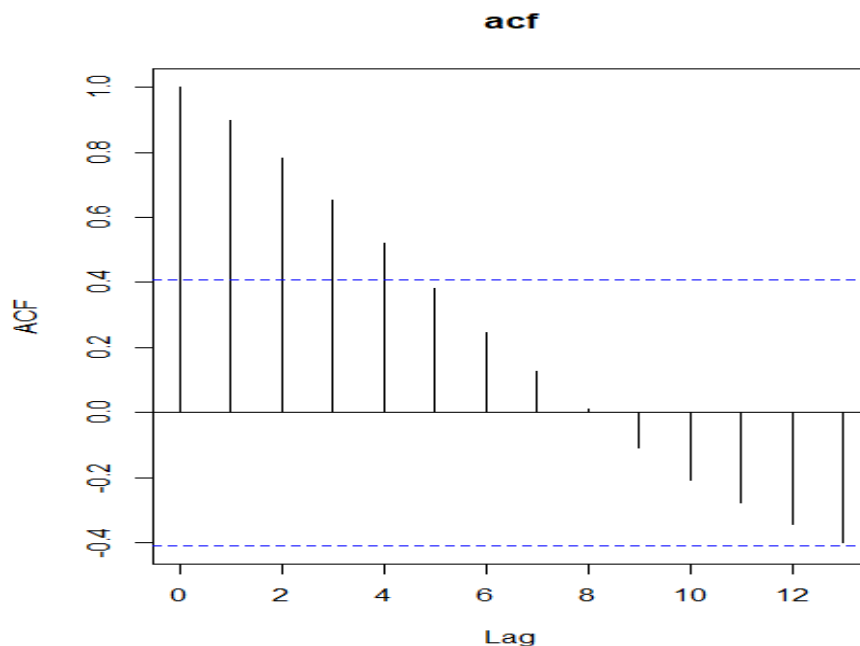


Figure 7.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 7.3), we can see that some of the points falls above the significance range (dotted horizontal blue lines), thus we can conclude that our data is not stationary (or non-stationary series).

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,1,1) is the best model that fits our data.

ARIMA (0,1,1) => $p=0$, $d=1$ and $q=1$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t .
- ❖ $d=1$ means that, the time series is not stationary and requires differencing once to make it stationary
- ❖ $q=1$ means that, this model takes into account the error term from one preceding/lagged value.

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.8784 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast () function from the “forecast” package.

The price of the Duck meat for the next 9 years (2022 to 2030) are given below and the figure 7.4 gives the graphical representation of the same, with a 95% level of confidence.

Years	Point Forecast	Lo 95	Hi 95
2022	380.8237	359.9719	401.6755
2023	394.2600	354.1529	434.3671
2024	407.6763	354.9432	460.4444
2025	421.1326	358.2347	484.0305
2026	434.5689	362.9453	506.1924
2027	448.0051	368.6092	527.4011
2028	461.4414	374.9689	547.9139
2029	474.8777	381.8655	567.8900
2030	488.3140	389.1926	587.4355

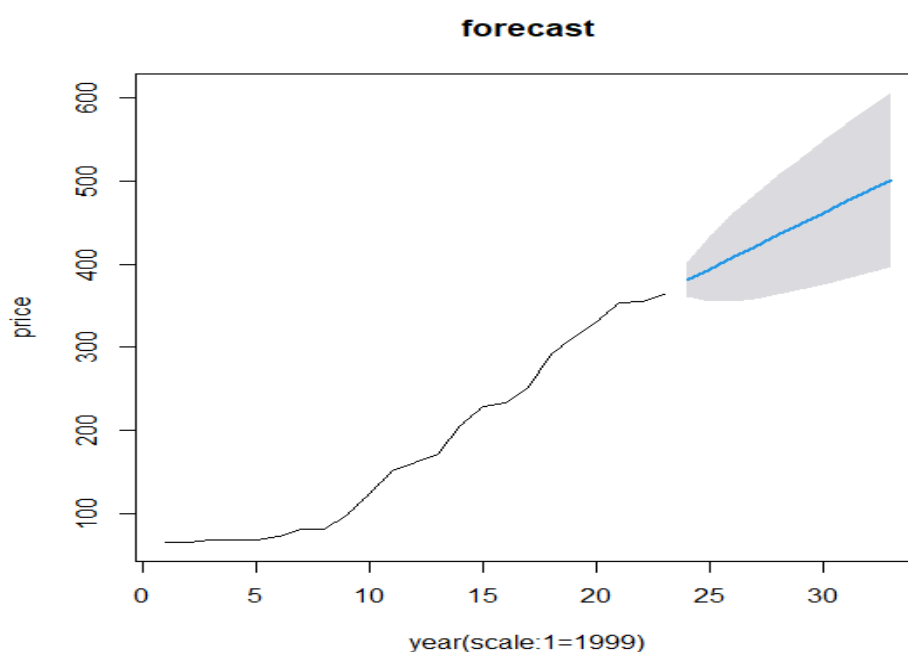


Figure 7.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ **Finding difference between the actual data and forecasted data**

To find the difference in the actual data and the forecasted data, we choose the price of Duck meat in Kerala from the year 1999 to 2017, and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	318.7251	330.6	11.8749
2019	344.6767	353.83	9.1533
2020	375.8511	354.17	21.6811
2021	391.4881	363.7	27.7811

There is only a slight difference in actual value and forecasted value.

➤ **RESULTS**

In the case of Duck meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,1). Also for the next 9 years the price of the meat increases.

6.8 Analysis of Quail Meat

❖ Trend

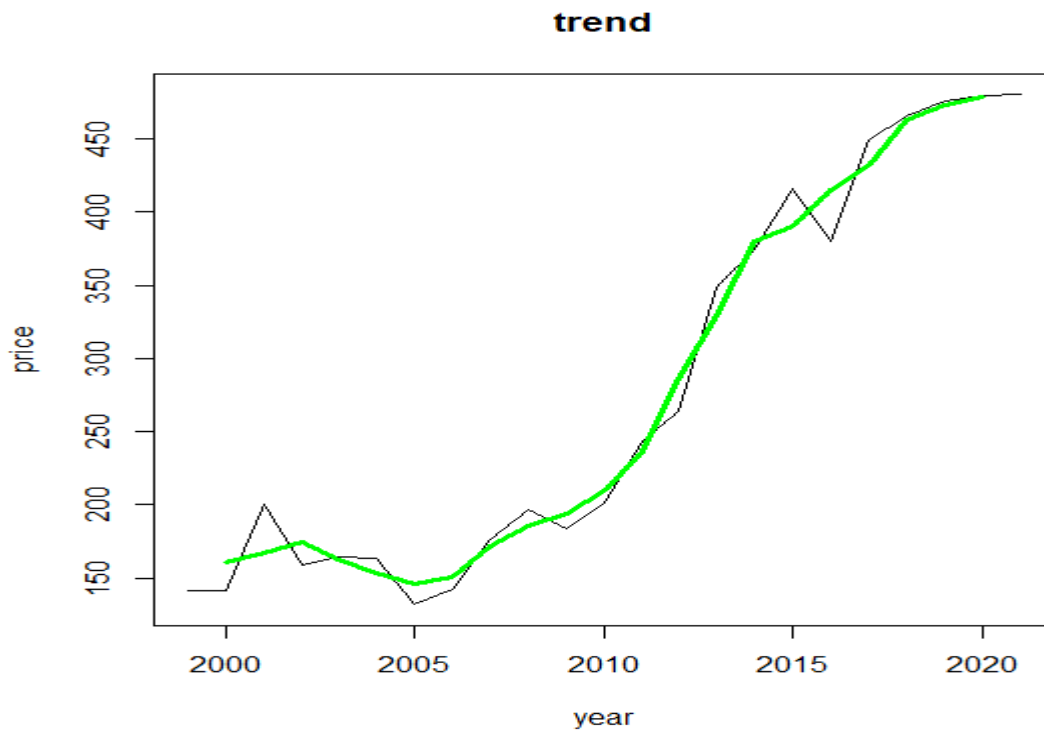


Figure 8.1

❖ Seasonality

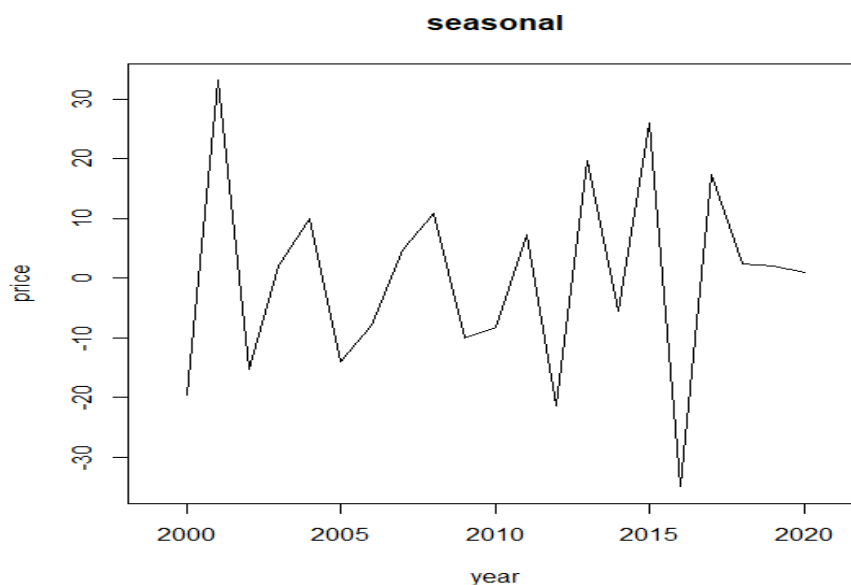


Figure 8.2

The above graph (fig 8.1) shows the TREND of the price distribution of Quail meat during the year 1999-2021. We can observe upward trend in the price. From this graph, we can see a clear inflation from the year 1999-2021.

Figure 8.2 shows the seasonality trend of the price distribution of Quail meat. From this it is clear that the seasonality is negligible.

❖ Auto-correlation Function

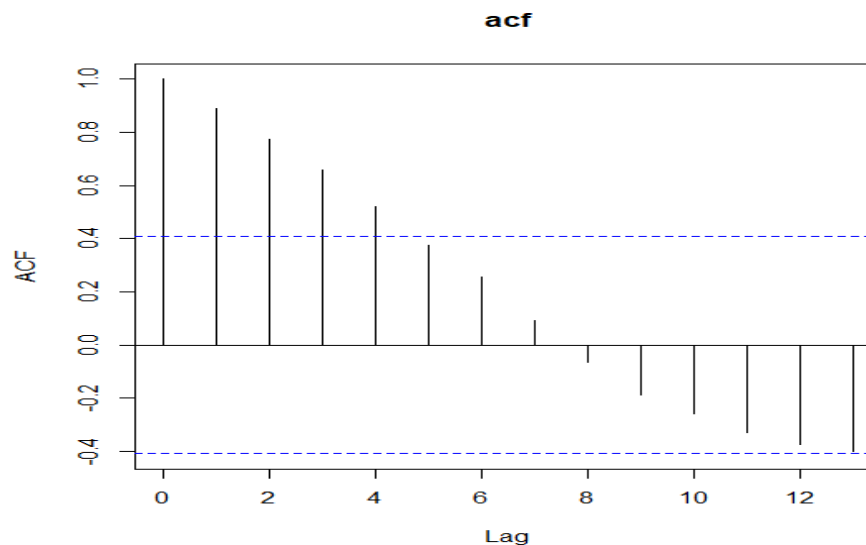


Figure 8.3

Here the Auto-correlation function (ACF) is used to determine the stationarity of the data. From the above graph (fig 8.3), we can see that some of the points falls above the significance range (dotted horizontal blue lines), thus we can conclude that our data is not stationary (or non-stationary series).

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (0,1,0) is the best model that fits our data.

ARIMA (0,1,0) => $p=0, d=1$ and $q=0$.

Where,

- ❖ $p=0$ means that, we need to take into account the Y value adjacent to the given time point t .
- ❖ $d=1$ means that, the time series is not stationary and requires differencing once to make it stationary.
- ❖ $q=0$ means that, there is no lagged forecast error in the prediction equation. (this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model)

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated Box.test() function is used. The observed p-value is 0.3916 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., our model is good.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use forecast () function from the “forecast” package.

The price of the Quail meat for the next 9 years (2022 to 2030) are given below and the figure 8.4 gives the graphical representation of the same, with a 95% level of confidence.

Year	Point Forecast	Lo 95	Hi 95
2022	495.7350	433.3449	558.1251
2023	511.1500	422.9170	599.3830
2024	526.5650	418.5021	634.6279
2025	541.9800	417.1997	666.7603
2026	557.3950	417.8864	696.9036
2027	572.8100	419.9860	725.6340
2028	588.2250	423.1562	753.2938
2029	603.6400	427.1740	780.1060
2030	619.0550	431.8846	806.2254

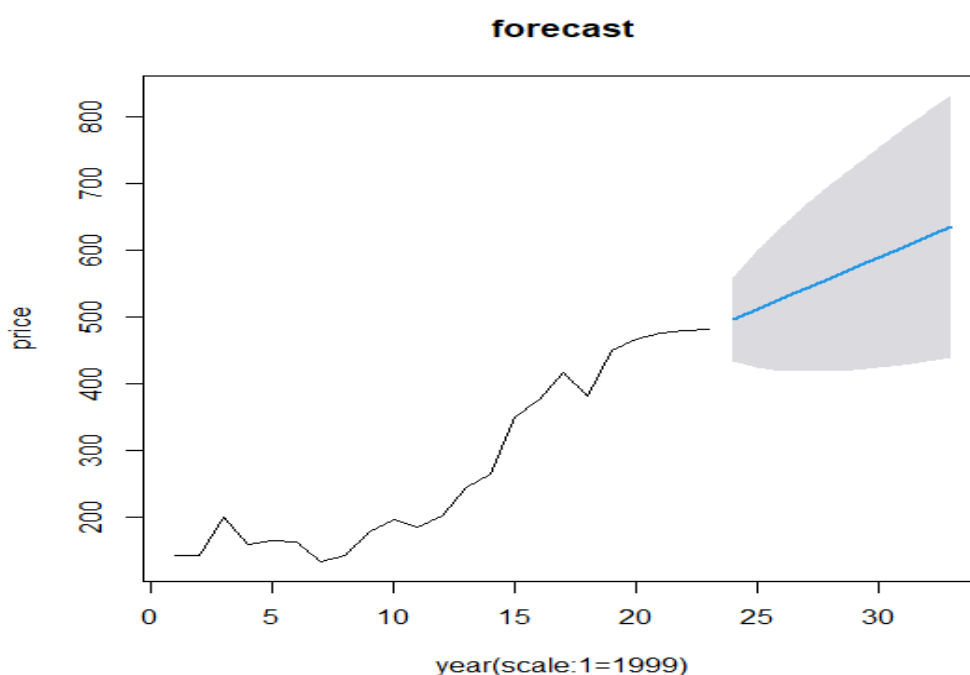


Figure 8.4

Here, the forecast value increases for the next 9 years i.e, there will be a clear inflation from the year 2022 to 2030.

➤ **Finding difference between the actual data and forecasted data**

To find the difference in the actual data and the forecasted data, we choose the price of Quail meat in Kerala from the year 1999 to 2017, and we forecasted the data for the next 4 years.

Thus the following are obtained:

Year	Forecasted Value	Original Value	Difference
2018	466.3011	465.93	0.3711
2019	483.4122	475.57	7.8422
2020	500.5233	479.37	21.1533
2021	517.6344	480.32	37.3144

There is only a slight difference in actual value and forecasted value.

➤ **RESULTS**

In the case of Quail meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,0). Also for the next 9 years the price of the meat increases.

6.9 Conclusion

Analysing the trend in the monthly market price of variety of meat and predicting the price for future.

Here we are provided with a data that contain the price of 8 different types of meat namely Broiler poultry meat, Desi poultry meat, Cattle meat, Buffalo meat, Goat meat, Pork meat, Duck meat, Quail meat. For each type of meat we have done analysis of trend, models are fitted and prices are predicted for next 8 years separately. Then the following conclusions are reached.

- In the case of Broiler poultry meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,0). Also for the next 9 years the price of broiler poultry meat increases.
- In the case of Desi poultry meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,0). Also for the next 9 years the price of desi poultry meat increases.
- In the case of Cattle meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,2,0). Also for the next 9 years the price of the meat increases.
- In the case of Buffalo meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,2,0). Also for the next 9 years the price of the meat increases.
- In the case of Goat meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,2,0). Also for the next 9 years the price of the meat increases.
- In the case of Pork meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,0). Also for the next 9 years the price of the meat increases.
- In the case of Duck meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA (0,1,1). Also for the next 9 years the price of the meat increases.
- In the case of Quail meat, we see a clear inflation from the year 1999-2021. The best model of this data is found to be ARIMA 0,1,0). Also for the next 9 years the price of the meat increases.

Chapter-7

Analysis Of Milk, Meat & Egg Production

7.1 Analysis of milk production

The annual milk production of cattle in Kerala from 2011 to 2020

Year	Milk production	No. of cases treated under Mastitis	Rainfall (mm)	Precipitation (mm/day)	Temperature (C)	Average feed consumption
2011	2581.98	96147	3035.1	5.05	22.48	5.67725
2012	2648.53	95376	2151.1	3.74	23.84	5.766667
2013	2483.56	94511	3255.4	5.55	23.04	7.233333
2014	2533.96	102191	3046.4	5.25	23.32	6.408333
2015	2507.89	102953	2600.6	4.6	21.99	15.93667
2016	2381.39	102404	1870.9	3.69	23.52	15.19889
2017	2442.54	108135	2664.9	5.49	22.38	14.78667
2018	2415.05	117700	3571.5	5.9	21.51	14.60667
2019	2403.27	115754	3118.1	5.95	25.51	18.25704
2020	2402.24	122622	2999	6.13	22.25	19.93598

- ✓ The above table contains the annual milk production of cattle from 2011 to 2020, the number of cases treated under Mastitis, amount of rainfall, precipitation, temperature and average feed consumption of the cattle in the respective years.

ANALYSIS

Here, we have to check the effect of various factors like disease (Mastitis), rainfall, precipitation, temperature and average feed consumption of cattle on the milk production of cattle. So, milk production is the response variable or dependent variable. The number of cases treated under Mastitis, amount of rainfall, precipitation, temperature and average feed consumption of the cattle are intercorrelated. i.e., a multicollinearity exists. To fix multicollinearity, here we are using principal component analysis.

To quantify how much the variance of the estimated regression coefficients is increased due to multicollinearity among the predictor variables, Variance Inflation Factor (VIF) is tested and the below inference is obtained.

Variables	VIF
no. of cases treated under Mastitis	6.221162
Rainfall	8.807253
Precipitation	10.264390
Temperature	1.115943
Average feed consumption	5.485335

Effect of disease (Mastitis), rainfall, and feed consumption shows a moderate multicollinearity, effect of precipitation shows a severe presence of multicollinearity and there is no effect in temperature.

Using this, a regression model is fitted. The equation of regression model is obtained as follows:

$$Y = 2652.893 - 37.119 * X_1 - 7.521 * X_2$$

Where,

Y = milk produced

X₁ = principal component

X₂ = temperature

The coefficients of the variables X1, X2 gives the amount of change that occur in the dependent variable Y due to a change in one unit in them. i.e., change in one unit in principal component will result in 37.119 change in milk production and likewise.

The standard error is variability to expect in coefficient, which captures sampling variability, so variation in intercept can be up to 501.756 and so on. The standard errors are obtained as:

	Std. Error
Intercept	501.756
pc1	15.495
temperature	21.807

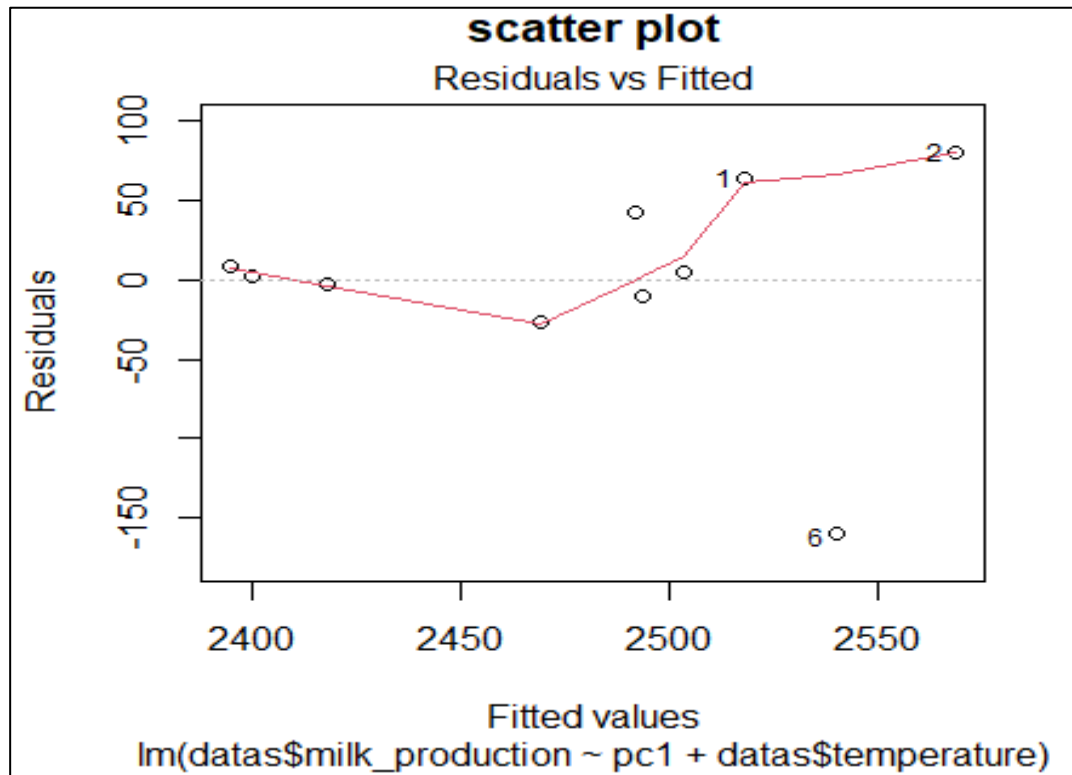
The Residual standard error of our model is 74.01 on 7 degrees of freedom, which means that our model can be off by an average of 74.01 while predicting the milk production.

While predicting the milk production using the principal component and temperature from 2011 to 2020, the following value are obtained.

Year	Milk production
2011	2518.257
2012	2569.010
2013	2494.006
2014	2492.207
2015	2503.505
2016	2540.340
2017	2469.700
2018	2418.186
2019	2395.080
2020	2400.120

Comparing this to the original data we can see that the predicted values and original values have only a small difference between them. i.e., our model is a good fit.

The following is the residual plot obtained,



Here the residuals appear on the y axis and the fitted values appear on the x axis. The residual = 0 line corresponds to the estimated regression line. We can see that the residuals bounce randomly around the 0 line. Which suggests that the assumption that the relationship is linear is reasonable.

➤ RESULT

The fitted regression equation is,

$$Y = 2652.893 - 37.119 * X1 - 7.521 * X2$$

Where,

Y = milk produced

X1 = principal component

X2 = temperature

7.2 Analysis of meat production

Poultry meat production in Kerala from 2011 to 2020

Year	Meat Production	No. of cases treated under various diseases	Rainfall (mm)	Precipitation (mm/day)	Temperature (C)
2011	425.57	111726	3035.1	5.05	22.48
2012	400.99	116209	2151.1	3.74	23.84
2013	416.12	120698	3255.4	5.55	23.04
2014	445.83	138456	3046.4	5.25	23.32
2015	466.04	151593	2600.6	4.6	21.99
2016	468.84	135396	1870.9	3.69	23.52
2017	468.88	173049	2664.9	5.49	22.38
2018	457.41	183703	3571.5	5.9	21.51
2019	452.71	184406	3118.1	5.95	25.51
2020	461.44	200919	2999	6.13	22.25

- ✓ The above table contains the annual meat production from 2011 to 2020, the number of cases of diseases treated, amount of rainfall, precipitation, and temperature in the respective years.

ANALYSIS

Here, we have to check the effect of various factors like diseases, rainfall, precipitation and temperature on the poultry meat production. So, meat production is the response variable or dependent variable. The number of cases treated under

various diseases, amount of rainfall, precipitation and temperature are intercorrelated. i.e., a multicollinearity exists. To fix multicollinearity, here we are using principal component analysis.

To quantify how much the variance of the estimated regression coefficients is increased due to multicollinearity among the predictor variables, Variance Inflation Factor (VIF) is tested and the below inference is obtained.

Variables	VIF
No. of cases treated under various diseases	3.304809
Rainfall	8.130041
Precipitation	12.505888
Temperature	1.133507

Effect of diseases shows a mild presence of multicollinearity, effect of rainfall shows a moderate multicollinearity, effect of precipitation shows a severe presence of multicollinearity and there is no effect in temperature.

Using this, a regression model is fitted. The equation of regression model is obtained as follows:

$$Y = 542.001 + 4.459 * X_1 - 4.160 * X_2$$

Where,

Y = milk produced

X_1 = principal component

X_2 = temperature

The coefficients of the variables X_1 , X_2 gives the amount of change that occur in the dependent variable Y due to a change in one unit in them. i.e., change in one unit in principal component will result in 4.459 change in meat production and likewise.

The standard error is variability to expect in coefficient, which captures sampling variability, so variation in intercept can be up to 172.191 and so on. The standard errors are obtained as:

	Std. Error
Intercept	172.191
pc1	5.664
temperature	7.484

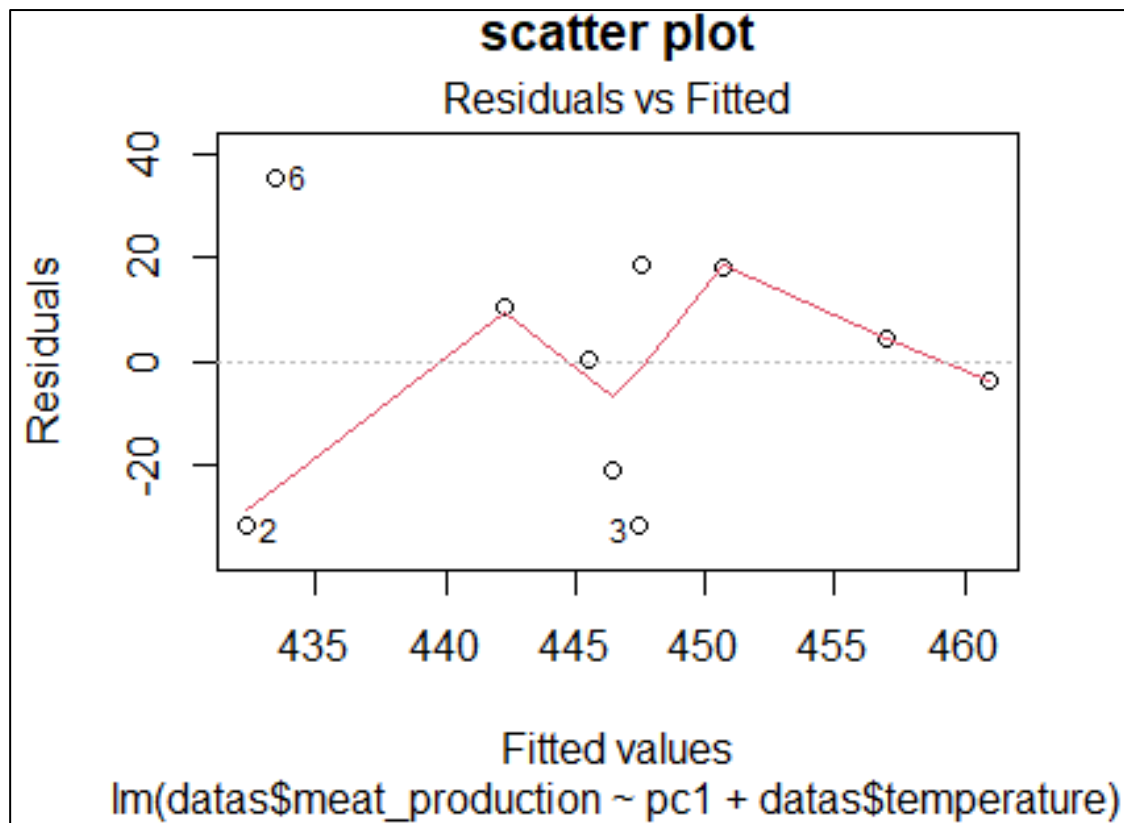
The Residual standard error of our model is 25.26 on 7 degrees of freedom, which means that our model can be off by an average of 25.26 while predicting the meat production.

While predicting the poultry meat production using the principal component and temperature from 2011 to 2020, the following value are obtained.

Year	Meat production
2011	446.4524
2012	432.3660
2013	447.4957
2014	445.5265
2015	447.5974
2016	433.4637
2017	450.7244
2018	460.9570
2019	442.2698
2020	456.9771

Comparing this to the original data we can see that the predicted values and original values have only a small difference between them. i.e., our model is a good fit.

The following is the residual plot obtained,



Here the residuals appear on the y axis and the fitted values appear on the x axis. The residual = 0 line corresponds to the estimated regression line. We can see that the residuals bounce randomly around the 0 line. Which suggests that the assumption that the relationship is linear is reasonable.

➤ RESULT

The fitted regression equation is,

$$Y = 542.001 + 4.459 \cdot X1 - 4.160 \cdot X2$$

Where,

Y = milk produced

X1 = principal component

X2 = temperature

7.3 Analysis of egg production

The annual egg production of Kerala from 2011 to 2020

year	Egg production (In lakh nos)	No. of cases treated under various diseases	Rainfall (mm)	Precipitation (mm/day)	Temperature (C)
2011	17049	111726	3035.1	5.05	22.48
2012	22376	116209	2151.1	3.74	23.84
2013	24770	120698	3255.4	5.55	23.04
2014	25036	138456	3046.4	5.25	23.32
2015	24425	151593	2600.6	4.6	21.99
2016	23444	135396	1870.9	3.69	23.52
2017	23481	173049	2664.9	5.49	22.38
2018	22906	183703	3571.5	5.9	21.51
2019	21845	184406	3118.1	5.95	25.51
2020	22135	200919	2999	6.13	22.25

- ✓ The above table contains the annual egg production of Kerala from 2011 to 2020, the number of cases treated under various diseases, amount of rainfall, precipitation and temperature in the respective years.

ANALYSIS

Here, we have to check the effect of various factors like disease, rainfall, precipitation and temperature on the egg production. So, egg production is the response variable or dependent variable. The number of cases treated under various diseases, amount

of rainfall, precipitation and temperature are intercorrelated. i.e., a multicollinearity exists. To fix multicollinearity, here we are using principal component analysis.

To quantify how much the variance of the estimated regression coefficients is increased due to multicollinearity among the predictor variables, Variance Inflation Factor (VIF) is tested and the below inference is obtained.

Variables	VIF
No. of cases treated under various diseases	3.304809
Rainfall	8.130041
Precipitation	12.505888
Temperature	1.133507

Effect of diseases shows a mild presence of multicollinearity, rainfall shows the presence of moderate multicollinearity, effect of precipitation shows a severe presence of multicollinearity and there is no effect in temperature.

Using this, a regression model is fitted. The equation of regression model is obtained as follows:

$$Y = 23525.60 - 12.09 * X_1 - 33.89 * X_2$$

Where,

Y = egg produced

X₁ = principal component

X₂ = temperature

The coefficients of the variables X₁, X₂ gives the amount of change that occur in the dependent variable Y due to a change in one unit in them. i.e., change in one unit in principal component will result in 12.09 change in egg production and likewise.

The standard error is variability to expect in coefficient, which captures sampling variability, so variation in intercept can be up to 17656.48 and so on. The standard errors are obtained as:

	Std. Error
Intercept	17656.48
pc1	580.84
temperature	767.38

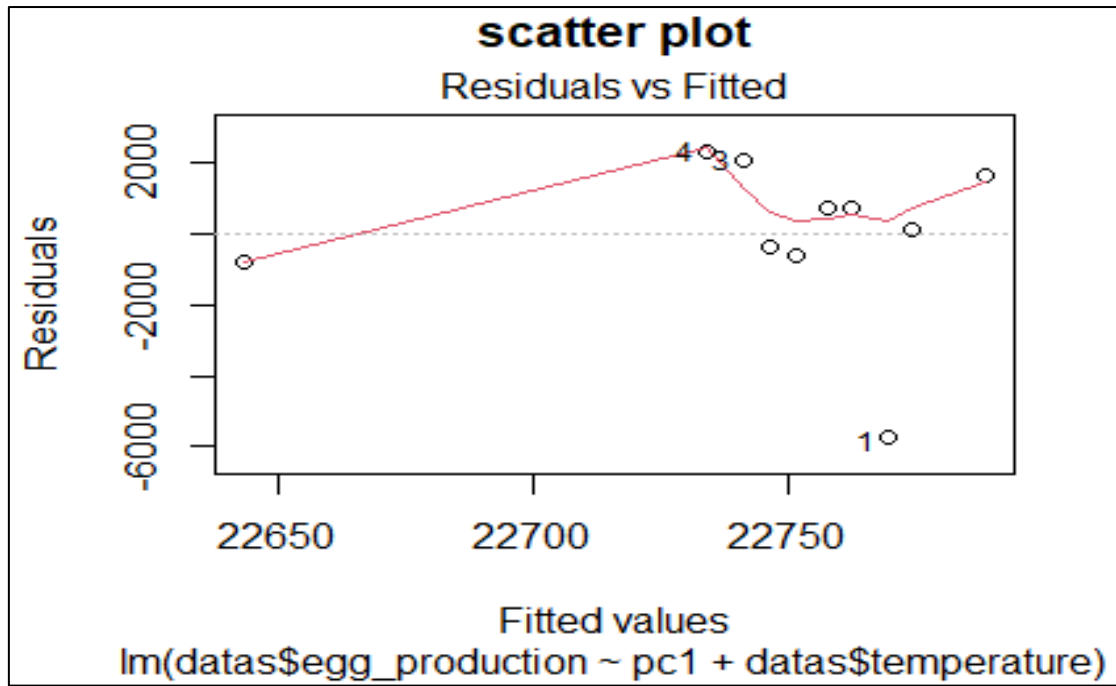
The Residual standard error of our model is 2591 on 7 degrees of freedom, which means that our model can be off by an average of 2591 while predicting the meat production.

While predicting the poultry meat production using the principal component and temperature from 2011 to 2020, the following value are obtained.

Year	Egg production
2011	22769.28
2012	22746.04
2013	22741.15
2014	22733.85
2015	22788.30
2016	22757.52
2017	22762.21
2018	22773.76
2019	22643.76
2020	22751.13

Comparing this to the original data we can see that the predicted values and original values have only a small difference between them. i.e., our model is a good fit.

The following is the residual plot obtained,



Here the residuals appear on the y axis and the fitted values appear on the x axis. The residual = 0 line corresponds to the estimated regression line. We can see that the residuals bounce randomly around the 0 line. Which suggests that the assumption that the relationship is linear is reasonable.

➤ RESULT

The fitted regression equation is,

$$Y = 23525.60 - 12.09 \cdot X_1 - 33.89 \cdot X_2$$

Where,

Y = milk produced

X₁ = principal component

X₂ = temperature

7.4 Conclusion

- We analyse the annual milk production of cattle in Kerala from 2011 to 2020. We use principal component analysis for study the relationship between the independent and dependent variables. The milk production is the dependent variable and the number of cases treated under Mastitis, amount of rainfall, precipitation, temperature and average feed consumption of the cattle are independent variables and we find there exist a multicollinearity according to the value of VIF.

High multicollinearity observed in effect of disease (Mastitis), rainfall, precipitation, average feed consumption.

We also predicted the milk production by principal component and temperature and we observe a small difference between the original data and the predicted values. Thus, we conclude that our model is a good fit.

By plotting the scatter plot, we analyse that the relationship is linear.

- We analyse the annual poultry meat production in Kerala from 2011 to 2020. We use principal component analysis for study the relationship between the independent and dependent variables. The poultry meat production is the dependent variable and the number of cases treated under various diseases, amount of rainfall, precipitation and temperature are independent variables and we find there exist a multicollinearity according to the value of VIF.

Effect of diseases shows a mild presence of multicollinearity, effect of rainfall shows a moderate multicollinearity, effect of precipitation shows a severe presence of multicollinearity and there is no effect in temperature.

We also predicted the poultry meat production by principal component and temperature and we observed a small difference between the original data and the predicted values. Thus, we conclude that our model is a good fit.

By plotting the scatter plot, we analyse that the relationship is linear.

- We analyse the annual egg production in Kerala from 2011 to 2020. We use principal component analysis for study the relationship between the independent and dependent variables. The egg production is the dependent

variable and the number of cases treated under various diseases, amount of rainfall, precipitation and temperature are independent variables and we find there exist a multicollinearity according to the value of VIF.

Effect of diseases shows a mild presence of multicollinearity, effect of rainfall shows a moderate multicollinearity, effect of precipitation shows a severe presence of multicollinearity and there is no effect in temperature.

We also predicted the egg production by principal component and temperature and we observe a small difference between the original data and the predicted values. Thus, conclude that our model is a good fit.

By plotting the scatter plot, we analyse that the relationship is linear.

Chapter-8

A Statistical Study On Production Of Meat In Kerala

8.1 The annual meat production of cattle in Kerala from 1992 to 2021

Year	Meat production
1992	32.98
1993	33.74
1994	34.52
1995	35.32
1996	36.11
1997	69.79
1998	71.76
1999	73.74
2000	82.46
2001	89.4
2002	95.42
2003	86.99
2004	86.89
2005	83.53
2006	76.3
2007	77.58
2008	124.1
2009	117.51
2010	124.55
2011	425.57
2012	400.99
2013	416.12
2014	445.83
2015	466.04
2016	468.84
2017	468.88
2018	457.41
2019	452.71
2020	461.44
2021	463.51

The above table contain the details of meat production of cattle in Kerala from 1992 to 2021.

8.2 Time series analysis for the meat production in Kerala

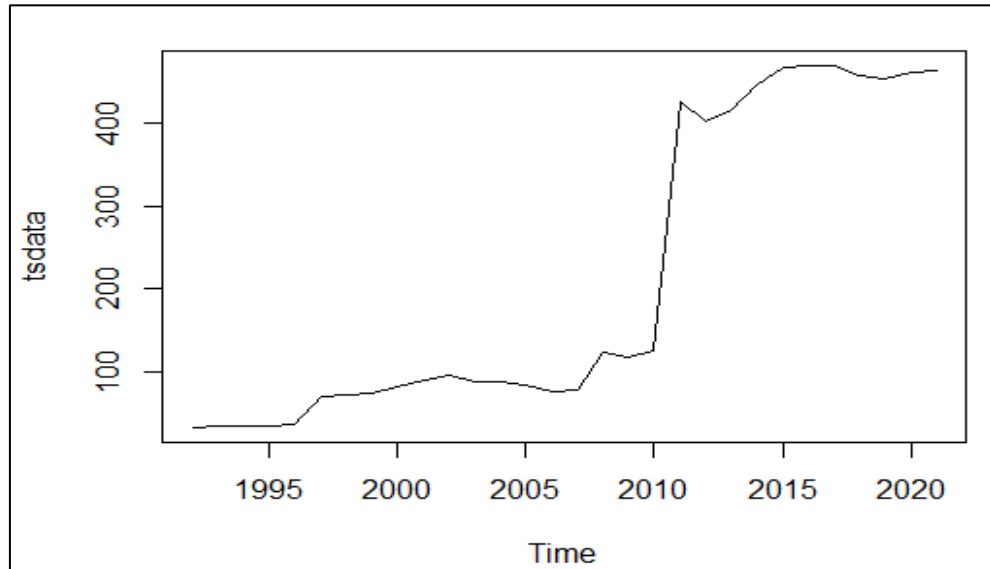


Figure 8.1

- The figure 8.1 show the meat production in Kerala of cattle from 1992 to 2020
In order to test the stationarity of the data, ACF test is used

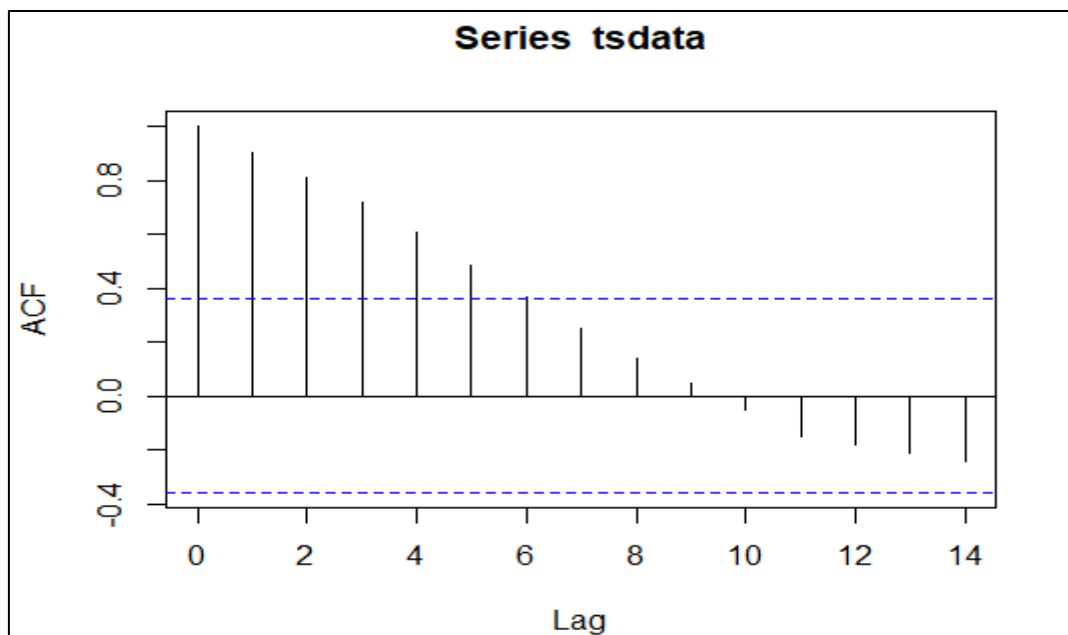


Figure 8.2

- By applying ACF test, figure 8.2 is obtained. From the figure 8.2, we can see that lot of points lies outside the error band which indicate that the data is not stationary.

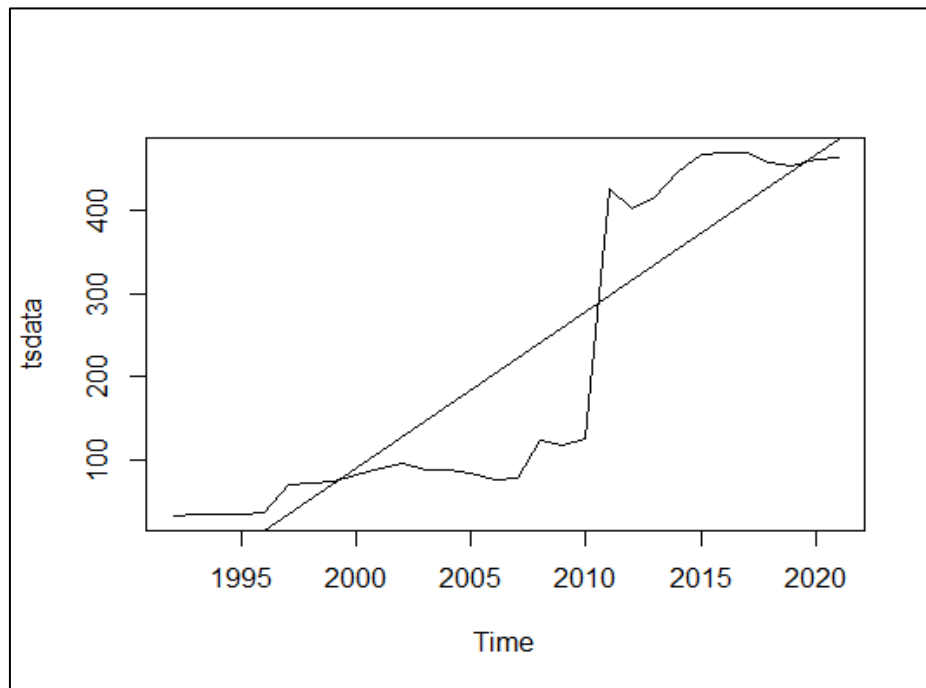


Figure 8.3

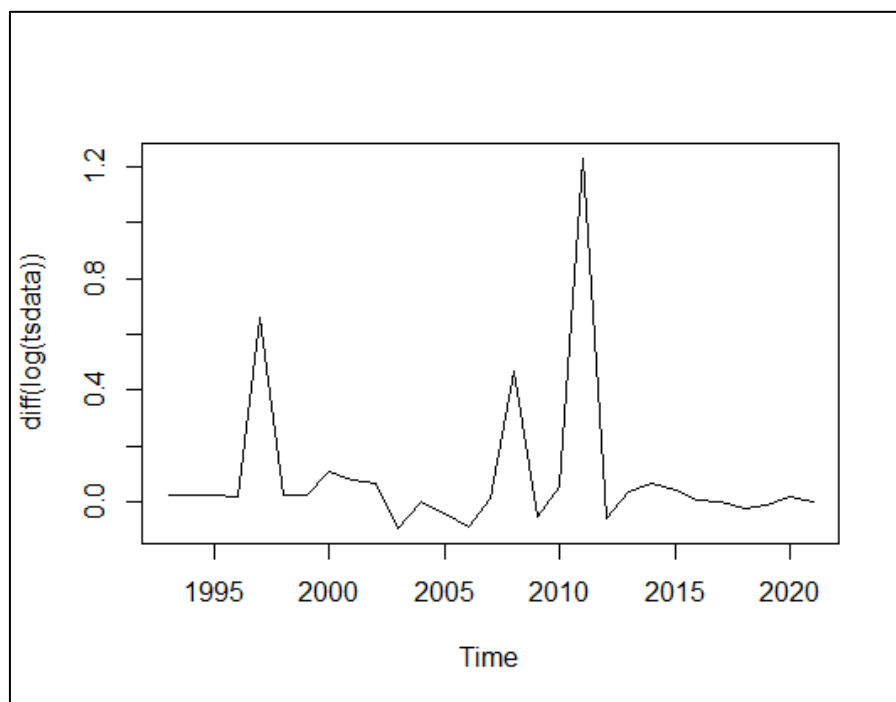


Figure 8.4

- The figure 8.3 gives the trend of the data and the figure 8.4 gives the seasonality of the data. Figure 8.3 indicates that there is a steady trend from 1992 to 1996 and an increase in trend from 1996 to 2021. From figure 8.4, we can conclude that the seasonality is negligible.

ACF differentiation and PACF differentiation

ACF differentiation is to remove any trend or seasonality present in the data, making it stationary. ACF plot can help to help to determine the order of the autoregressive (AR) component in an ARIMA model. If the ACF decays exponentially or gradually, it suggests an AR component.

The PACF plot helps to determine the order of the moving average (MA) component in an ARIMA model. If the PACF cuts off after a certain lag, it suggests a moving average component

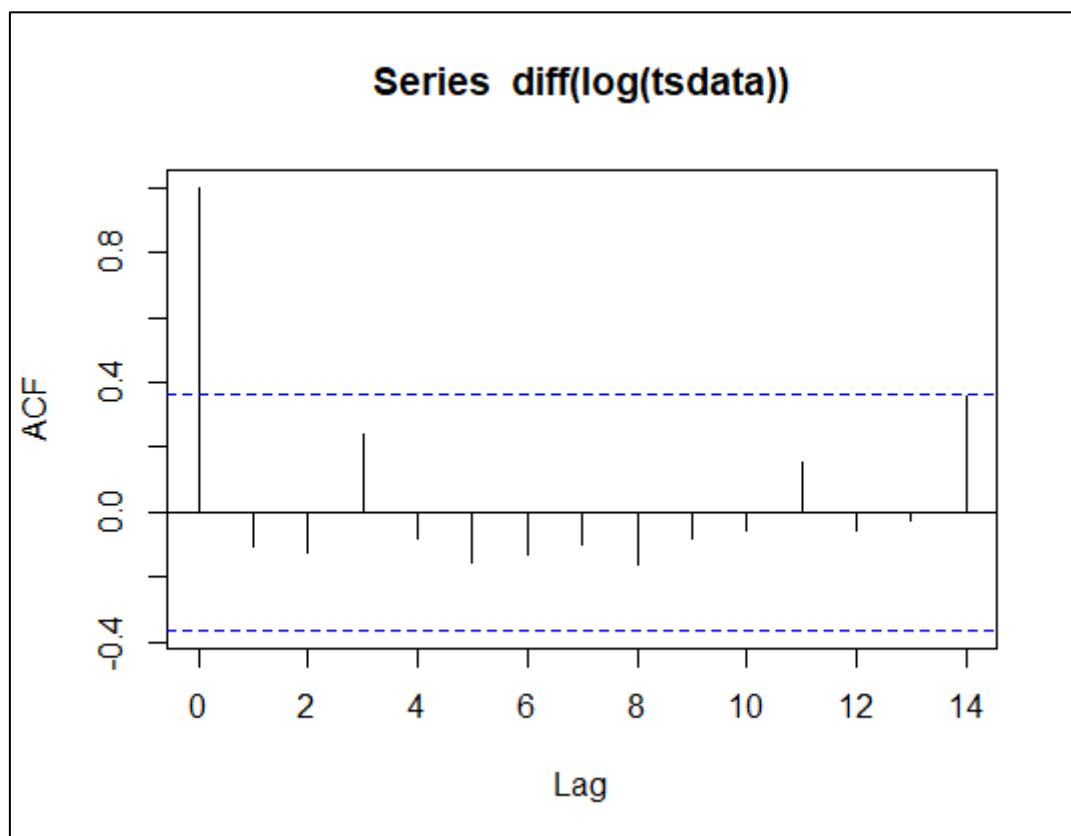


Figure 8.5

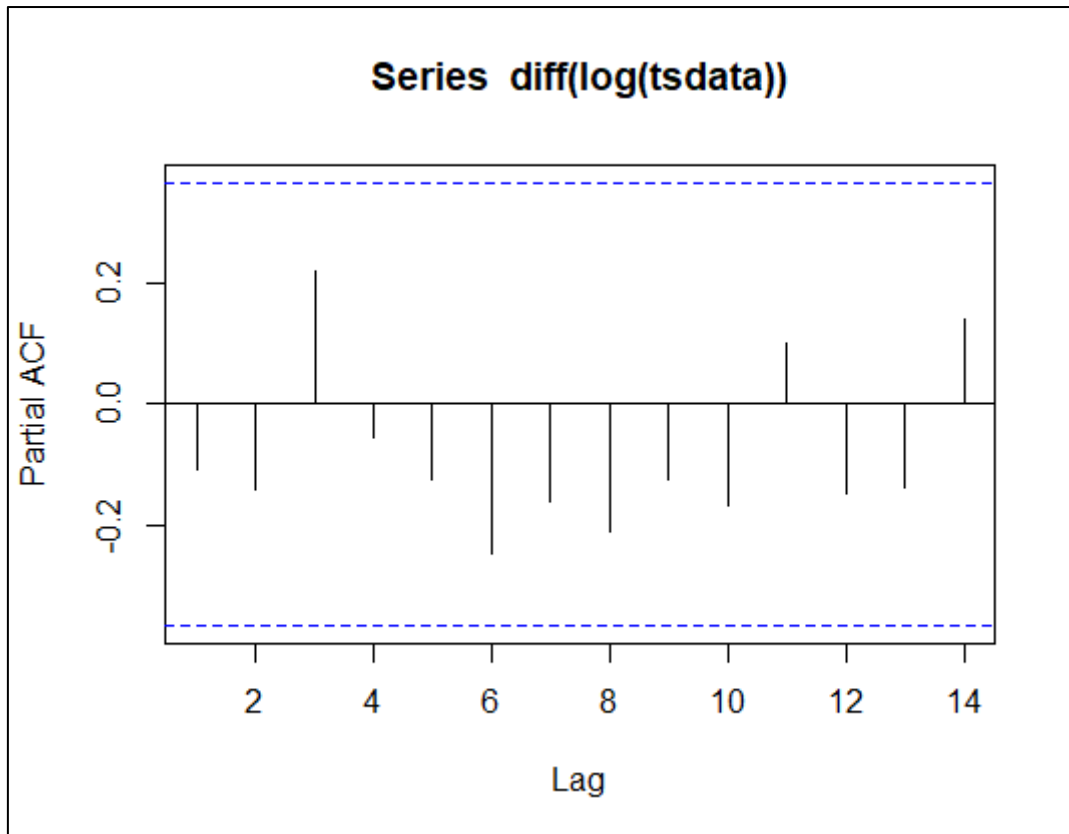


Figure 8.6

➤ Fitting an ARIMA model

From the figure 8.5 and figure 8.6, the ACF differentiation and PACF differentiation tells us that the model that best fits the data is ARIMA (2,1,0). i.e., $p=2$, $d=1$ and $q=0$.

- $p=2$ tells that the current value of the time series is related to the two immediately preceding value.
- $d=1$ tells that the time series is not stationary and requires differencing once to make it stationary.
- $q=0$ tells us that this model does not have any moving average component, and any autocorrelation in the data is due to the autoregressive (AR) component of the ARIMA model

□ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated, `Box.test()` function is used. The observed p-value = 0.9743 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e., our model is good.

➤ Forecasting with ARIMA

We use the ARIMA to forecast future time points in the series. For this we use forecast() function from the “forecast” package. The meat production for the next 8 years (2022 to 2029) is given below and the figure 8.7 gives the graphical representation of the same, with a 95% confidence.

Year	Meat production (forecasted)
2022	460.8641
2023	463.5445
2024	465.6151
2025	466.1215
2026	467.5845
2027	469.1353
2028	470.3395
2029	471.6822

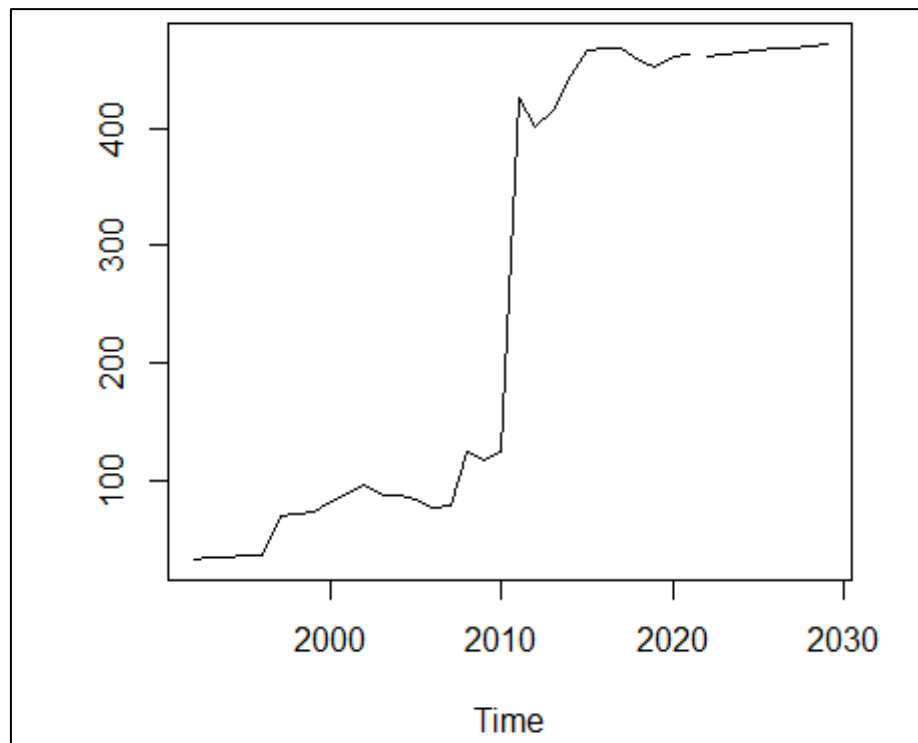


Figure 8.7

➤ Finding difference between the actual data and forecasted data

To find the difference between the actual data and the forecasted data, we choose the meat production of cattle in Kerala from 1992 to 2017 and forecasted for the coming 2 years.

Year	Actual data	Forecasted data	Difference
2018	457.41	485.1489	27.7389
2019	452.71	494.6639	41.9539
2020	461.44	500.7583	39.3183
2021	463.51	511.7834	48.2734

➤ RESULTS

- ☐ The best model that fits the data is ARIMA (2,1,0).
- ☐ For the next 8 years (2022 to 2029), meat production of cattle in Kerala will increase.
- ☐ There is a slight difference between the actual data and the forecasted data.

8.3 Conclusion

We analyse the trends in the annual meat production of cattle in Kerala from the year 1992 to 2021, and predict the rate of prediction for future.

Here we have a sample of annual meat production of cattle in Kerala from the year 1992 to 2021. From the data we analyse the trend, model is fitted and the price and production of meat are predicted for the next 8 years separately. Then the following conclusions are reached.

In the case of cattle, we could clearly see an increasing trend in the data. ARIMA (2,1,0) is the best model that fits the data. Also, for the next 8 years, the model gave a steady increase in the price of cow milk.

Chapter-9

A Statistical Study On Disease Treated In Cattle

9.1 The total diseases treated (cattle) in Kerala from 1966 to 2021

Year	Total cases treated
1966	1430594
1967	1627001
1968	1642049
1969	1705122
1970	1663653
1971	1883660
1972	1777272
1973	1877677
1974	1955040
1975	2190063
1976	2254128
1977	2638064
1978	2712842
1979	2790237
1980	3009565
1981	3156885
1982	3411279
1983	3300939
1984	3395647
1985	4717040
1986	3317222
1987	3725201
1988	4147022
1989	3621295
1990	3866712
1991	3783036

1992	3290580
1993	3329500
1994	3400886
1995	3441264
1996	3851009
1997	4257629
1998	3893319
1999	3738038
2000	3884731
2001	4161224
2002	4924597
2003	4628952
2004	4879854
2005	5028750
2006	5259660
2007	5015907
2008	4873206
2009	4663331
2010	4287006
2011	4829694
2012	4814817
2013	5078368
2014	5452171
2015	5669572
2016	5674920
2017	6252149
2018	6666862
2019	6634211
2020	7742731
2021	8278918

- ✓ The above table contain the details of total diseases treated (cattle) in Kerala from 1966 to 2021.

9.2 Time series analysis for the total disease treated in Kerala

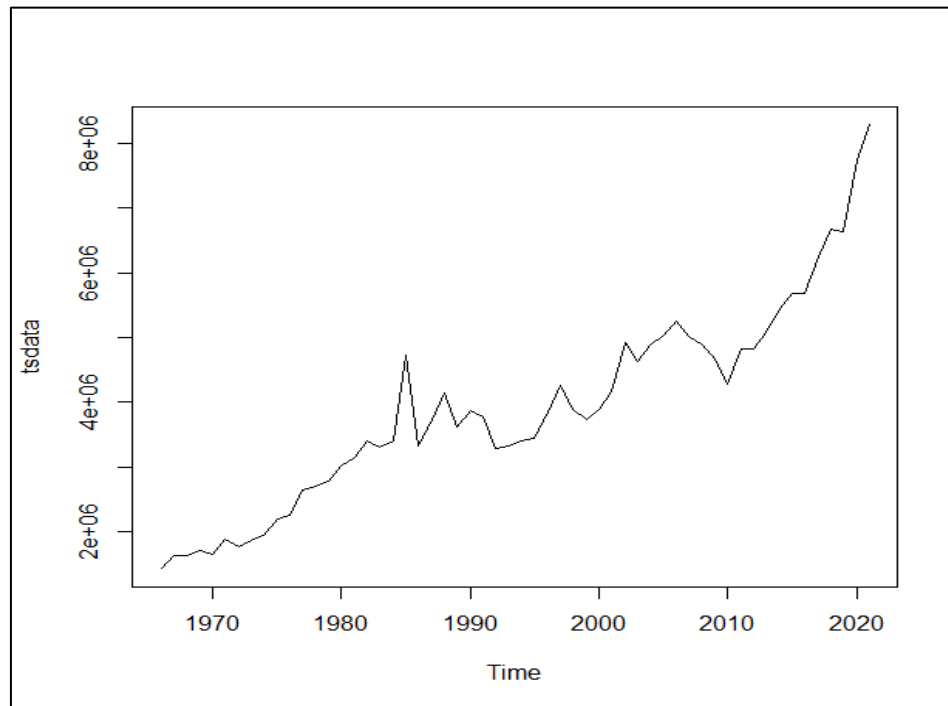


Figure 9.1

The figure 9.1 shows the total diseases treated in Kerala of cattle from 1966 to 2021. In order to test the stationarity of the data, ACF test is used.

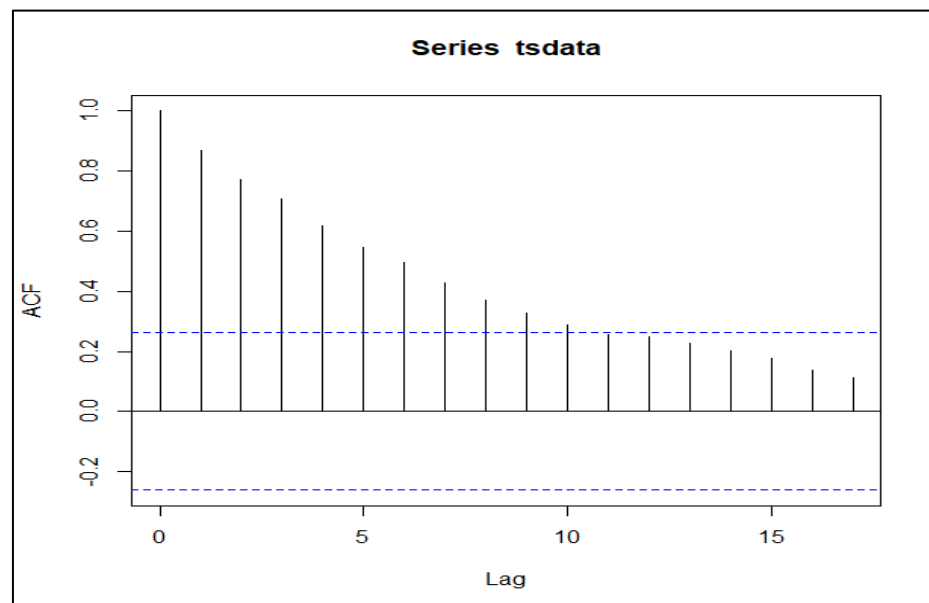


Figure 9.2

By applying ACF test, figure 9.2 is obtained. From the figure 9.2, we can see that lot of points lies outside the error band which indicate that the data is not stationary.

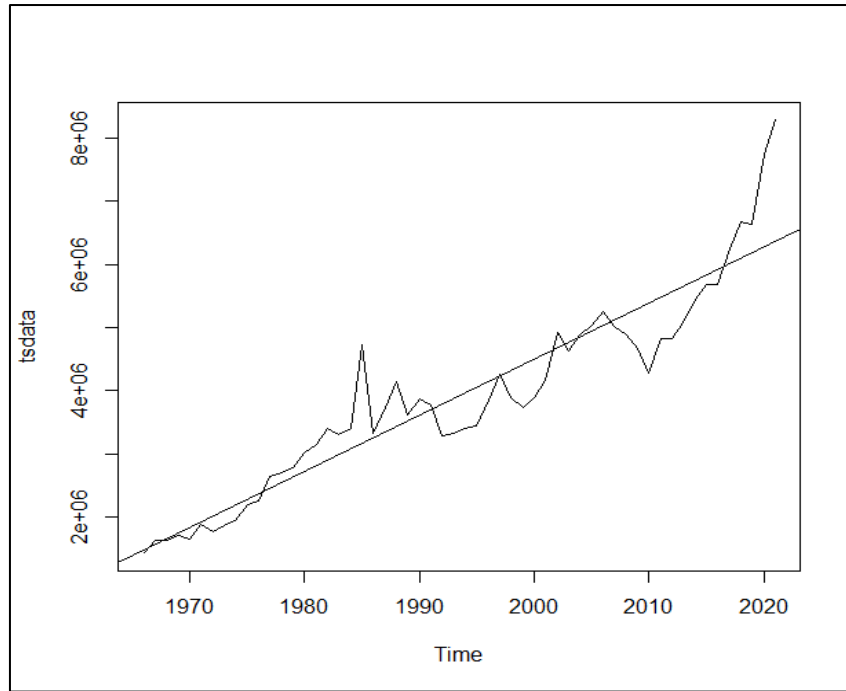


Figure 9.3

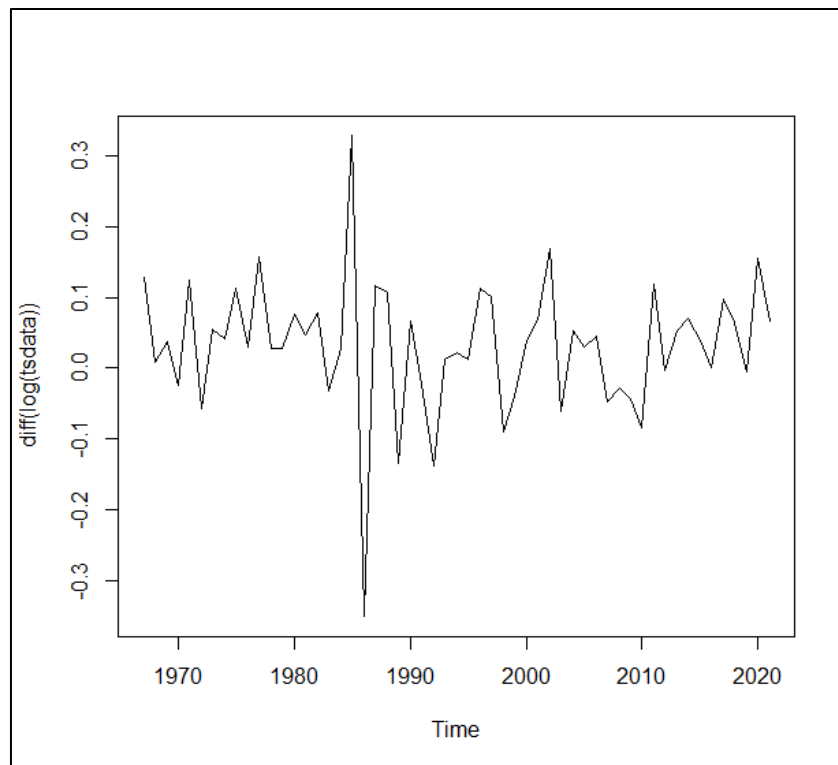


Figure 9.4

The figure 9.3 gives the trend of the data and the figure 9.4 gives the seasonality of the data. Figure 9.3 indicates that there is an upward trend from 1966 to 2021. From figure 9.4, we can conclude that the seasonality is negligible.

ACF differentiation and PACF differentiation:

ACF differentiation is to remove any trend or seasonality present in the data, making it stationary. ACF plot can help to help to determine the order of the autoregressive (AR) component in an ARIMA model. If the ACF decays exponentially or gradually, it suggests an AR component.

The PACF plot helps to determine the order of the moving average (MA) component in an ARIMA model. If the PACF cuts off after a certain lag, it suggests a moving average component.

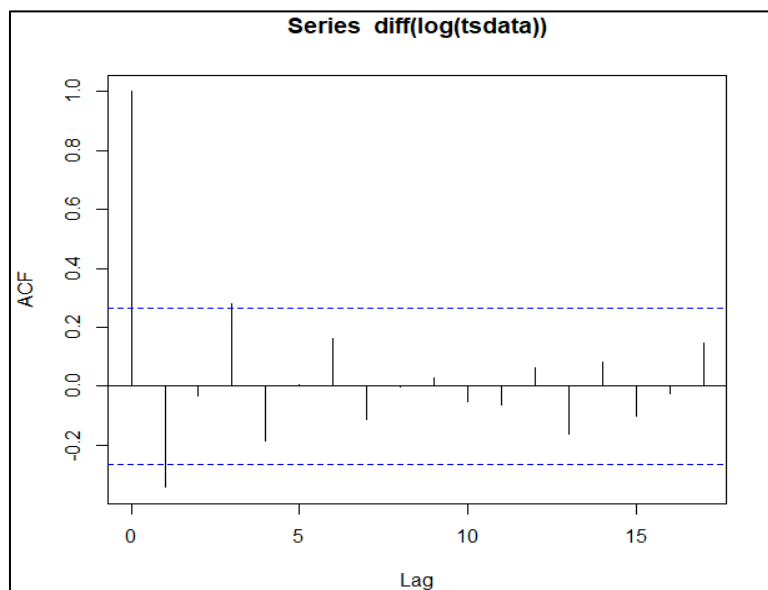


Figure 9.5

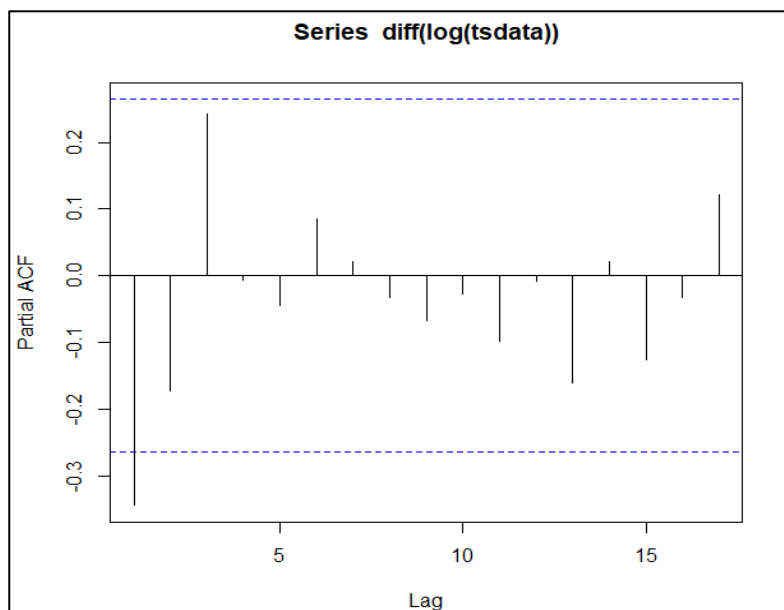


Figure 9.6

➤ Fitting an ARIMA model

From the figure 9.5 and figure 9.6, the ACF differentiation and PACF differentiation tells us that the model that best fits the data is ARIMA (2,1,0). i.e., $p=2$, $d=1$ and $q=0$.

- $p=2$ tells that, the order of the auto-regressive (AR) component. In this case the model considers the two most recent lagged values of the time series.
- $d=1$ tells that the time series is not stationary and requires differencing once to make it stationary.
- $q=0$ tells us that, there is no lagged forecast error in the prediction equation. (This model does not have any moving average component, and any auto-correlation in the data which is due to the auto-regressive (AR) component of the ARIMA model).

□ Testing the distribution of errors in ARIMA model

In order to check if the successive forecast errors are correlated, `Box.test()` function is used. The observed p-value = 0.6073 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors. i.e., our model is good.

➤ Forecasting with ARIMA

We use the ARIMA to forecast future time points in the series. For this we use `forecast()` function from the “forecast” package. The chances of occurrence of diseases in cattle for the next 9 years (2022 to 2030) is given below and the figure 9.7 gives the graphical representation of the same, with a 95% confidence.

Year	Point Forecast
2022	8662911
2023	9601060
2024	10324773
2025	11076936
2026	12043157
2027	12991937
2028	14017356
2029	15164924
2030	16377405

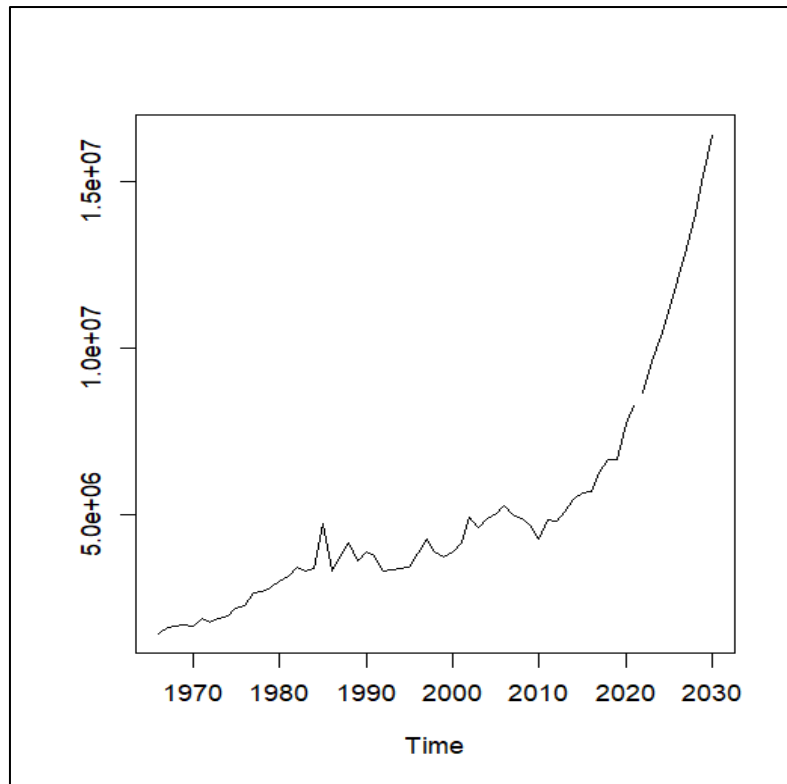


Figure 9.7

➤ Finding difference between the actual data and the forecasted data

To find the difference between the actual data and the forecasted data, we choose the total disease treated in cattle from 1966 to 2017 and forecasted for the coming 4 years.

Year	Actual data	Forecasted data	Difference
2018	6666860	6432799	234061
2019	6634211	6661407	27196
2020	7742731	7083954	658777
2021	8278918	7378115	900803

➤ RESULTS

- ☐ The best model that fits the data is ARIMA (2,1,0).
- ☐ For the next 9 years (2022 to 2030), chances of occurrence of diseases in cattle will increase.
- ☐ There is a slight difference between the actual data and the forecasted data.

9.3 Conclusion

We analyse the trend of the Total Diseases treated in cattle in Kerala from the year 1966 to 2021 and predict the chances of occurrence of diseases in the upcoming 9 years.

Here we are provided with a sample data which includes the total diseases treaded in Cattle in Kerala from 1966 to 2021. For this data, trends are analysed, models are fitted, and the chances of occurrence of the diseases are predicted for the next 9 years separately. Then the following conclusions is obtained.

- We can clearly see an upward trend in the occurrence of diseases in cattle from 1966 to 2021. We can also say that the data is not stationary.
- ARIMA (2,1,0) is the best model that fits the data.
- For the next 9 years the model gives a steady increase in the chances of occurrence of diseases in cattle.
- There is only a slight difference between the actual data and the forecasted data.

Chapter-10

A Statistical Study On Season Wise Milk Production In Kerala

10.1 Season wise and category wise milk production of Kerala since 1990 (in 000 tonnes)

Year	Cattle			Total	Buffalo			Total	Goat			Total	Grand Total
	Summer	Rainy	Winter		Summer	Rainy	Winter		Summer	Rainy	Winter		
1990-91	455.11	540.18	496.56	1491.85	34.07	36.25	30.86	101.18	30.93	35.72	29.89	96.54	1689.57
1991-92	481.5	571.12	523.1	1575.69	35.76	38.16	32.96	106.88	32.06	37.33	32.57	101.96	1784.53
1992-93	517.13	602.29	555.367	1674.79	36.47	38.32	34.51	109.3	33.3	37.98	34.06	105.34	1889.43
1993-94	564.87	625.76	590.91	1781.54	36.67	38.55	35.01	110.23	34.98	39.11	35.47	109.56	2001.33
1994-95	601.85	665.67	629.6	1897.12	35.93	37.84	34.82	108.59	35.75	39.85	36.94	112.54	2118.25
1995-96	648.79	684.92	658.6	1992.26	28.89	29.58	26.82	85.29	36.12	40.75	37.8	114.67	2192.22
1996-97	683.46	700.15	674.2	2057.83	26.97	29	25.85	81.82	38.33	42.1	38.01	118.44	2258.09
1997-98	716.13	725.52	705.8	2147.48	24.65	25.82	23.58	74.05	39.87	42.73	38.8	121.4	2342.93
1998-99	745.31	762.06	712.3	2219.67	24.99	26.01	23.8	74.8	44.06	38.34	43.17	125.57	2420.04
1999-2000	722.24	873.77	731.5	2327.52	21.65	26.67	21.71	70.03	39.2	48	40.52	127.72	2525.27
2000-01	762.6	872.42	784.8	2419.83	19.9	22.84	20.71	63.45	38.87	43.87	39.38	122.12	2605.4
2001-02	784.47	909.66	843.5	2537.61	20.12	22.53	20.34	62.99	38.52	41.83	36.93	117.28	2717.88
2002-03	725.62	804.63	734.5	2264.71	15.77	16.57	15.69	48.03	36.15	37.35	32.74	106.24	2418.98
2003-04	649.18	703.97	638.8	1991.99	12.34	14.61	12.84	39.79	26.65	27.95	24.17	78.77	2110.55
2004-05	630.35	683.85	598.2	1912.43	11.86	14.24	11.83	37.93	25.06	27.1	22.7	74.86	2025.22
2005-06	629.11	685.02	633.4	1947.52	10.79	13.58	12.1	36.47	24.63	28.46	26.12	79.21	2063.2
2006-07	631.56	689.51	681.9	2002.94	10.16	10.99	8.7	29.85	26.25	30.16	29.68	86.09	2118.88
2007-08	687.31	734.91	701.2	2123.39	9.36	9.82	8.17	27.35	29.24	34.14	33.52	96.9	2247.64
2008-09	718.34	802.47	790.3	2311.13	10.2	10.39	15.72	36.31	33.58	34.84	34.67	103.09	2450.53
2009-10	741.47	816.69	808.2	2366.36	12.82	15.46	16.17	44.45	41.91	43.56	40.83	126.3	2537.11
2010-11	754.66	875.51	882.1	2512.3	6.96	8.6	5.43	20.99	33.64	45.48	30.93	110.05	2643.34
2011-12	777.89	898.74	905.36	2581.98	8.97	10.61	7.44	27.02	32.7	44.53	29.99	107.22	2716.22
2012-13	842.12	943.45	863	2648.53	9.01	14.21	14.39	37.61	36.65	35.53	33.29	105.47	2791.61
2013-14	851.58	844.69	787.3	2483.56	20.95	21.5	15.6	58.05	35.3	38.91	37.86	112.07	2653.68

2014-15	866.11	860.73	807.1	2533.96	21.83	21.77	18.81	62.41	37.73	39.21	37.86	114.8	2711.17
2015-16	788.13	867.11	852.7	2507.89	4.53	4.95	3.47	12.95	42.61	42.99	43.28	128.88	2649.72
2016-17	772.99	826.88	781.5	2381.39	4.55	4.18	3.48	12.21	43.05	40.92	42.72	126.69	2520.29
2017-18	778.86	842.87	820.8	2442.54	4.22	4.65	3.8	12.67	38.96	41.54	40.27	120.77	2575.98
2018-19	790.2	834.91	789.9	2415.05	4.37	4.01	3.79	12.17	38.99	42.11	40.36	121.46	2548.68
2019-20	790.24	845.09	767.9	2403.27	3.03	4.16	4.76	11.95	40.68	42.67	45.78	129.13	2544.35
2020-21	759.66	857.13	785.5	2402.24	3.75	2.9	5.78	12.43	43.8	34.8	40.61	119.21	2533.88
2021-22	760.09	860.33	775.6	2395.97	4.11	3.12	5.98	13.21	44.07	36.87	42.34	123.28	2532.46

10.2 Season wise data of milk production (Cattle, Buffalo, Goat) in Kerala during 1990 to 2021

Serial number	Milk Production (In 000 Tonnes)
1	520.11
2	612.15
3	557.31
4	549.32
5	646.61
6	588.6
7	586.9
8	678.59
9	623.94
10	636.52
11	703.42
12	661.39
13	673.53
14	743.36
15	701.36
16	713.8
17	755.25
18	723.17
19	748.76
20	771.25
21	738.08
22	780.65
23	794.07
24	768.21
25	814.36
26	826.41
27	779.27

28	783.09
29	948.44
30	793.74
31	821.37
32	939.13
33	844.9
34	843.11
35	974.02
36	900.75
37	777.54
38	858.55
39	782.89
40	688.17
41	746.53
42	675.85
43	667.27
44	725.19
45	632.76
46	664.53
47	727.06
48	671.61
49	667.97
50	730.66
51	720.25
52	725.91
53	778.87
54	742.86
55	762.12
56	847.7
57	840.71
58	796.2
59	875.71
60	865.2
61	795.26
62	929.59
63	918.49
64	819.56
65	953.88
66	942.79
67	887.78
68	993.19
69	910.64
70	907.83
71	905.1
72	840.75
73	925.67

74	921.71
75	863.79
76	835.27
77	915.05
78	899.4
79	820.59
80	871.98
81	827.72
82	822.04
83	889.06
84	864.88
85	833.56
86	881.03
87	834.09
88	833.95
89	891.92
90	818.48
91	807.21
92	894.83
93	831.84
94	808.27
95	900.32
96	823.87

Here we consider 3 seasons namely Summer, Rainy, Winter respectively.

10.3 Time series analysis of the season wise milk production in Kerala

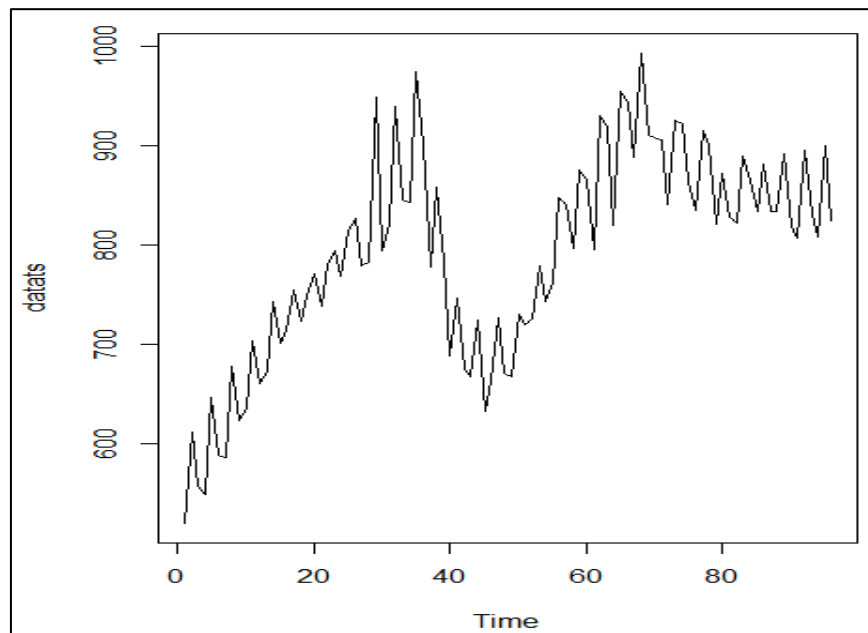


Figure 10.1

The figure 10.1 shows the season wise Milk production (Cattle, Buffalo, Goat) in Kerala from 1992 to 2020 in Summer, Rainy and Winter respectively. In order to test the stationarity of the data, ACF test is used,

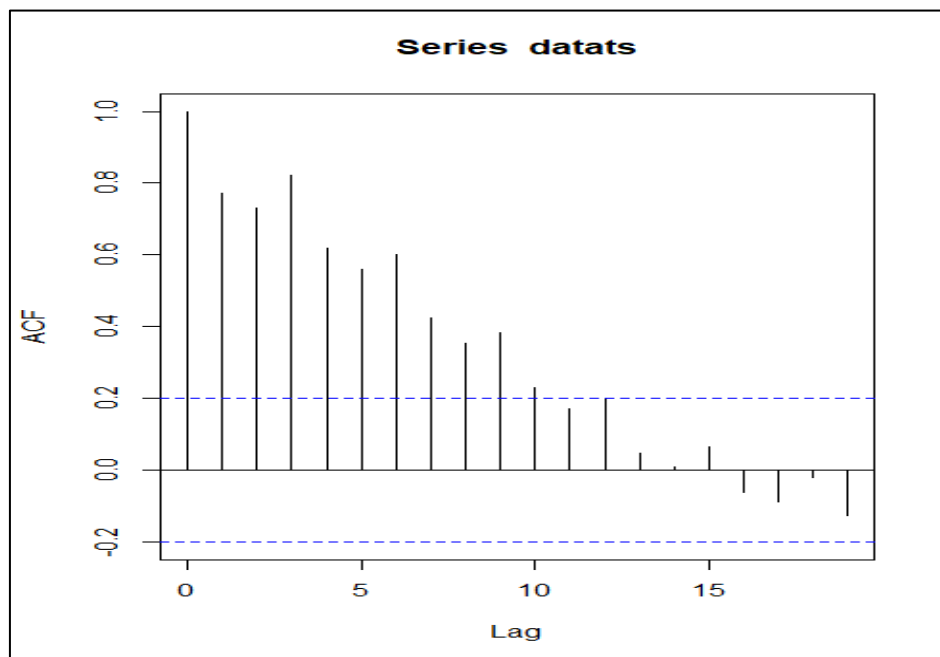


Figure 10.2

By applying ACF test, figure 10.2 is obtained. From the figure 10.2, we can see that lot of points lies outside the error band which indicate that the data is not stationary.

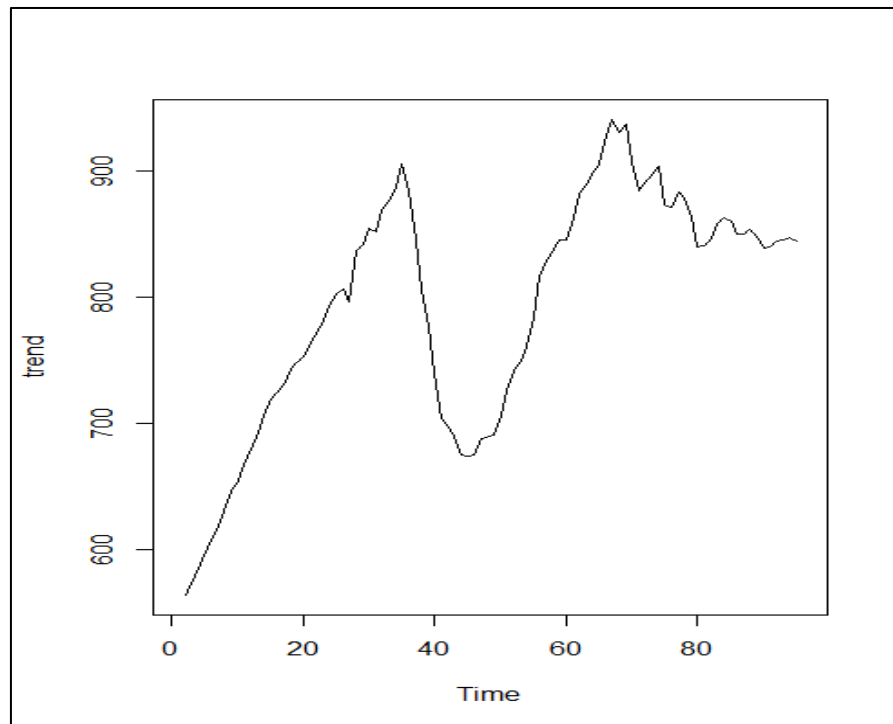


Figure 10.3

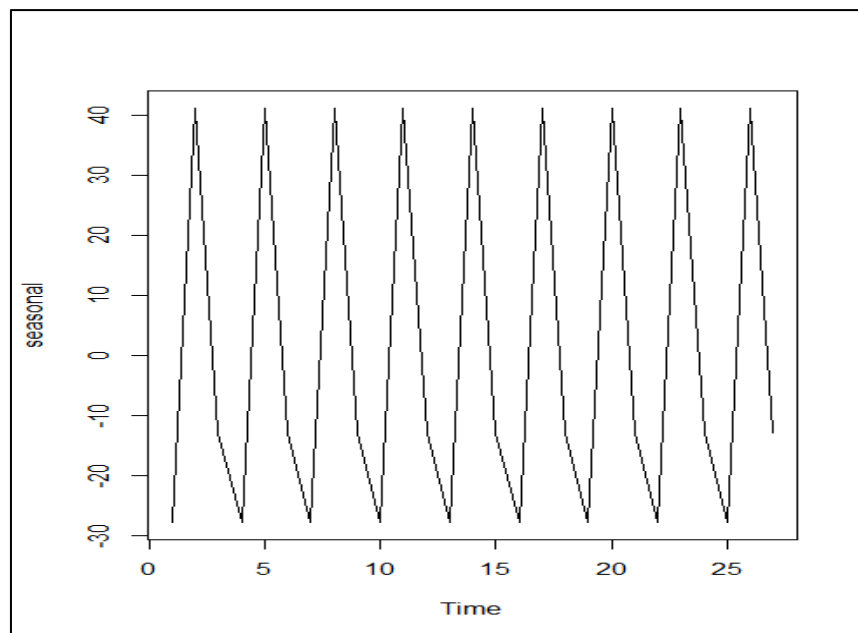


Figure 10.4

The figure 10.3 gives the trend of the data and the figure 10.4 gives the seasonality of the data. Figure 10.3 indicates that there is both upward and downward trend. That is milk production is clearly affected by different seasons.

➤ Fitting of ARIMA model

Using the `auto.arima()` function a model is fitted to the given data. The output tells us that, ARIMA (3,1,1) is the best model that fits our data.

ARIMA (3,1,1) $\Rightarrow$ $p=3$, $d=1$ and $q=1$.

Where,

- $p=3$ means that, the current value of the time series is related to the three immediately preceding value.
- $d=1$ means that, the time series is not stationary and requires differencing once to make it stationary.
- $q=1$ means that, this model takes into account the error term from one preceding/lagged value.

□ Testing the distribution of errors in ARIMA model

In order to check the successive forecast errors are correlated `Box.test()` function is used. The observed p-value is 0.7178 which is greater than 0.05, suggesting that there are no significant autocorrelations between successive errors, i.e., **our model is good**.

➤ Forecasting with ARIMA model

ARIMA models provide another approach to time series forecasting. For this we use `forecast()` function from the “forecast” package.

The seasonality change of milk production of Cattle, Buffalo, Goat for the next 3 years in 3 different seasons are given below and figure 5 gives the graphical representation of the same, with a 95% level of confidence.

Serial Number	Forecast	Lo 95	Hi 95
97	810.0353	739.4714	880.5993
98	885.9288	807.0492	964.8083
99	827.7442	743.7838	911.7046
100	812.8842	694.5330	931.2353
101	875.1017	744.6683	1005.5350
102	831.0152	692.9132	969.1173
103	816.0580	652.1617	979.9543
104	866.8168	690.2024	1043.4313
105	833.5971	648.5106	1018.6835

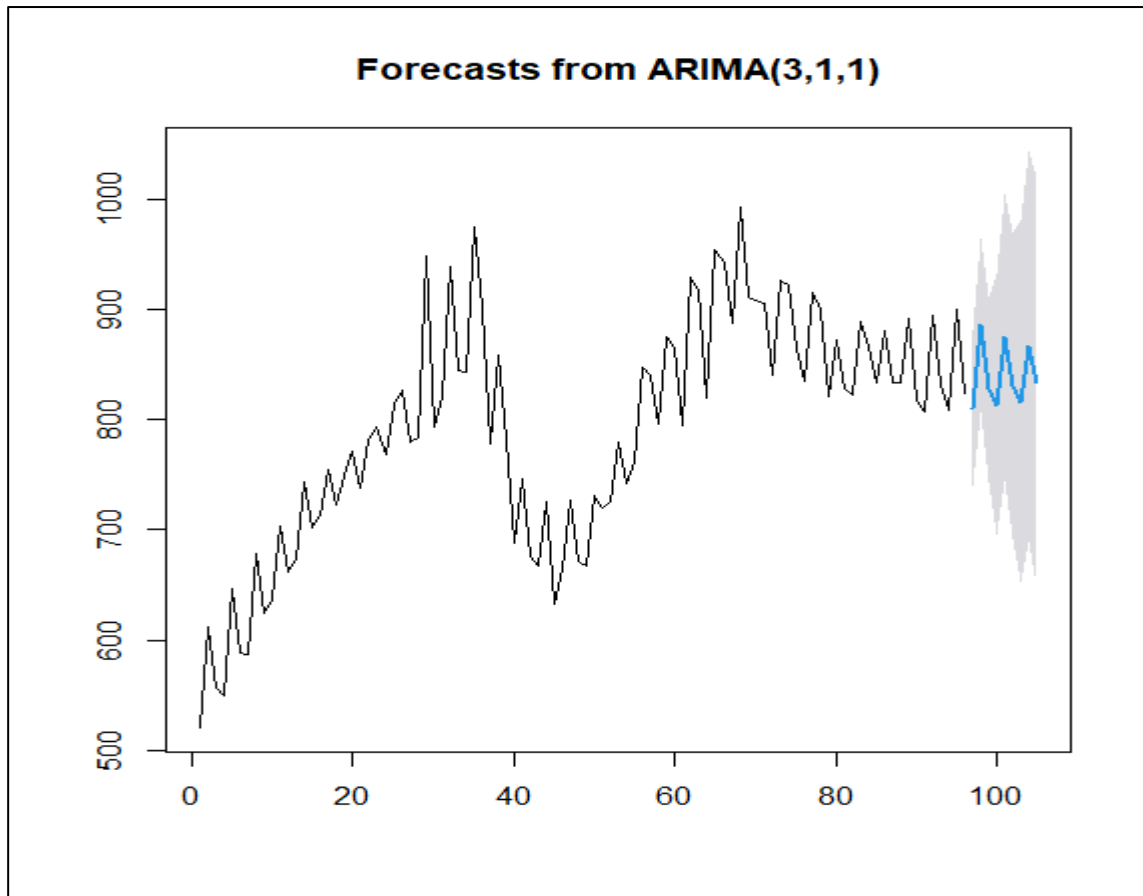


Figure 10.5

Here, the figure 10.5 shows the forecasted graph of season wise milk production in the next 3 years for the three different seasons i.e., from this graph it is clear that milk production of Cattle, Buffalo, Goat is clearly affected by seasons (Summer, Winter and Rainy).

➤ Finding difference between the actual data and forecasted data

To find the difference in the actual data and the forecasted data we choose the season wise milk production of Cattle, Buffalo and Goat, and we forecasted the data for the 4 years. Thus, the following are obtained:

Serial Number	Forecasted Value	Original Value	Difference
93	826.5105	831.84	5.3295
94	812.5864	808.27	4.3164
95	883.6323	900.32	16.6877
96	832.3391	823.87	8.4691

There is only a slight difference in actual value and forecasted value.

➤ RESULTS

- The best model that fits the data is ARIMA (3,1,1).
- From the forecasted values of milk production in the next 3 years for the 3 different seasons, it is clear that the milk production of Cattle, Buffalo, Goat is clearly affected by seasons (Summer, Winter and Rainy).
- There is a slight difference between the actual data and the forecasted data.

10.4 Conclusion

We analyse the trend of the season wise total milk production (Cattle, Buffalo and Goat) from the year 1990 to 2021 for the three different seasons such as Summer, Rainy and Winter respectively, and predict the season wise milk production for the next three years.

Here we are provided with a sample data of season wise milk production (Cattle, Buffalo and Goat) from the year 1990 to 2021 for the three different seasons such as Summer, Rainy and Winter respectively. For this data the trend is analysed, models are fitted, and the season wise milk production is predicted for the next 3 years separately. Then the following conclusions are reached.

- We can see both upward and downward trend in the data. From this we can say that the milk production is clearly affected by the seasonal changes, i.e., the milk production is maximum in the Rainy season and minimum in summer season. We can also say that the data is not stationary.
- ARIMA (3,1,1) is the best model that fits the data.
- From the forecasted graph, we can say that also for the upcoming three years the milk production of Cattle, Buffalo and Goat will be affected by the seasonal changes.
- There is a slight difference between the actual data and the forecasted data

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